Algorithms, Error Correction, Learning, and Numerical Linear Algebra

Foundational Quantum Computation

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CHAPTER 1

Quantum States, Operations, and Measurements

1.1 Hilbert spaces and pure states

Definition 1.1 (Qubit and computational basis)**.** A **qubit** is a normalized vector in the two-dimensional complex Hilbert space $\mathcal{H}_2 = \mathbb{C}^2$, modulo the equivalence relation $|\psi\rangle \sim e^{i\gamma}|\psi\rangle$. In the orthonormal computational basis $\{|0\rangle, |1\rangle\}$,

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad |\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The inner product is conjugate-linear in its first argument: $\langle u|v\rangle = u^\dagger v$.

Factoring out global phase gives the Bloch-sphere parametrization

$$|\psi(\vartheta, \varphi)\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle, \qquad 0 \le \vartheta \le \pi, \quad 0 \le \varphi < 2\pi.$$

Its Bloch vector is $\mathbf{r} = (\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$. Orthogonal pure states occupy antipodal points. The X- and Y-basis eigenstates are

$$|\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}, \qquad |\pm i\rangle = \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}}.$$

Relative phases influence interference, even when the resulting measurement probabilities agree. The following identities will be used repeatedly:

$$e^{i\varphi} = \cos\varphi + i\sin\varphi, \quad e^{i\pi/2} = i, \quad e^{i\pi/4} = \frac{1+i}{\sqrt{2}}, \quad e^{i\pi} = -1.$$

1.2 Linear operators, spectral structure, and quantum gates

A linear operator U is **unitary** if $U^\dagger U = UU^\dagger = I$, so $U^{-1} = U^\dagger$ and $\|U|\psi\rangle\| = \||\psi\rangle\|$. A Hermitian operator $A = A^\dagger$ has real eigenvalues and admits a spectral decomposition $A = \sum_j \lambda_j P_j$. For a unitary operator every eigenvalue has modulus one: $\lambda_j = e^{2\pi i\phi_j}$. Hermiticity does not imply unitarity; for example

$$A = \begin{pmatrix} 1 & i \\ -i & 2 \end{pmatrix} = A^\dagger$$

is not unitary. If a Hermitian operator is an involution, $A^2 = I$, then $P_\pm = (I \pm A)/2$ are its mutually orthogonal spectral projections.

The standard one-qubit gates are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

$$S = \mathrm{diag}(1, i), \qquad T = \mathrm{diag}(1, e^{i\pi/4}), \qquad S^2 = Z, \quad T^2 = S, \quad S^4 = T^8 = I.$$

The Pauli matrices satisfy $X^2 = Y^2 = Z^2 = I$, $XY = iZ$, $YX = -iZ$, $XZ = -ZX$, and

$$HXH = Z, \qquad HZH = X, \qquad HYH = -Y.$$

Consequently $X|0\rangle = |1\rangle$, $Y|0\rangle = i|1\rangle$, $Y|1\rangle = -i|0\rangle$, $Z|+\rangle = |-\rangle$, and $H|0\rangle = |+\rangle$.

Definition 1.2 (Rotation convention)**.** For $P \in \{X, Y, Z\}$, define

$$R_P(\theta) := e^{-i\theta P/2} = \cos\frac{\theta}{2} I - i \sin\frac{\theta}{2} P.$$

In particular,

$$R_X(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \quad R_Y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix},$$

$$R_Z(\theta) = \mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2}).$$

The phase gate $P(\theta) = \mathrm{diag}(1, e^{i\theta})$ is $e^{i\theta/2} R_Z(\theta)$.

A global phase on an **uncontrolled** gate is physically irrelevant; controlling that gate can convert the formerly global phase into an observable relative phase. For instance, the real rotation $R_Y(\pi/2) = 2^{-1/2} \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ satisfies $R_Y(\pi/2)^2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, its fourth power is $-I$, and its eighth power is I. It is unitary but is not a Hermitian involution.

1.3 Composite systems, reversible gates, and entanglement

For n qubits the state space is $\mathcal{H}_2^{\otimes n}$, of dimension $N = 2^n$. We use **most-significant-bit-first** ordering:

$$|x_1 \cdots x_n\rangle = |x_1\rangle \otimes \cdots \otimes |x_n\rangle, \qquad x = \sum_{j=1}^{n} 2^{n-j} x_j.$$

Products obey $(A \otimes B)(C \otimes D) = AC \otimes BD$, and for $A = (a_{ij})$ the Kronecker product $A \otimes B$ has blocks $a_{ij}B$. In particular $X \otimes Y = \left(\begin{smallmatrix} 0 & Y \\ Y & 0 \end{smallmatrix}\right)$.

Controlled reversible operations include

$$\begin{aligned} \mathrm{CNOT}\,|a,b\rangle &= |a, a\oplus b\rangle, \\ CZ|a,b\rangle &= (-1)^{ab}|a,b\rangle, \\ \mathrm{SWAP}\,|a,b\rangle &= |b,a\rangle, \\ \mathrm{CCNOT}\,|a,b,c\rangle &= |a,b,c\oplus ab\rangle, \end{aligned}$$

where $a,b,c\in\{0,1\}$. More generally, $C(U)=|0\rangle\langle 0|\otimes I+|1\rangle\langle 1|\otimes U$. The identity $CZ=(I\otimes H)\,\mathrm{CNOT}\,(I\otimes H)$ follows by conjugating the target. SWAP can be realized by $\mathrm{CNOT}_{1\to 2}\,\mathrm{CNOT}_{2\to 1}\,\mathrm{CNOT}_{1\to 2}$ in circuit order.

Definition 1.3 (Separability and Bell states)**.** A bipartite pure state is separable when $|\psi\rangle_{AB}=|u\rangle_A\otimes|v\rangle_B$; otherwise it is entangled. The Bell basis consists of

$$|\Phi^{\pm}\rangle=\frac{|00\rangle\pm|11\rangle}{\sqrt{2}},\qquad |\Psi^{\pm}\rangle=\frac{|01\rangle\pm|10\rangle}{\sqrt{2}}.$$

For $|\psi\rangle=a|00\rangle+b|01\rangle+c|10\rangle+d|11\rangle$, separability is equivalent to $ad-bc=0$.

A Bell pair is produced from $|00\rangle$ by H_1 followed by $\mathrm{CNOT}_{1\to 2}$. Its reduced density matrices satisfy $\mathrm{Tr}_B(|\Phi^+\rangle\langle\Phi^+|)=I/2$ and likewise on B; the joint state is pure although each marginal is maximally mixed.

1.4 Measurements, observables, and statistical estimation

Definition 1.4 (Projective and generalized measurement)**.** A projective measurement is a family $\{P_m\}$ satisfying $P_m=P_m^\dagger=P_m^2$, $P_mP_{m'}=0$ for $m\neq m'$, and $\sum_m P_m=I$. Its outcome probability and conditional state are

$$\Pr(m)=\langle\psi|P_m|\psi\rangle,\qquad |\psi_m\rangle=\frac{P_m|\psi\rangle}{\sqrt{\Pr(m)}}$$

when $\Pr(m)>0$. A general measurement uses Kraus operators M_m with $\sum_m M_m^\dagger M_m=I$; then $\Pr(m)=\langle\psi|M_m^\dagger M_m|\psi\rangle$ and the conditional state is $M_m|\psi\rangle/\sqrt{\Pr(m)}$.

The expectation of a Hermitian observable $O=\sum_m \lambda_m P_m$ is

$$\mathbb{E}[O]=\langle\psi|O|\psi\rangle=\sum_m \lambda_m\,\Pr(m).$$

For instance, measuring $|+\rangle$ in the computational basis yields 0 and 1 with probability $1/2$ each; conditioning on outcome 0 produces $|0\rangle$. For an observable with outcomes in $\{+1,-1\}$ and mean f, the empirical mean of S independent shots has variance $(1-f^2)/S$.

Proposition 1.5 (No-cloning theorem)**.** *No unitary operation can clone every unknown pure state of a Hilbert space of dimension greater than one.*

Proof. Suppose $U(|\phi\rangle|0\rangle) = |\phi\rangle|\phi\rangle$ and analogously for $|\psi\rangle$. Preservation of inner products requires $\langle\phi|\psi\rangle = \langle\phi|\psi\rangle^2$. Two distinct nonorthogonal normalized states have inner product neither zero nor one, a contradiction. □

For an entangled pair, a local trace-preserving operation does not change the remote reduced density matrix if the remote outcome is not conditioned on a communicated classical message; this is the no-signalling principle.

1.5 Quantum teleportation and interaction-free measurement

Let $|\psi\rangle_1 = \alpha|0\rangle + \beta|1\rangle$ and let qubits $(2,3)$ share $|\Phi^+\rangle$. Apply CNOT $1 \to 2$ and H on qubit 1. The resulting state is

$$\frac{1}{2}\big(|00\rangle_{12}|\psi\rangle_3 + |01\rangle_{12}X|\psi\rangle_3 + |10\rangle_{12}Z|\psi\rangle_3 + |11\rangle_{12}XZ|\psi\rangle_3\big).$$

Thus outcome (m_1, m_2) leaves Bob's state $X^{m_2}Z^{m_1}|\psi\rangle$ (up to phase), corrected by $Z^{m_1}X^{m_2}$. The two classical bits are essential; the protocol neither clones the input nor permits superluminal signalling.

In the idealized Elitzur–Vaidman interferometric test, an unobstructed two-path interferometer sends a single photon to the bright port with certainty. A live perfectly absorbing object in one arm yields, on one trial, absorption with probability $1/2$, an unambiguous nonabsorbing dark-port event with probability $1/4$, and an inconclusive bright-port event with probability $1/4$. Repeating only following inconclusive events yields eventual success probability

$$\sum_{j=0}^{\infty}\left(\frac{1}{4}\right)^j\frac{1}{4} = \frac{1}{3}.$$

A reversible path-marking CNOT illustrates interference loss but does not simulate the absorption branch; the latter must be included separately.

1.6 Geometric, algebraic, and operational derivations

1.6.1 The Bloch representation from density operators

For an arbitrary single-qubit density operator ρ, Hermiticity and unit trace imply a unique expansion

$$\rho = \frac{1}{2}(I + r_xX + r_yY + r_zZ), \qquad r_j = \mathrm{Tr}(\rho\sigma_j) \in \mathbb{R}.$$

Indeed I, X, Y, Z form an orthogonal basis of the real vector space of Hermitian 2×2 matrices under the Hilbert–Schmidt pairing: $\operatorname{Tr}(\sigma_j\sigma_k) = 2\delta_{jk}$ and $\operatorname{Tr}(I\sigma_j) = 0$. The characteristic polynomial is

$$\det(\rho - \lambda I) = \left(\lambda - \frac{1}{2}\right)^2 - \frac{\|\mathbf{r}\|_2^2}{4},$$

so the eigenvalues are $(1 \pm \|\mathbf{r}\|_2)/2$. Consequently $\rho \succeq 0$ if and only if $\|\mathbf{r}\|_2 \le 1$, and the state is pure if and only if $\|\mathbf{r}\|_2 = 1$. Moreover

$$\operatorname{Tr}(\rho^2) = \frac{1 + \|\mathbf{r}\|_2^2}{2}.$$

These equalities identify the entire Bloch ball, not merely its spherical boundary. They also make the distinction between a statistical mixture and coherent superposition quantitative.

Example 1.6 (A relative phase invisible to computational-basis measurements)**.** Consider $|\psi_\phi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$. Measuring Z gives probabilities $1/2, 1/2$ independently of ϕ. Yet $\langle X\rangle = \cos\phi$ and $\langle Y\rangle = \sin\phi$. Thus the choice $\phi = 0$ produces $|+\rangle$ with deterministic X outcome $+1$, whereas $\phi = \pi$ produces $|-\rangle$ with deterministic X outcome -1. In particular, agreement of computational-basis statistics does not imply equality of quantum states.

1.6.2 Unitary gates, eigenvectors, and controlled global phases

Proposition 1.7. *If $U \in \mathbb{C}^{d\times d}$ is unitary and $U|v\rangle = \lambda|v\rangle$ for a nonzero vector, then $|\lambda| = 1$. Conversely, a normal matrix with an orthonormal eigenbasis and unit-modulus eigenvalues is unitary.*

Proof. Norm preservation gives $\|v\|_2 = \|Uv\|_2 = |\lambda|\|v\|_2$; since $v \neq 0$, $|\lambda| = 1$. Conversely, if $U = V\operatorname{diag}(\lambda_j)V^\dagger$ with V unitary, then $U^\dagger U = V\operatorname{diag}(|\lambda_j|^2)V^\dagger = I$. □

The Pauli commutation relations can be checked directly from the matrices, but it is more useful to retain their product structure:

$$\sigma_j\sigma_k = \delta_{jk}I + i\sum_\ell \varepsilon_{jk\ell}\sigma_\ell, \quad [\sigma_j,\sigma_k] = 2i\sum_\ell \varepsilon_{jk\ell}\sigma_\ell, \quad \{\sigma_j,\sigma_k\} = 2\delta_{jk}I.$$

For instance $XY = iZ$ whereas $YX = -iZ$. Therefore $e^{-i\theta Z/2}Xe^{i\theta Z/2} = \cos\theta X + \sin\theta Y$. This follows either from the 2×2 matrices or by differentiating $F(\theta) = R_Z(\theta)^\dagger X R_Z(\theta)$ and solving the resulting coupled linear equations.

An important subtlety concerns $e^{i\gamma}U$. As isolated gates, U and $e^{i\gamma}U$ induce the same map on pure-state rays. As *controlled* operations they differ:

$$C(e^{i\gamma}U) = |0\rangle\langle 0| \otimes I + e^{i\gamma}|1\rangle\langle 1| \otimes U.$$

On input $|+\rangle|v\rangle$ with $U|v\rangle = |v\rangle$, the control becomes $(|0\rangle + e^{i\gamma}|1\rangle)/\sqrt{2}$, which is physically distinguishable from $|+\rangle$ when $\gamma \not\equiv 0 \pmod{2\pi}$. Consequently phases must not be discarded prematurely when constructing controlled gates in phase estimation.

Example 1.8 (Computing powers of a real unitary)**.** Let $R = R_Y(\pi/2) = 2^{-1/2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$. Multiplying twice yields $R^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; hence $R^4 = -I$ and $R^8 = I$. The state $R^4|0\rangle = -|0\rangle$ is indistinguishable from $|0\rangle$ without a reference branch, but controlled-R^4 generates a measurable relative phase. A gate may be unitary without being Hermitian and can have finite order only up to a global phase.

1.6.3 Tensor-product ordering and computational matrices

For $|a\rangle = \alpha|0\rangle + \beta|1\rangle$ and $|b\rangle = \gamma|0\rangle + \delta|1\rangle$,

$$|a\rangle \otimes |b\rangle = (\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta)^T.$$

This formula fixes the ordering $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. The controlled-NOT matrix with first-qubit control is

$$\mathrm{CNOT}_{1\to 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

On the other hand $\mathrm{CNOT}_{2\to 1}$ interchanges $|01\rangle$ and $|11\rangle$. The matrices are related by SWAP conjugation, not by equality. For a general input $\sum_{a,b} \gamma_{ab}|ab\rangle$, CNOT maps the amplitude at $|a, b\rangle$ to the basis vector $|a, b \oplus a\rangle$; no amplitude disappears because the underlying map is a permutation.

The three-CNOT SWAP identity follows on arbitrary computational basis inputs, not only on a chosen example:

$$(a, b) \mapsto (a, a \oplus b) \mapsto (b, a \oplus b) \mapsto (b, a).$$

Since computational basis vectors span the space, equality on all four basis states establishes equality of the corresponding unitaries. This is a general method of proving identities between reversible Boolean circuits.

1.6.4 Separability by rank and partial traces

Write an arbitrary two-qubit pure state as $|\psi\rangle = \sum_{i,j=0}^{1} M_{ij}|i\rangle|j\rangle$ with $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $|\psi\rangle = |u\rangle \otimes |v\rangle$ then $M = uv^T$, so $\operatorname{rank} M = 1$ and $\det M = ad - bc = 0$. Conversely $\det M = 0$ and $M \neq 0$ imply $M = uv^T$ and hence separability. This is the two-dimensional instance of the Schmidt-rank criterion.

The reduced density operator of the first subsystem is

$$\rho_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|) = MM^\dagger,$$

whose eigenvalues are the squared Schmidt coefficients. For the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, $M = I/\sqrt{2}$, so $\rho_A = I/2$. Conversely, if $|\psi\rangle = |00\rangle$ then $M = \mathrm{diag}(1, 0)$ and $\rho_A = |0\rangle\langle 0|$. The mixedness of a subsystem of a *pure bipartite state* therefore witnesses entanglement, while local mixedness alone does not certify that a general bipartite mixed state is entangled.

Example 1.9 (A determinant test with a complex phase)**.** Take $|\psi\rangle = (|00\rangle + i|01\rangle + |10\rangle + i|11\rangle)/2$. Then $M = \frac{1}{2}\begin{pmatrix} 1 & i \\ 1 & i \end{pmatrix}$ has determinant zero, and indeed $|\psi\rangle = |+\rangle \otimes (|0\rangle + i|1\rangle)/\sqrt{2}$. Replacing the final coefficient i by $-i$ gives $\det M = -i/2 \neq 0$, yielding an entangled state. The test is algebraic and does not depend on the choice of global phase.

1.6.5 The general measurement update and its normalization

A projective measurement $\{P_j\}$ satisfies $P_j = P_j^\dagger = P_j^2$, $P_j P_k = \delta_{jk} P_j$, and $\sum_j P_j = I$. For a pure input, the j-th outcome occurs with probability $p_j = \|P_j|\psi\rangle\|_2^2$, and the normalized conditional output is $P_j|\psi\rangle/\sqrt{p_j}$ when $p_j > 0$. For a mixed input, $p_j = \mathrm{Tr}(P_j\rho)$ and $\rho_j = P_j\rho P_j/p_j$. The unconditional post-measurement state is $\sum_j P_j\rho P_j$.

For a general POVM $\{E_j\}$ one requires $E_j \succeq 0$ and $\sum_j E_j = I$. The POVM determines probabilities $\mathrm{Tr}(E_j\rho)$, but not the output state. To specify the latter one supplies a measurement instrument with Kraus operators $M_{j,\mu}$ satisfying $E_j = \sum_\mu M_{j,\mu}^\dagger M_{j,\mu}$. Thus it is mathematically incorrect to infer a unique state update from a POVM element alone.

Example 1.10 (Measuring one qubit of an entangled pair)**.** For $|\Phi^+\rangle$, a Z measurement on the first qubit has outcomes 0 and 1 with probability $1/2$ each, leaving $|00\rangle$ and $|11\rangle$ respectively. Ignoring the outcome gives $\rho' = (|00\rangle\langle 00| + |11\rangle\langle 11|)/2$, which is classically correlated but separable. The reduced state on the second qubit is $I/2$ both before and after the uncommunicated measurement. This is a concrete manifestation of no-signalling.

1.6.6 Teleportation with every conditional branch displayed

Let the unknown input be $|\psi\rangle_1 = \alpha|0\rangle + \beta|1\rangle$ and let the shared entanglement be $|\Phi^+\rangle_{23}$. Applying $\mathrm{CNOT}_{1\to 2}$ and then H_1 yields

$$\begin{aligned}(H_1 \otimes I_{23})\,\mathrm{CNOT}_{1\to 2}(|\psi\rangle_1|\Phi^+\rangle_{23}) \\ = \frac{1}{2}\left(|00\rangle_{12}|\psi\rangle_3 + |01\rangle_{12}X|\psi\rangle_3 + |10\rangle_{12}Z|\psi\rangle_3 + |11\rangle_{12}XZ|\psi\rangle_3\right).\end{aligned}$$

To verify this identity directly, expand the input into $(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle)/\sqrt{2}$, flip the second qubit conditional on the first, and use $H|0\rangle = (|0\rangle+|1\rangle)/\sqrt{2}$ and $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. Grouping terms by the first two bits gives the displayed branches. All four outcomes have probability $1/4$, independent of α and β, and applying $Z^{m_1}X^{m_2}$ (up to an immaterial global phase in the 11 branch) recovers the input.

Teleportation does not allow superluminal signalling: prior to receiving m_1, m_2, Bob's state is the mixture $\frac{1}{4}\sum_{m_1,m_2} X^{m_2}Z^{m_1}\rho Z^{m_1}X^{m_2} = I/2$, independent of the input. The classical message is essential for reconstruction.

1.6.7 No-cloning and its inner-product obstruction

Theorem 1.11 (No-cloning for arbitrary pure states)**.** *No fixed unitary and fixed blank state can copy every unknown pure state.*

Proof. Suppose $U|\psi\rangle|b\rangle = |\psi\rangle|\psi\rangle$ and $U|\phi\rangle|b\rangle = |\phi\rangle|\phi\rangle$. Preservation of the inner product implies $\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$. Hence the overlap must be 0 or 1. Nonorthogonal, nonidentical states (for example $|0\rangle$ and $|+\rangle$) violate this condition. □

The proof does not prohibit copying an *orthogonal known set*, for instance $|0\rangle, |1\rangle$ by CNOT. It prohibits a universal physical copying procedure acting correctly on every unknown superposition. This distinction motivates syndrome measurements rather than direct repeated observations of the logical amplitudes in error correction.

1.6.8 Interaction-free measurement as an interferometric calculation

A balanced Mach–Zehnder interferometer can be represented abstractly by a Hadamard beam splitter acting on paths $|0\rangle, |1\rangle$. With both paths open, two beam splitters implement $H^2 = I$ and return $|0\rangle$ deterministically. With a perfectly absorbing object in path $|1\rangle$, the first splitter yields $|+\rangle$; absorption occurs with probability $1/2$. Conditional on survival, the state collapses to $|0\rangle$ with probability $1/2$, after which the final splitter gives a dark-port click with conditional probability $1/2$. The unconditional probability of detecting the object without absorption is therefore $(1/2)(1/2) = 1/4$. Conditioned on the nonabsorbing outcomes, this is not the same as a probability of $1/2$; specifying the conditioning event is essential.

1.7 Further elementary calculations and operational identities

1.7.1 Controlled matrices in the computational basis

For an arbitrary one-qubit matrix $U = \left(\begin{smallmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{smallmatrix}\right)$, a controlled-$U$ with the first wire as control has block-diagonal form

$$C(U) = \begin{pmatrix} I_2 & 0 \\ 0 & U \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & u_{00} & u_{01} \\ 0 & 0 & u_{10} & u_{11} \end{pmatrix}.$$

This form uses the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Setting $U = X$ gives CNOT. Setting $U = Z$ gives the diagonal controlled-Z matrix $\mathrm{diag}(1, 1, 1, -1)$, which is symmetric under exchange of the two wires despite its controlled-unitary representation. Setting $U = Y$ produces a controlled-Y that introduces complex amplitudes in the subspace with control value one. Setting $U = H$ gives a controlled-Hadamard with a nontrivial 2×2 unitary block. For arbitrary complex U, unitarity of $C(U)$ is equivalent to unitarity of U, because $C(U)^\dagger C(U) = \mathrm{diag}(I_2, U^\dagger U)$.

Example 1.12 (Entanglement from a controlled gate)**.** Prepare $|+\rangle|0\rangle = (|00\rangle + |10\rangle)/\sqrt{2}$. CNOT produces $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. If the control were instead $|0\rangle$, the same gate would leave the product state $|00\rangle$ unchanged. Thus the gate's potential to entangle does not imply that it entangles every input. Entanglement is a property of the resulting state relative to a bipartition, not a binary property inferred from the circuit symbol alone.

1.7.2 Explicit Pauli conjugations

The identities $HXH = Z$, $HZH = X$, and $HYH = -Y$ can be proved with matrix multiplication. For example,

$$HXH = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Since $Y = iXZ$, conjugation gives $HYH = i(HXH)(HZH) = iZX = -iXZ = -Y$. The same technique works for a multi-qubit Clifford circuit: conjugating a Pauli string one gate at a time produces another Pauli string, up to phase. This algebraic closure is why Clifford circuits admit efficient Pauli-frame tracking and stabilizer simulation.

A rotation identity illustrates the distinction between conjugation and left multiplication. One has

$$R_Z(\theta) X R_Z(\theta)^\dagger = \cos\theta\, X + \sin\theta\, Y,$$

whereas $XR_Z(\theta)$ is not in general a linear combination of X and Y with the same coefficients. From $XR_Z(\theta) = R_Z(-\theta)X$, moving a Pauli through a noncommuting rotation changes the sign of its angle. This relation reappears directly in adaptive one-way quantum computing.

1.7.3 The Born rule for noncomputational projectors

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, measuring in the Hadamard basis $\{|+\rangle, |-\rangle\}$ gives

$$p_+ = \left|\frac{\alpha+\beta}{\sqrt{2}}\right|^2, \qquad p_- = \left|\frac{\alpha-\beta}{\sqrt{2}}\right|^2.$$

Because $|\alpha|^2+|\beta|^2 = 1$, the two probabilities sum to one. These expressions contain the interference term $2\,\mathrm{Re}(\alpha^*\beta)$ with opposite signs. In particular, states with the same computational-basis probabilities but different relative phases may give different Hadamard-basis statistics. For $\alpha = 1/\sqrt{2}$ and $\beta = i/\sqrt{2}$, both probabilities equal one half, whereas $\beta = 1/\sqrt{2}$ makes the plus outcome deterministic.

Example 1.13 (Measuring a rank-one projector)**.** Let $P_+ = |+\rangle\langle+| = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ and input $|0\rangle$. The probability of the + outcome is $\langle 0|P_+|0\rangle = 1/2$. The unnormalized post-measurement state is $P_+|0\rangle = |+\rangle/\sqrt{2}$; dividing by $\sqrt{1/2}$ yields $|+\rangle$. The probability is obtained from the squared norm of the *unnormalized* branch, whereas the post-measurement state requires an additional normalization. This difference is the same in projective measurements, teleportation, and matrix block-encoding postselection.

1.7.4 Quantum teleportation as a Pauli correction table

The full teleportation identity may be organized by two measured classical bits (m_1, m_2):

(m_1, m_2)	uncorrected state on receiver	recovery
00	$\|\psi\rangle$	I
01	$X\|\psi\rangle$	X
10	$Z\|\psi\rangle$	Z
11	$XZ\|\psi\rangle$	ZX

The last recovery is ZX because $(ZX)(XZ) = I$, whereas XZ and ZX differ only by a global negative sign. If one instead applies XZ in the final branch, the result is $-|\psi\rangle$, physically equivalent for an unconditionally recovered isolated state. The table's bit labels are tied to the order of the Hadamard and CNOT operations; another convention may interchange which measured bit controls the X or Z correction.

Before receiving the classical bits, the receiver has density matrix

$$\rho_B = \frac{1}{4}(\rho + X\rho X + Z\rho Z + XZ\rho ZX) = \frac{I}{2}.$$

The equality follows by expanding $\rho = (I + \mathbf{r}\cdot\boldsymbol{\sigma})/2$: averaging over the four Pauli conjugations cancels each traceless Pauli component. Because the receiver's marginal is independent of the input, the protocol cannot transmit information without the classical message even though the shared Bell pair is entangled.

1.7.5 Uncertainty of a Pauli measurement

For a qubit density matrix $\rho = (I + r_x X + r_y Y + r_z Z)/2$, a measurement of Z produces outcomes $+1$ and -1 with probabilities $(1+r_z)/2$ and $(1-r_z)/2$. The outcome variance is

$$\mathrm{Var}_\rho(Z) = \mathrm{Tr}(\rho Z^2) - [\mathrm{Tr}(\rho Z)]^2 = 1 - r_z^2.$$

A pure Z eigenstate has zero measurement variance, whereas $|+\rangle$ has variance one. For X and Z on a pure qubit, the Bloch constraint $r_x^2 + r_y^2 + r_z^2 = 1$ implies

$$\mathrm{Var}(X) + \mathrm{Var}(Z) = 2 - r_x^2 - r_z^2 = 1 + r_y^2 \geq 1.$$

This elementary uncertainty relation follows from the Bloch-sphere geometry and is independent of a particular experimental apparatus. It also underlies the finite-shot error formulas used later for variational expectation estimation.

1.7.6 Distinguishing global and relative phase numerically

Compare $|\psi_1\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ with $|\psi_2\rangle = i(|0\rangle + |1\rangle)/\sqrt{2}$. They differ by global phase and represent the same pure state; every POVM gives identical probabilities. Compare instead $|\psi_3\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}$. Then $|\langle\psi_1|\psi_3\rangle|^2 = |(1+i)/2|^2 = 1/2$, so they are not the same ray. Measuring X distinguishes them statistically: $|\psi_1\rangle$ gives outcome $+1$ with certainty, while $|\psi_3\rangle$ gives ± 1 with equal probabilities. This gives an operational criterion for deciding whether two phase expressions differ physically: inspect whether the phase is common to every computational amplitude or changes relative coefficients.

CHAPTER 2

Reversible Oracles and Boolean Fourier Sampling

2.1 Oracle models and phase kickback

Definition 2.1 (Standard and phase oracles)**.** For $f : \mathbb{F}_2^n \to \mathbb{F}_2$, the reversible evaluation oracle and phase oracle are

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle, \qquad O_f|x\rangle = (-1)^{f(x)}|x\rangle.$$

The evaluation oracle is unitary and satisfies $U_f^2 = I$. Its phase version is obtained by initializing the target to $|-\rangle$: $U_f(|x\rangle|-\rangle) = (-1)^{f(x)}|x\rangle|-\rangle$.

The linearity of a quantum oracle gives $U_f(\sum_x a_x|x\rangle|0\rangle) = \sum_x a_x|x\rangle|f(x)\rangle$ but does not make the full truth table accessible by one measurement. The algorithms below use interference to expose global properties.

The Walsh–Hadamard transform is

$$H^{\otimes n}|x\rangle = 2^{-n/2}\sum_{z\in\mathbb{F}_2^n}(-1)^{x\cdot z}|z\rangle, \qquad x\cdot z = \bigoplus_{j=1}^{n} x_j z_j.$$

Its characters satisfy $\sum_x (-1)^{x\cdot z} = 2^n \mathbf{1}_{\{z=0^n\}}$, the orthogonality identity underlying Deutsch–Jozsa and Simon sampling.

2.2 Deutsch's one-bit promise problem

Definition 2.2 (Deutsch problem)**.** Given $f : \{0,1\} \to \{0,1\}$ with the promise that f is either constant or balanced, determine which case holds by querying its reversible oracle.

The four Boolean functions are $f(x) = 0, 1, x$, and $x \oplus 1$. Their oracles can respectively be realized by I, $I \otimes X$, CNOT, and $(I \otimes X)\,\mathrm{CNOT}$. Prepare $|+\rangle|-\rangle$, apply U_f, and apply H to the input. Just before measurement, its first qubit is

$$\frac{(-1)^{f(0)} + (-1)^{f(1)}}{2}|0\rangle + \frac{(-1)^{f(0)} - (-1)^{f(1)}}{2}|1\rangle.$$

Therefore outcome 0 occurs exactly for constant functions, and outcome 1 exactly for balanced functions. The algorithm needs one oracle query; the output register remains $|-\rangle$.

2.3 Deutsch–Jozsa and zero-frequency detection

Definition 2.3 (Deutsch–Jozsa problem)**.** For $f : \mathbb{F}_2^n \to \mathbb{F}_2$, promised either constant or balanced (exactly 2^{n-1} zeros and 2^{n-1} ones), distinguish the two cases.

Apply $H^{\otimes n}$, then O_f, then $H^{\otimes n}$, beginning at $|0^n\rangle$. The amplitude of result z is the normalized Boolean Fourier coefficient

$$\widehat{g}(z) = 2^{-n} \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x)+x\cdot z}, \qquad g(x) = (-1)^{f(x)}.$$

In particular,

$$\widehat{g}(0^n) = 2^{-n}\left(|f^{-1}(0)| - |f^{-1}(1)|\right).$$

For constant f the zero string occurs with probability one; for balanced f its probability is zero. The promise is indispensable: an arbitrary nonconstant function need not be balanced.

For functions of two input bits, all affine $f(x_1, x_2) = c \oplus a_1 x_1 \oplus a_2 x_2$ are implemented by a target X if $c = 1$ and one CNOT from each input j with $a_j = 1$. The eight affine truth tables correspond bijectively to $(c, a_1, a_2) \in \mathbb{F}_2^3$; this construction exhibits the Boolean-oracle equivalences without enumerating individual circuits.

2.4 Simon's hidden-XOR problem

Definition 2.4 (Simon promise problem)**.** Given a black-box function $f : \mathbb{F}_2^n \to \mathcal{Y}$ promised to admit a nonzero $s \in \mathbb{F}_2^n$ such that

$$f(x) = f(y) \quad \Longleftrightarrow \quad y = x \text{ or } y = x \oplus s,$$

recover the hidden XOR period s.

Initialize $|0^n\rangle|0\rangle$, apply $H^{\otimes n}$ to the input register, and query U_f. The resulting state is

$$2^{-n/2} \sum_x |x\rangle|f(x)\rangle.$$

Measuring the output register and obtaining $f(x_0)$ leaves the normalized input state

$$\frac{|x_0\rangle + |x_0 \oplus s\rangle}{\sqrt{2}}.$$

A second Walsh–Hadamard transform gives

$$\frac{1}{\sqrt{2^{n+1}}} \sum_{z \in \mathbb{F}_2^n} (-1)^{x_0\cdot z}\left(1 + (-1)^{s\cdot z}\right)|z\rangle.$$

Consequently the measurement distribution is

$$\Pr(Z = z) = \begin{cases} 2^{-(n-1)}, & s \cdot z = 0, \\ 0, & s \cdot z = 1. \end{cases}$$

The same distribution follows **without** measuring the output register. Pairing x with $x \oplus s$ in

$$2^{-n} \sum_{z,x} (-1)^{z \cdot x} |z\rangle |f(x)\rangle$$

introduces the factor $1 + (-1)^{z \cdot s}$, and tracing out the output yields the stated marginal probabilities.

Repeated samples $z^{(1)}, z^{(2)}, \ldots$ give a linear system $Zs = 0$ over $\mathbb{F}_2$. Once the row rank is $n - 1$, the nullspace contains precisely 0 and s. This occurs with high probability after $O(n)$ oracle invocations; Gaussian elimination gives polynomial classical postprocessing. The sample 0^n is allowed but supplies no new equation.

Example 2.5 (Three-bit Simon reconstruction)**.** If $s = 101$, then $z_1 \oplus z_3 = 0$, and the allowed uniformly distributed outcomes are $000, 010, 101, 111$. Conversely, measured rows 010 and 111 imply $s_2 = 0$ and $s_1 \oplus s_2 \oplus s_3 = 0$. Under the nonzero-period promise, $s = 101$.

Deutsch–Jozsa determines whether a specific Walsh coefficient vanishes, whereas Simon sampling recovers a hidden linear subspace from randomly obtained orthogonality constraints. Both use character orthogonality, but the classical information extracted per oracle invocation is fundamentally different.

2.5 Oracle constructions and Fourier-sampling proofs

2.5.1 Oracle reversibility and the phase-kickback identity

For any Boolean function $f : \{0,1\}^n \to \{0,1\}$, define

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle.$$

This is a permutation of the computational basis; applying it twice restores the input, so $U_f^2 = I$ and $U_f^\dagger = U_f$. The analogous mapping $|x\rangle \mapsto |f(x)\rangle$ is not necessarily reversible, because distinct inputs may share the same function value. The target register is not an arbitrary embellishment: it supplies a reversible embedding of a general classical function.

Prepare the target in the -1 eigenstate of X, namely $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. Then

$$U_f(|x\rangle|-\rangle) = |x\rangle X^{f(x)}|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle.$$

By linearity, for an arbitrary input $\sum_x \alpha_x |x\rangle$, the output is $\sum_x (-1)^{f(x)} \alpha_x |x\rangle|-\rangle$. The target factors out unchanged, while the function is encoded in relative phases. This is the precise form of phase kickback used by Deutsch–Jozsa and Simon-related Fourier analyses.

Example 2.6 (An explicit parity oracle)**.** For $f(x_1, x_2) = x_1 \oplus x_2$, implement U_f by two CNOT gates, one from each input bit to the same target. On the state $(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|-\rangle/2$, the effect is

$$\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)|-\rangle.$$

The signs can be calculated either from the truth table or from $(-1)^{x_1+x_2}$. Applying $H^{\otimes 2}$ to the input register now yields $|11\rangle$ deterministically, because the phase function is the Walsh character indexed by $(1, 1)$.

2.5.2 Walsh characters and their orthogonality

Define $\chi_s(x) = (-1)^{s\cdot x}$ where $s \cdot x = \bigoplus_{j=1}^{n} s_j x_j$ is an inner product over $\mathbb{F}_2$. The characters satisfy

$$\frac{1}{2^n} \sum_{x\in\{0,1\}^n} \chi_s(x)\chi_t(x) = \delta_{s,t}.$$

Proof. The summand is $(-1)^{(s\oplus t)\cdot x}$ and the sum factorizes:

$$\sum_{x\in\{0,1\}^n} (-1)^{(s\oplus t)\cdot x} = \prod_{j=1}^{n} \left(1 + (-1)^{s_j \oplus t_j}\right).$$

If $s = t$, every factor equals 2, yielding 2^n; otherwise at least one factor is zero. □

The normalized Fourier coefficients of a real Boolean-sign function $g(x) = (-1)^{f(x)}$ are

$$\widehat{g}(s) = 2^{-n} \sum_{x} g(x)\chi_s(x).$$

The identity $H^{\otimes n}|x\rangle = 2^{-n/2} \sum_s \chi_s(x)|s\rangle$ gives

$$H^{\otimes n} \frac{1}{2^{n/2}} \sum_{x} g(x)|x\rangle = \sum_{s} \widehat{g}(s)|s\rangle.$$

Parseval's identity $\sum_s \widehat{g}(s)^2 = \mathbb{E}_x[g(x)^2] = 1$ immediately shows that the Fourier coefficients squared define a probability distribution. This connection between orthogonal characters and measurement probabilities is the mathematical mechanism behind Fourier sampling.

2.5.3 Deutsch's promise problem with all four functions

When $n = 1$, there are four Boolean functions. They are $f_0(x) = 0$, $f_1(x) = 1$, $f_2(x) = x$, and $f_3(x) = x \oplus 1$. The first two are constant, the latter two balanced. Starting with $|0\rangle|1\rangle$, applying $H \otimes H$, querying U_f, and applying H to the input yields

$$\sum_{z\in\{0,1\}} \left(\frac{1}{2} \sum_{x\in\{0,1\}} (-1)^{f(x)+xz} \right) |z\rangle|-\rangle.$$

For constant f, the $z = 0$ amplitude is ± 1 and the $z = 1$ amplitude vanishes. For balanced f, the $z = 0$ amplitude vanishes and $z = 1$ has amplitude ± 1. Thus one query decides the promise problem exactly. One should not confuse this with learning the full truth table: the global phase distinguishes neither constant function from the other nor the two balanced functions.

For two input bits, an affine function is $f(x) = a_1 x_1 \oplus a_2 x_2 \oplus b$. There are $2^3 = 8$ such functions. The reversible implementation uses one CNOT for every nonzero a_j and an X on the target if $b = 1$. The phase-oracle operator on the input is

$$O_f = (-1)^b Z_1^{a_1} Z_2^{a_2}.$$

The constant offset b contributes only a global sign in a stand-alone phase oracle, but is meaningful in the target-bit reversible oracle. The eight circuits can therefore be represented by four choices of the Z pattern, with each pattern having two global-phase variants. This carefully separates the eight affine truth tables from phase-oracle equivalence classes.

2.5.4 Deutsch–Jozsa: exact amplitude calculation

Under the promise that f is either constant or balanced, the state immediately before the final measurement is

$$\sum_{z \in \{0,1\}^n} A_z |z\rangle, \qquad A_z = 2^{-n} \sum_x (-1)^{f(x)+x\cdot z}.$$

At $z = 0^n$ this specializes to

$$A_{0^n} = \frac{\#\{x : f(x) = 0\} - \#\{x : f(x) = 1\}}{2^n}.$$

A constant function produces $|A_{0^n}| = 1$, whereas a balanced function yields $A_{0^n} = 0$. In the constant case all other amplitudes vanish by Walsh orthogonality. Therefore measuring 0^n is certain for constant inputs and impossible for balanced inputs. This is an *exact* promise-problem decision, not a high-probability heuristic.

Example 2.7 (A two-input balanced function)**.** Let $f(x_1, x_2) = x_1$. The phase-encoded uniform superposition has amplitudes $(1, 1, -1, -1)/2$ in lexicographic order. After $H^{\otimes 2}$, the outcome is $|10\rangle$ with probability one, as $(-1)^{f(x)} = \chi_{10}(x)$. If instead $f(x) = x_1 \oplus x_2$, the outcome is $|11\rangle$. Both are balanced, although their nonzero Fourier frequencies differ. Deutsch–Jozsa asks only whether the zero-frequency component vanishes.

2.5.5 Simon's collision promise and uniform hyperplane sampling

Definition 2.8 (Simon's hidden-XOR promise)**.** A function $f : \mathbb{F}_2^n \to \mathcal{Y}$ satisfies the Simon promise if there exists a nonzero $s \in \mathbb{F}_2^n$ such that

$$f(x) = f(y) \iff y = x \text{ or } y = x \oplus s.$$

The problem is to recover s using an oracle that computes f reversibly.

The initial state, after Hadamards on the query register and one oracle call, is

$$|\Psi\rangle = \frac{1}{2^{n/2}} \sum_{x \in \mathbb{F}_2^n} |x\rangle |f(x)\rangle.$$

Measuring the output register returns an image value associated with exactly the pair $\{x_0, x_0 \oplus s\}$, leaving the normalized query state $(|x_0\rangle + |x_0 \oplus s\rangle)/\sqrt{2}$. Its Walsh transform is

$$\frac{1}{\sqrt{2^{n+1}}} \sum_{y \in \mathbb{F}_2^n} (-1)^{x_0 \cdot y} \left(1 + (-1)^{s \cdot y}\right) |y\rangle.$$

If $s \cdot y = 1$, the factor in parentheses vanishes. If $s \cdot y = 0$, its magnitude is 2. Since precisely 2^{n-1} strings belong to the hyperplane $s^{\perp}$, the measurement distribution is uniform on this hyperplane:

$$\Pr(Y = y) = \begin{cases} 2^{-(n-1)}, & s \cdot y = 0, \\ 0, & s \cdot y = 1. \end{cases}$$

2.5.6 The unmeasured-register proof and orthogonality cancellation

Measuring the output register is optional. After the final Hadamards, the joint state is

$$|\Phi\rangle = 2^{-n} \sum_{y,x} (-1)^{x \cdot y} |y\rangle |f(x)\rangle.$$

The probability of measuring a fixed y in the first register is the squared norm of its output-vector coefficient:

$$\begin{aligned} \Pr(Y = y) &= 2^{-2n} \sum_{x,x'} (-1)^{(x \oplus x') \cdot y} \langle f(x') | f(x) \rangle \\ &= 2^{-2n} \sum_{x} \left(1 + (-1)^{s \cdot y}\right) = 2^{-n} \left(1 + (-1)^{s \cdot y}\right). \end{aligned}$$

Here the orthogonality $\langle f(x') | f(x) \rangle = \mathbf{1}_{f(x')=f(x)}$ is valid because the function values are encoded in orthogonal computational-basis states. The promise reduces the surviving pairs to $x' = x$ and $x' = x \oplus s$. The result agrees exactly with the measured-output derivation. This is a useful general principle: a local measurement may be deferred when only the final marginal statistics are required.

2.5.7 Recovery as Gaussian elimination over $\mathbb{F}_2$

Each observed y supplies a linear equation $y \cdot s = 0$ over $\mathbb{F}_2$. After obtaining rows $y^{(1)}, \ldots, y^{(m)}$, construct a binary matrix $Y \in \mathbb{F}_2^{m \times n}$ and solve $Ys = 0$. If $\operatorname{rank} Y = n - 1$, its nullspace has dimension one and the promise $s \neq 0$ singles out the correct vector. The sampling problem is therefore not to estimate Fourier amplitudes, but to collect enough linearly independent equations.

Example 2.9 (An explicit three-bit hidden string)**.** Suppose $s = 110$. The allowed samples satisfy $y_1 \oplus y_2 = 0$, so $s^\perp = \{000, 001, 110, 111\}$. If measurements produce $y^{(1)} = 001$ and $y^{(2)} = 110$, the equations are $s_3 = 0$ and $s_1 \oplus s_2 = 0$. Their solution space is $\{000, 110\}$; the nonzero promise yields $s = 110$. Samples $000, 000, 001$ would not suffice: repeated samples do not increase matrix rank.

Proposition 2.10 (A simple sample bound)**.** *After $m = n - 1 + t$ independent uniform samples from the $(n-1)$-dimensional hyperplane $s^\perp$, the probability that they fail to span $s^\perp$ is at most 2^{-t}.*

Proof. If the rows do not span $s^\perp$, there is a nonzero vector in its dual orthogonal to all rows. A fixed nonzero dual vector has zero dot product with a uniformly random row with probability $1/2$. By the union bound over at most $2^{n-1} - 1$ nonzero dual vectors, the failure probability is at most $(2^{n-1} - 1)2^{-m} < 2^{-t}$. □

Thus $O(n + \log(1/\delta))$ queries suffice to determine the hidden string with failure probability at most δ. The proof is conditional on the exact collision promise; arbitrary periodic-looking functions need not exhibit this distribution.

2.6 Boolean-oracle enumerations and additional Fourier calculations

2.6.1 Every affine Boolean oracle on two input bits

An affine Boolean function on two bits has the form $f(x_1, x_2) = a_1x_1 \oplus a_2x_2 \oplus b$ for $a_1, a_2, b \in \mathbb{F}_2$. The eight possible truth tables, in input order $(00, 01, 10, 11)$, are

a_1	a_2	b	$(f(00), f(01), f(10), f(11))$	reversible target action
0	0	0	0000	I
0	0	1	1111	X_t
0	1	0	0101	$\mathrm{CNOT}_{2\to t}$
0	1	1	1010	$X_t\,\mathrm{CNOT}_{2\to t}$
1	0	0	0011	$\mathrm{CNOT}_{1\to t}$
1	0	1	1100	$X_t\,\mathrm{CNOT}_{1\to t}$
1	1	0	0110	$\mathrm{CNOT}_{1\to t}\,\mathrm{CNOT}_{2\to t}$
1	1	1	1001	$X_t\,\mathrm{CNOT}_{1\to t}\,\mathrm{CNOT}_{2\to t}$

The matrix product order is not intended to replace the explicit circuit-time convention; here the listed gates act on a common target and commute, so their relative order does not affect the output. For an input $|x_1x_2\rangle|y\rangle$, the effect is exactly $|x_1x_2\rangle|y \oplus f(x_1, x_2)\rangle$. For the target initialized in $|-\rangle$, the effective phase oracle is $(-1)^b Z_1^{a_1} Z_2^{a_2}$, so complementing the function by flipping b multiplies every query-register amplitude by a common sign. It is important not to infer that the two underlying classical truth tables are identical simply because the induced one-register phase operators differ only by global phase.

2.6.2 Deutsch–Jozsa amplitudes for a nonlinear balanced example

In two input bits every balanced Boolean function has exactly two ones and two zeros. Take the non-affine-looking expression $f(x_1, x_2) = x_1 \oplus x_2 \oplus x_1x_2$. Its truth table is $(0, 1, 1, 1)$, so it is *not* balanced: the number of ones is three. The Deutsch–Jozsa promise does not apply, and

$$A_{00} = \frac{1}{4}(1 - 1 - 1 - 1) = -\frac{1}{2},$$

so measuring 00 occurs with probability 1/4. This example illustrates a critical limitation: the Deutsch–Jozsa circuit still produces a well-defined measurement distribution for every oracle, but its exact decision guarantee holds only under the constant-or-balanced promise. A nonzero 00 probability does not imply that the function is constant; it only implies that the function has nonzero zero-frequency Fourier coefficient.

For an actual balanced example take $f(x_1, x_2) = x_1$. The sign vector is $(1, 1, -1, -1)$ and $A_{00} = 0$ exactly. Hadamard transformation yields the deterministic output 10. Thus the oracle's algebraic structure determines which nonzero Fourier outcome occurs, but the promise decision needs only distinguish whether the zero outcome is possible.

2.6.3 Fourier sampling as a distribution rather than full coefficient recovery

Let $g : \mathbb{F}_2^n \to \{-1, +1\}$ and define $\widehat{g}(s) = 2^{-n} \sum_x g(x)(-1)^{s \cdot x}$. One use of a suitable phase oracle and a final Hadamard transform produces s with probability $|\widehat{g}(s)|^2$. Repeated sampling estimates *probability mass* in regions of Fourier space; it does not generally reveal the signs of Fourier coefficients, because the Born rule squares amplitudes. For example g and $-g$ have coefficients differing by sign at every frequency but induce exactly the same Fourier-sampling distribution. Recovering a signed coefficient requires an interference or Hadamard-test-type procedure with a phase reference, not merely measuring the Fourier register.

For the parity function $g(x) = (-1)^{a \cdot x}$, character orthogonality gives $\widehat{g}(s) = \delta_{s,a}$ and Fourier sampling is deterministic. More general Boolean functions may have many nonzero coefficients. Parseval's identity ensures that the squared coefficients sum to one, but it imposes no requirement that the distribution be uniform over its support. A statement about observing every frequency with equal probability requires a separate structural assumption, as in Simon's hidden-XOR promise.

2.6.4 A four-bit Simon recovery computation

Suppose the hidden XOR string is $s = 1010$. The valid measurement outcomes satisfy $y_1 \oplus y_3 = 0$. Independent samples might be $y^{(1)} = 0100$, $y^{(2)} = 1010$, and $y^{(3)} = 0001$.

They yield the equations

$$s_2 = 0, \qquad s_1 \oplus s_3 = 0, \qquad s_4 = 0.$$

The nullspace consists of 0000 and 1010, so the nonzero hidden-string promise identifies $s = 1010$. A sample 1110 is valid because $y_1 \oplus y_3 = 1 \oplus 1 = 0$; by contrast 1000 has $y_1 \oplus y_3 = 1$ and is impossible under the ideal promise. This distinction is useful when checking purported experimental results: an ideal Simon circuit assigns exactly zero probability to strings outside the hyperplane.

The quantum measurement records are equations, not guesses of the hidden string. Seeing a sample equal to the hidden string can occur for some s, but the algorithm's correctness comes from solving the full nullspace constraints, not from assuming any individual sampled y equals s.

2.6.5 A finite-field rank calculation with dependent rows

Let $n = 4$, and suppose the collected Simon rows are

$$Y = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Over $\mathbb{F}_2$, the fourth row is the XOR of the first two, so it does not increase rank. The first three rows are independent, and solving $Ys = 0$ gives $s_2 = 0$, $s_1 = s_3$, and $s_2 = s_4$, hence $s = (t, 0, t, 0)$. The nonzero promise yields 1010. The example shows why the number of observed strings is not by itself the relevant statistic; the rank of their binary matrix determines whether the hidden vector is uniquely recoverable.

CHAPTER 3

Quantum Fourier Transform and Phase Estimation

3.1 Fourier transform on the cyclic group

Definition 3.1 (Quantum Fourier transform)**.** For $N = 2^n$ and $\omega_N = e^{2\pi i/N}$, define

$$F_N|x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega_N^{xy} |y\rangle, \qquad F_N^\dagger|y\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \omega_N^{-xy} |x\rangle.$$

We consistently use the **positive exponent** for F_N and the negative exponent for $F_N^\dagger$.

The root-of-unity identity $\sum_{y=0}^{N-1} \omega_N^{(x-x')y} = N\delta_{x,x'}$ proves $F_N^\dagger F_N = I$. For $N = 2$, $F_2 = H$. A quantum Fourier transform evolves amplitudes coherently; it does not output all coefficients of a classical discrete Fourier transform.

For $x = x_1 \dots x_n$ in binary, the product form is

$$F_{2^n}|x\rangle = 2^{-n/2} \bigotimes_{k=1}^{n} \big(|0\rangle + e^{2\pi i x/2^k}|1\rangle\big).$$

Here $e^{2\pi i x/2^k}$ depends only on the fractional part of the binary expansion of $x/2^k$. Define the QFT controlled-phase gate

$$P_k = \mathrm{diag}(1, e^{2\pi i/2^k}), \qquad k \geq 2,$$

so $P_2 = S$ and $P_3 = T$. This P_k is not the same notation as a Pauli-axis rotation $R_P(\theta)$.

A conventional n-qubit implementation uses n Hadamard gates, $n(n-1)/2$ controlled-phase gates, and an optional output bit reversal. Without the final SWAP network, the outputs are in reversed order. For three qubits the state **before** reversing the first and last wires is

$$\frac{1}{\sqrt{8}}(|0\rangle + e^{2\pi i x/8}|1\rangle) \otimes (|0\rangle + e^{2\pi i x/4}|1\rangle) \otimes (|0\rangle + e^{2\pi i x/2}|1\rangle).$$

After reversal the factor ordering coincides with the formal definition of $F_8|x\rangle$.

Example 3.2 (Fourier transform of a basis state)**.** For $x = 6$ and $N = 8$, $\omega_8^{xy} = e^{3\pi i y/2} = (-i)^y$. Thus

$$F_8|6\rangle = \frac{1}{\sqrt{8}} \sum_{y=0}^{7} (-i)^y |y\rangle = \frac{1}{\sqrt{8}}(|0\rangle - i|1\rangle - |2\rangle + i|3\rangle + |4\rangle - i|5\rangle - |6\rangle + i|7\rangle).$$

The corresponding product-state phases are $e^{2\pi i 6/2} = 1$, $e^{2\pi i 6/4} = -1$, and $e^{2\pi i 6/8} = -i$ in natural output order.

3.2 Periodic combs and Fourier peaks

Let $|\psi\rangle = L^{-1/2} \sum_{\ell=0}^{L-1} |x_0 + \ell r\rangle$, with all terms distinct and in $\{0, \dots, N-1\}$. Its inverse-QFT amplitude at y is

$$\langle y|F_N^\dagger|\psi\rangle = \frac{e^{-2\pi i x_0 y/N}}{\sqrt{LN}} \sum_{\ell=0}^{L-1} e^{-2\pi i \ell r y/N}.$$

For $q = e^{-2\pi i r y/N} \neq 1$, the geometric sum is $(1-q^L)/(1-q)$; otherwise it equals L. The distribution peaks near $y \approx jN/r$, not necessarily exactly at integers of that form if $r \nmid N$. The phase determined by x_0 disappears from the final outcome probabilities.

3.3 Eigenphase estimation

Definition 3.3 (Quantum phase estimation (QPE))**.** Given a unitary U and a normalized eigenvector $|u\rangle$ with $U|u\rangle = e^{2\pi i\phi}|u\rangle$, $\phi \in [0,1)$, QPE estimates ϕ to t binary digits using t control qubits, controlled powers U^{2^j}, and an inverse QFT.

Set $Q = 2^t$. Starting with $|0^t\rangle|u\rangle$, Hadamard gates and controlled powers produce

$$\frac{1}{\sqrt{Q}} \sum_{k=0}^{Q-1} e^{2\pi i k\phi}|k\rangle|u\rangle = \bigotimes_{j=0}^{t-1} \frac{|0\rangle + e^{2\pi i 2^j \phi}|1\rangle}{\sqrt{2}} \otimes |u\rangle,$$

where factor ordering follows the chosen control-wire convention. Applying $F_Q^\dagger$ yields the probability amplitude

$$A_y = \frac{1}{Q} \sum_{k=0}^{Q-1} e^{2\pi i k(\phi - y/Q)}.$$

If $\delta = \phi - y/Q \notin \mathbb{Z}$, then

$$\Pr(Y = y) = \frac{\sin^2(\pi Q\delta)}{Q^2 \sin^2(\pi\delta)}.$$

For $\delta \in \mathbb{Z}$ the probability is 1. In particular $\phi = m/Q$ is recovered exactly as $y = m$. For input $|\psi\rangle = \sum_s c_s|u_s\rangle$, the output distribution is the eigenphase mixture $\Pr(Y = y) = \sum_s |c_s|^2 \Pr(Y = y \mid \phi_s)$, assuming the $|u_s\rangle$ form an orthonormal eigenbasis.

Example 3.4 (A two-dimensional phase-estimation calculation)**.** Consider the unitary

$$U = \frac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

The eigenvectors $|+\rangle$ and $|-\rangle$ have eigenvalues 1 and i, so their eigenphases are 0 and $1/4$. With $t = 3$ controls, input $|-\rangle$ produces $y = 2 = (010)_2$ with certainty; input $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ produces 000 or 010, each with probability $1/2$. The Y-eigenstates $|\pm i\rangle$ are not eigenstates of this U.

The gate-level cost depends on implementing powers of U rather than on control-register length alone. In oracle analyses, a query to U^{2^j} must not automatically be counted as one elementary gate.

3.4 Exact Fourier identities and finite-register calculations

3.4.1 Unitarity of the discrete Fourier transform

Let $N = 2^n$ and $\omega_N = e^{2\pi i/N}$. The quantum Fourier transform is the unitary

$$F_N = \frac{1}{\sqrt{N}} \sum_{x,y=0}^{N-1} \omega_N^{xy} |y\rangle\langle x|, \qquad F_N|x\rangle = \frac{1}{\sqrt{N}} \sum_y \omega_N^{xy} |y\rangle.$$

Its adjoint uses the negative exponent. Direct multiplication shows

$$\langle x'|F_N^\dagger F_N|x\rangle = \frac{1}{N} \sum_{y=0}^{N-1} \omega_N^{(x-x')y} = \delta_{x,x'}.$$

The last equality is the finite geometric-series identity. When $x \neq x' \pmod N$, the ratio $r = \omega_N^{x-x'}$ satisfies $r \neq 1$ and $r^N = 1$, so $\sum_{y=0}^{N-1} r^y = (1 - r^N)/(1 - r) = 0$. This proves $F_N^\dagger F_N = I$ rather than merely asserting orthogonality.

Example 3.5 (The complete $N = 4$ matrix)**.** Writing $\omega_4 = i$, the Fourier matrix is

$$F_4 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}.$$

For input $|3\rangle$, its output is $(|0\rangle - i|1\rangle - |2\rangle + i|3\rangle)/2$. Every basis outcome has probability $1/4$, but the phases are essential for any subsequent inverse transform or interference experiment. Applying $F_4^\dagger$ recovers $|3\rangle$ exactly.

3.4.2 Binary fractions and the product-state form

Represent $x = x_1 x_2 \cdots x_n$ in binary, with x_1 most significant. The QFT phase for output $y = y_1 \cdots y_n$ can be expressed as a sum of binary fractional phases. The identity

$$F_{2^n}|x\rangle = \frac{1}{2^{n/2}} \bigotimes_{j=1}^{n} \left(|0\rangle + e^{2\pi i\, 0.x_{n-j+1}\cdots x_n} |1\rangle \right)$$

is written in the *bit-reversed output order* convention; final SWAP gates restore the standard most-significant-first order. Here $0.x_j \cdots x_n = \sum_{k=j}^{n} x_k 2^{-(k-j+1)}$. Its proof is an elementary factorization of $\sum_y e^{2\pi i x y/2^n}|y\rangle$ into a product over independent output bits, using $y = \sum_j 2^{n-j} y_j$. Each factor corresponds to a Hadamard followed by controlled phases contributed by later input bits.

The controlled phase gate appearing at separation $k-1$ is $R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$, with the controlled version applying the phase only to $|11\rangle$. A depth/count statement must specify the implementation model: the exact textbook circuit has n Hadamards and $n(n-1)/2$ controlled phase rotations, plus at most $\lfloor n/2 \rfloor$ SWAPs. Small controlled rotations can be omitted only when accepting an approximation whose error is then analyzed separately.

3.4.3 Eight-point phase tables and a complete basis-state computation

For $N = 8$, the product xy mod 8 determines the root-of-unity factor ω_8^{xy}. For fixed $x = 6$, the row $6y$ mod 8 is

$$0, 6, 4, 2, 0, 6, 4, 2 \quad (y = 0, 1, \ldots, 7).$$

Therefore

$$\begin{aligned} F_8|6\rangle &= \frac{1}{\sqrt{8}} \sum_{y=0}^{7} e^{2\pi i (6y)/8}|y\rangle \\ &= \frac{1}{\sqrt{8}} \left(|0\rangle - i|1\rangle - |2\rangle + i|3\rangle + |4\rangle - i|5\rangle - |6\rangle + i|7\rangle\right). \end{aligned}$$

It is easy to mistake the phase at $y = 1$: $e^{3\pi i/2} = -i$, not $+i$. The probability distribution is uniform despite the alternating phases. A direct consistency check is $\sum_y |\omega_8^{6y}/\sqrt{8}|^2 = 8(1/8) = 1$.

Example 3.6 (A superposition whose Fourier output is sparse)**.** Consider $(|0\rangle+|4\rangle)/\sqrt{2}$ in dimension eight. Its Fourier amplitude at y is

$$A_y = \frac{1}{4}\left(1 + e^{\pi i y}\right) = \frac{1}{4}(1 + (-1)^y).$$

Thus $A_y = 1/2$ for even y and $A_y = 0$ for odd y. Each of $0, 2, 4, 6$ occurs with probability $1/4$. This calculation illustrates how relative phase, rather than the mere magnitudes of amplitudes, carries the information that Fourier measurement extracts.

3.4.4 Periodic combs and the exact geometric-sum amplitude

Suppose an input register of size $Q = 2^t$ is prepared in the normalized arithmetic progression

$$|\psi_{x_0,r}\rangle = \frac{1}{\sqrt{L}} \sum_{\ell=0}^{L-1} |x_0 + \ell r\rangle, \quad L = \#\{\ell \geq 0 : x_0 + \ell r < Q\}.$$

After applying $F_Q^\dagger$, the amplitude at y is

$$A_y = \frac{e^{-2\pi i x_0 y/Q}}{\sqrt{QL}} \sum_{\ell=0}^{L-1} e^{-2\pi i \ell r y/Q}.$$

If ry/Q is an integer, all summands align and $|A_y|^2 = L/Q$. Otherwise the exact probability is

$$\Pr(Y = y) = \frac{1}{QL} \left| \frac{1 - e^{-2\pi i L r y/Q}}{1 - e^{-2\pi i r y/Q}} \right|^2 = \frac{1}{QL} \frac{\sin^2(\pi L r y/Q)}{\sin^2(\pi r y/Q)},$$

where the removable singularities are filled by continuity. Peaks occur when ry/Q is close to an integer, equivalently y/Q close to j/r. This is the formula underlying the spectral peaks used in order finding. The phase factor involving x_0 affects amplitudes but cancels from probabilities. The approximation $\Pr(Y \approx jQ/r)$ must not be confused with an exact uniform distribution unless $r \mid Q$ in the relevant idealized configuration.

3.4.5 Phase estimation as Fourier analysis of eigenvalues

Let $U|u\rangle = e^{2\pi i \phi}|u\rangle$, with $\phi \in [0, 1)$. Write $Q = 2^t$. Preparing $|+\rangle^{\otimes t}|u\rangle$ and applying controlled U^{2^j} produces

$$\frac{1}{\sqrt{Q}} \sum_{k=0}^{Q-1} e^{2\pi i k \phi} |k\rangle |u\rangle.$$

Applying $F_Q^\dagger$ yields

$$\sum_{y=0}^{Q-1} A_y |y\rangle |u\rangle, \qquad A_y = \frac{1}{Q} \sum_{k=0}^{Q-1} e^{2\pi i k(\phi - y/Q)}.$$

If $\phi = m/Q$ for some integer m, the sum is 1 when $y = m$ and zero otherwise. This proves exact phase recovery for representable binary phases. For arbitrary ϕ, write $\delta_y = \phi - y/Q$; the probability is

$$\Pr(Y = y) = \frac{1}{Q^2} \frac{\sin^2(\pi Q \delta_y)}{\sin^2(\pi \delta_y)},$$

with the usual continuous extension. A finite register cannot in general return an exact irrational phase, even when the eigenstate is prepared perfectly.

Proposition 3.7 (A constant-probability nearest-integer guarantee)**.** *Let $y_\star$ be a nearest integer to $Q\phi$ modulo Q. Then phase estimation measures $y_\star$ with probability at least $4/\pi^2$.*

Proof. Choose $\delta = \phi - y_\star/Q$, so $|\delta| \le 1/(2Q)$. For $0 \le x \le \pi/2$, $\sin x \ge 2x/\pi$, while $|\sin x| \le |x|$ for all real x. Hence

$$\frac{\sin^2(\pi Q\delta)}{Q^2 \sin^2(\pi\delta)} \ge \frac{(2Q|\delta|)^2}{Q^2(\pi|\delta|)^2} = \frac{4}{\pi^2}.$$

At $\delta = 0$ the probability equals one. The estimate is a lower bound, not an exact success probability for every phase. □

3.4.6 A two-by-two spectral decomposition and a three-bit readout

Let

$$U = \frac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

The vectors $|+\rangle = (1,1)^T/\sqrt{2}$ and $|-\rangle = (1,-1)^T/\sqrt{2}$ are eigenvectors: by direct multiplication $U|+\rangle = |+\rangle$ and $U|-\rangle = i|-\rangle$. The eigenphases are 0 and $1/4$. With $t = 3$, $Q = 8$, the phase $1/4$ is represented exactly by the integer $Q/4 = 2$, so the output register becomes $|010\rangle$ with probability one when the eigenstate is $|-\rangle$. For $|+\rangle$ it becomes $|000\rangle$. If the system input is instead $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$, linearity gives

$$\frac{1}{\sqrt{2}}\big(|000\rangle|+\rangle + |010\rangle|-\rangle\big).$$

Measuring only the phase register yields 000 or 010, each with probability $1/2$. The system is not guaranteed to remain in its original $|0\rangle$ state after this measurement: the phase-estimation circuit correlates different eigenspaces with different output labels.

3.4.7 Circuit convention checks

For an n-qubit Fourier circuit, the output bit reversal may be implemented by SWAP gates or handled by relabeling the classical output. This choice changes the raw measured bit string but not the mathematical Fourier transform once the convention is specified. The controlled powers in phase estimation must be assigned consistently to significance: applying controlled U^{2^j} to counting bit j and then an inverse QFT in the corresponding register order is equivalent to reversed wire assignment followed by matching SWAPs. An apparently incorrect phase readout often results from mixing the binary order of wires with the arithmetic integer labeling, rather than from the Fourier algebra itself.

3.5 Gate-level Fourier circuits and intermediate-state calculations

3.5.1 The exact three-qubit QFT without the final bit reversal

A three-qubit QFT can be performed by a Hadamard on the first wire, controlled phase rotations from later wires to the first, a Hadamard and controlled phase on the second, and a final Hadamard on the third. Under the most-significant-bit-first ordering, the output of this sequence has reversed wire significance until the outer wires are swapped. The relevant phase gates are

$$R_2 = \mathrm{diag}(1, e^{i\pi/2}), \qquad R_3 = \mathrm{diag}(1, e^{i\pi/4}).$$

For a computational basis input $|x_1x_2x_3\rangle$, the first wire starts in $|x_1\rangle$. Applying its Hadamard gives $[|0\rangle + (-1)^{x_1}|1\rangle]/\sqrt{2}$. A controlled-$R_2$ driven by x_2 multiplies the $|1\rangle$ coefficient by $e^{2\pi i x_2/4}$, and controlled-R_3 driven by x_3 multiplies it by $e^{2\pi i x_3/8}$. Hence the first wire becomes

$$\frac{|0\rangle + e^{2\pi i(4x_1+2x_2+x_3)/8}|1\rangle}{\sqrt{2}} = \frac{|0\rangle + e^{2\pi i\, 0.x_1x_2x_3}|1\rangle}{\sqrt{2}}.$$

The second wire, after its Hadamard and controlled-R_2 driven by the third, becomes $(|0\rangle + e^{2\pi i\, 0.x_2x_3}|1\rangle)/\sqrt{2}$. The third wire becomes $(|0\rangle + e^{2\pi i\, 0.x_3}|1\rangle)/\sqrt{2}$. These three factors appear from most detailed to least detailed fraction before the final SWAP; after swapping first and third wires, their order matches the conventional $F_8|x\rangle$ state.

Example 3.8 (The input $|110\rangle$ at every major stage)**.** Let $(x_1, x_2, x_3) = (1, 1, 0)$. The first Hadamard gives $|-\rangle_1|10\rangle_{23}$. The controlled-$R_2$ from wire 2 applies i to the $|1\rangle_1$ coefficient, giving $(|0\rangle - i|1\rangle)_1|10\rangle_{23}/\sqrt{2}$. The controlled-$R_3$ from wire 3 is inactive because $x_3 = 0$. The second Hadamard turns $|1\rangle_2$ into $|-\rangle_2$, and the subsequent controlled-R_2 from wire 3 is also inactive. The final Hadamard on wire 3 turns $|0\rangle_3$ into $|+\rangle_3$. The un-swapped output is

$$\frac{|0\rangle - i|1\rangle}{\sqrt{2}} \otimes |-\rangle \otimes |+\rangle.$$

Swapping wires 1 and 3 gives

$$|+\rangle \otimes |-\rangle \otimes \frac{|0\rangle - i|1\rangle}{\sqrt{2}}.$$

Expanding this product in the standard output integer basis gives precisely $F_8|6\rangle = 8^{-1/2}\sum_{y=0}^{7}(-i)^y|y\rangle$. In particular $|101\rangle$ $(y = 5)$ has amplitude $-i/\sqrt{8}$. The computation simultaneously verifies the circuit phase convention and the output-bit-order rule.

3.5.2 Finite-resolution phase-estimation failure as a distribution

The phase-estimation amplitude is a Dirichlet-kernel sample. If $Q\phi = m + \eta$ with $m \in \mathbb{Z}$ and fractional displacement $\eta \in [-1/2, 1/2]$, then the nearest integer m has probability

$$P_m = \frac{1}{Q^2}\frac{\sin^2(\pi\eta)}{\sin^2(\pi\eta/Q)}.$$

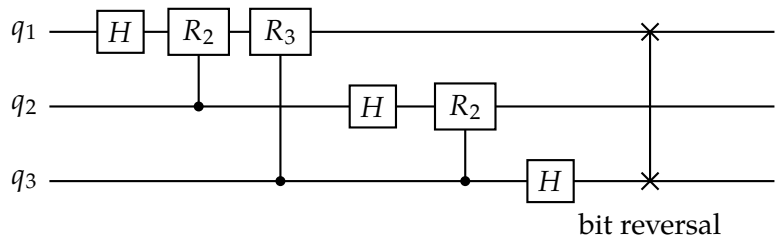


Figure 3.1: An editable three-qubit QFT decomposition: H and controlled phase gates followed by a final wire reversal.

For Q large and fixed η, this approaches $[\sin(\pi\eta)/(\pi\eta)]^2$, interpreted continuously at $\eta = 0$. The worst displacement to the nearest integer is approximately $\eta = \pm 1/2$, yielding the familiar lower bound $4/\pi^2$. If the eigenphase is exactly representable, $\eta = 0$ and the distribution collapses to a delta function. The same circuit therefore ranges from deterministic phase recovery to a broad but sharply centered distribution, depending on arithmetic alignment with the counting register.

If the system input is a superposition $\sum_j \beta_j |u_j\rangle$ of eigenvectors with phases ϕ_j, the final joint state is $\sum_{j,y} \beta_j A_y(\phi_j)|y\rangle|u_j\rangle$. The phase-register marginal is the mixture $\Pr(Y = y) = \sum_j |\beta_j|^2 |A_y(\phi_j)|^2$ when the eigenvectors are orthogonal. Interference between different eigenvectors does not appear in this marginal, because the corresponding system states are orthogonal. This is the formal reason that eigenphase measurements reveal a spectral distribution weighted by the input's eigenbasis components.

3.5.3 The geometric series at an approximate Fourier peak

For a periodic state $L^{-1/2} \sum_{\ell=0}^{L-1} |x_0 + \ell r\rangle$, let $\delta = ry/Q - j$ be the offset of ry/Q from the closest integer j. The probability of measuring y is

$$P(y) = \frac{1}{QL} \left| \sum_{\ell=0}^{L-1} e^{-2\pi i \ell \delta} \right|^2 = \frac{1}{QL} \frac{\sin^2(\pi L \delta)}{\sin^2(\pi \delta)}.$$

The limit at $\delta = 0$ is L/Q because both numerator and denominator vanish quadratically and their ratio tends to L^2. The primary lobe has width of order $1/L$ in the variable δ, so longer periodic superpositions produce narrower Fourier peaks. This explains the computational role of a sufficiently large counting register: it can separate nearby rational phases more reliably. The identity also clarifies why an observed y need not be a multiple of Q/r when Q/r is not an integer.

CHAPTER 4

Period Finding, Integer Factorization, and Amplitude Amplification

4.1 The multiplicative order and factoring reduction

Definition 4.1 (Multiplicative order)**.** For coprime integers a, N, the order of a modulo N is

$$r = \mathrm{ord}_N(a) := \min\{k \geq 1 : a^k \equiv 1 \pmod{N}\}.$$

Thus $f(x) = a^x \bmod N$ is periodic with least positive period r.

A quantum order-finding procedure is one component of Shor's factoring algorithm. Given composite N, first choose an integer $1 < a < N$. If $d = \gcd(a, N)$ satisfies $1 < d < N$, factorization has already succeeded. Otherwise compute $r = \mathrm{ord}_N(a)$. If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then

$$(a^{r/2} - 1)(a^{r/2} + 1) = a^r - 1 \equiv 0 \pmod{N}$$

provides candidate nontrivial divisors $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$. Odd r or $a^{r/2} \equiv -1 \pmod{N}$ requires a fresh attempt.

4.2 Two equivalent quantum order-finding constructions

One construction uses a reversible arithmetic oracle to prepare

$$\frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle |a^x \bmod N\rangle, \qquad Q = 2^t.$$

Measuring the second register leaves an arithmetic progression $x_0, x_0 + r, \ldots$ in the first. The inverse QFT produces Fourier peaks near $y \approx jQ/r$, as derived in the preceding chapter.

An equivalent formulation uses modular multiplication $U_a|z\rangle = |az \bmod N\rangle$ on the invertible residue classes. Its eigenvectors are

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} |a^k \bmod N\rangle, \qquad U_a|u_s\rangle = e^{2\pi i s/r}|u_s\rangle.$$

Since $|1\rangle = r^{-1/2} \sum_{s=0}^{r-1} |u_s\rangle$, QPE on U_a with target $|1\rangle$ samples phases s/r. The periodic-comb and eigenphase interpretations are the same Fourier mechanism.

If $r \mid Q$, the inverse-QFT outcome is exactly uniform on $y = jQ/r$ for $j = 0, \ldots, r-1$. If $r \nmid Q$, the distribution concentrates around these locations. Moreover, if $\gcd(j, r) > 1$, reducing j/r hides part of r; additional samples or candidate-multiple checks may be necessary.

4.3 Recovering periods using continued fractions

For $\xi = y/Q$, write its finite simple continued fraction $[a_0; a_1, \ldots, a_m]$. The convergents p_j/q_j are obtained using

$$p_{-2} = 0, \quad p_{-1} = 1, \quad p_j = a_j p_{j-1} + p_{j-2},$$
$$q_{-2} = 1, \quad q_{-1} = 0, \quad q_j = a_j q_{j-1} + q_{j-2}.$$

A promising denominator $q_j < N$ must be validated with $a^{q_j} \equiv 1 \pmod{N}$ (or by testing an appropriate multiple). The observed integer y is a frequency label, not the order itself.

Example 4.2 (Order eight modulo seventeen)**.** The sequence $2^x \bmod 17$ takes the values

$$1,\ 2,\ 4,\ 8,\ 16,\ 15,\ 13,\ 9$$

before repeating with period $r = 8$. With $Q = 16$, conditioning on output 4 leaves $(|2\rangle + |10\rangle)/\sqrt{2}$. The inverse-QFT amplitude is

$$A_y = \frac{1}{\sqrt{32}} e^{-\pi i y/4}[1 + (-1)^y].$$

Hence each even $y \in \{0, 2, \ldots, 14\}$ occurs with probability $1/8$, and odd outcomes are impossible. In particular $A_2 = -i/\sqrt{8}$. This is an example of order finding over a prime modulus, not factorization of the prime.

Example 4.3 (Nonintegral peaks and continued fractions)**.** For $N = 7, a = 2$, the order is 3, and with $Q = 16$ the peaks lie near $0, 16/3, 32/3$. Measurement $y = 5$ gives $5/16 = [0; 3, 5]$ with convergent $1/3$, whose denominator is verified by $2^3 \equiv 1 \pmod{7}$. Likewise $y = 11$ gives $11/16 = [0; 1, 2, 5]$ and convergent $2/3$.

For $N = 31, a = 2$, the order is 5. Given $Q = 32$ and $y = 13$, the expansion $13/32 = [0; 2, 2, 6]$ has convergent $2/5$, whose denominator passes $2^5 \equiv 1 \pmod{31}$.

Example 4.4 (Actual factoring instances)**.** For $N = 15$ and $a = 7$, the modular powers are $1, 7, 4, 13, 1$, so $r = 4$. Taking $Q = 16$ and observing a second-register value corresponding to an arithmetic progression of period four yields Fourier labels $\{0, 4, 8, 12\}$, each with probability $1/4$. The observation $12/16 = 3/4$ recovers $r = 4$, and

$$\gcd(7^2 - 1, 15) = 3, \qquad \gcd(7^2 + 1, 15) = 5.$$

For $N = 21$ and $a = 2$, the order is 6 because $2^6 \equiv 1 \pmod{21}$. Then $\gcd(2^3 - 1, 21) = 7$ and $\gcd(2^3 + 1, 21) = 3$.

4.4 The unstructured search problem

Definition 4.5 (Unique marked-item search)**.** Let $f : \{0,1\}^n \to \{0,1\}$ have exactly one marked input w. Given coherent access to $O_f|x\rangle = (-1)^{f(x)}|x\rangle$, produce w with high probability while minimizing oracle queries.

Put $N = 2^n$ and $|s\rangle = N^{-1/2} \sum_x |x\rangle$. The phase oracle is $O_f = I - 2|w\rangle\langle w|$. The diffusion operator and Grover iterate are

$$D = 2|s\rangle\langle s| - I, \qquad G = DO_f.$$

If a state has computational-basis amplitudes a_x with arithmetic mean $\bar{a} = N^{-1} \sum_x a_x$, then D maps $a_x \mapsto 2\bar{a} - a_x$. It is a reflection about the uniform state.

Define $|w^\perp\rangle = (N-1)^{-1/2} \sum_{x \neq w} |x\rangle$, and $\sin\theta = N^{-1/2}$. The initial state is

$$|s\rangle = \cos\theta|w^\perp\rangle + \sin\theta|w\rangle.$$

The product of the two reflections rotates by 2θ, hence

$$G^k|s\rangle = \cos((2k+1)\theta)|w^\perp\rangle + \sin((2k+1)\theta)|w\rangle, \quad \Pr(w) = \sin^2((2k+1)\theta).$$

The ideal real-valued iteration count is $k_* = \pi/(4\theta) - 1/2$, with $\theta = \arcsin(N^{-1/2})$; choosing an appropriate nearby nonnegative integer gives $\Theta(\sqrt{N})$ oracle queries. Excess iterations cause over-rotation. For M known marked items, replace $\sin\theta$ by $\sqrt{M/N}$ and obtain $\Theta(\sqrt{N/M})$ queries in the nontrivial regime.

4.5 Diffusion-circuit decomposition

Because $|s\rangle = H^{\otimes n}|0^n\rangle$,

$$D = H^{\otimes n}(2|0^n\rangle\langle 0^n| - I)H^{\otimes n}.$$

Let $C^{n-1}Z = I - 2|1^n\rangle\langle 1^n|$. Then

$$D = -H^{\otimes n}X^{\otimes n}C^{n-1}ZX^{\otimes n}H^{\otimes n}.$$

The minus sign is a global phase for an ordinary Grover iterate. To implement O_w, conjugate $C^{n-1}Z$ by X gates on every wire where the target word w has bit zero. Multi-controlled gates may require ancillas and elementary-gate decompositions; oracle query counts are not physical gate counts.

Example 4.6 (Exact two-qubit Grover search)**.** Let $N = 4$ and $w = 11$. The initial amplitudes are all $1/2$. The phase oracle changes them to $(1/2, 1/2, 1/2, -1/2)$, whose mean is $1/4$. Diffusion maps the amplitudes to $(0, 0, 0, 1)$, so a single iteration gives $|11\rangle$ with probability one. The geometric calculation agrees: $\theta = \arcsin(1/2) = \pi/6$, and $(2 \cdot 1 + 1)\theta = \pi/2$.

The classical unstructured black-box problem requires $\Theta(N/M)$ queries in the usual marked-set regime, whereas amplitude amplification uses $\Theta(\sqrt{N/M})$ when M is known. The speedup is quadratic, not exponential.

4.6 Fully worked order-finding and search calculations

4.6.1 Reduction of factoring to a multiplicative order

Fix an odd composite integer N and choose an integer a with $1 < a < N$. If $d = \gcd(a, N) > 1$ and $d < N$, Euclid's algorithm already returns a nontrivial factor. Otherwise a lies in the multiplicative group $(\mathbb{Z}/N\mathbb{Z})^\times$ and possesses a finite order

$$r = \operatorname{ord}_N(a) = \min\{k \geq 1 : a^k \equiv 1 \pmod{N}\}.$$

Suppose r is even. Put $z = a^{r/2} \bmod N$. Then $z^2 \equiv 1 \pmod{N}$, and N divides $(z-1)(z+1)$. If $z \not\equiv -1 \pmod{N}$, and r is the order, then also $z \not\equiv 1 \pmod{N}$; therefore a nontrivial square root of unity has been found. For suitable composite N, $\gcd(z-1, N)$ and $\gcd(z+1, N)$ yield nontrivial factors. The failure cases r odd and $z \equiv -1$ explain the need to try additional values of a. For even N, the factor 2 is available classically, so one customarily restricts the quantum order-finding reduction to odd inputs.

Example 4.7 (Factoring fifteen)**.** Choose $N = 15$ and $a = 7$. Modulo 15 the powers are $7^0 \equiv 1, 7^1 \equiv 7, 7^2 \equiv 4, 7^3 \equiv 13$, and $7^4 \equiv 1$. Hence $r = 4$. The half-period value is $z = 7^2 \equiv 4$, and

$$\gcd(z-1, 15) = \gcd(3, 15) = 3, \quad \gcd(z+1, 15) = \gcd(5, 15) = 5.$$

The factorization $15 = 3 \cdot 5$ is recovered. The quantum subroutine is needed only to find the order; modular exponentiation and the two gcd calculations are classical.

Example 4.8 (Factoring twenty-one)**.** For $N = 21$, $a = 2$, the successive residues are $1, 2, 4, 8, 16, 11, 1$, giving order $r = 6$. The half-period residue is $2^3 \equiv 8$, so $\gcd(8-1, 21) = 7$ and $\gcd(8+1, 21) = 3$. Both examples expose the same structural reduction even though their periods differ.

4.6.2 The modular multiplication operator and its eigenvectors

For $\gcd(a, N) = 1$, define a permutation unitary on the residue basis by $M_a|x\rangle = |ax \bmod N\rangle$ for $0 \leq x < N$, extended unitarily on unused computational basis states when the register dimension exceeds N. Restricting to the orbit of 1, the states

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} |a^k \bmod N\rangle, \quad s = 0, \ldots, r-1,$$

are eigenvectors satisfying $M_a|u_s\rangle = e^{2\pi i s/r}|u_s\rangle$. The equality follows by replacing k with $k+1$ modulo r. The basis state $|1\rangle$ decomposes as $r^{-1/2} \sum_s |u_s\rangle$. Thus quantum phase estimation applied to M_a with system input $|1\rangle$ observes eigenphases s/r with equal weights $1/r$, before considering finite-register resolution.

This derivation relates the circuit with controlled powers of modular multiplication to the apparent alternative that computes $a^x \bmod N$ in a second register. They implement the same periodic structure, though intermediate entanglement and the role of the second register can be presented differently. Neither construction assumes that the order r is already known; it is recovered from measured rational approximations.

4.6.3 An exact inverse-QFT computation with modulus seventeen

The following calculation is an *order-finding illustration*, not an attempt to factor the prime number 17. For $a = 2 \bmod 17$, the residues at exponents $0, \ldots, 7$ are

$$1, 2, 4, 8, 16, 15, 13, 9,$$

and $2^8 \equiv 1 \pmod{17}$, so the order is 8. Let $Q = 16$. Conditioning on observing the output value 4 from the modular-exponentiation register produces the normalized counting-register state

$$|\psi\rangle = (|2\rangle + |10\rangle)/\sqrt{2}.$$

The inverse Fourier transform gives an amplitude

$$\begin{aligned} A_y &= \frac{1}{\sqrt{32}} \left(e^{-2\pi i(2)y/16} + e^{-2\pi i(10)y/16} \right) \\ &= \frac{e^{-\pi i y/4}}{\sqrt{32}} \left(1 + (-1)^y\right). \end{aligned}$$

When y is odd the two terms cancel. For even y, the magnitude is $2/\sqrt{32} = 1/\sqrt{8}$, so every even outcome $y \in \{0, 2, 4, 6, 8, 10, 12, 14\}$ has probability $1/8$. The full state is

$$F_{16}^{\dagger}|\psi\rangle = \frac{1}{\sqrt{8}} \sum_{j=0}^{7} e^{-\pi i j/2} |2j\rangle.$$

For example $A_0 = 1/\sqrt{8}$, $A_2 = -i/\sqrt{8}$, and $A_4 = -1/\sqrt{8}$. These phase values matter to the state but disappear from the final measurement probabilities. The identity $8 \cdot (1/8) = 1$ checks normalization.

4.6.4 Continued fractions when the period does not divide the register size

For $N = 7$, $a = 2$, one has $2^1 \equiv 2$, $2^2 \equiv 4$, $2^3 \equiv 1 \pmod{7}$, so $r = 3$. If $Q = 16$, the ideal peak locations are 0, $16/3 \approx 5.333$, and $32/3 \approx 10.667$. Suppose phase estimation measures $y = 5$. Then

$$\frac{y}{Q} = \frac{5}{16} = [0; 3, 5],$$

whose convergents are $0/1$, $1/3$, and $5/16$. Testing the candidate denominator 3 gives $2^3 \equiv 1 \pmod 7$, so the order is indeed 3. With $y = 11$, the expansion is $11/16 = [0; 1, 2, 5]$ and the convergent $2/3$ again identifies the order. The fraction y/Q is not itself s/r because the register resolution is finite.

Example 4.9 (A different rational approximation)**.** For $N = 31$ and $a = 2$, $2^5 = 32 \equiv 1 \pmod{31}$ and the order is 5. Suppose $Q = 32$ and $y = 13$. Then $13/32 = 0.40625$, close to $2/5 = 0.4$. Euclidean division gives $32 = 2 \cdot 13 + 6$, $13 = 2 \cdot 6 + 1$, and $6 = 6 \cdot 1$, so

$$\frac{13}{32} = [0; 2, 2, 6], \qquad \frac{0}{1}, \frac{1}{2}, \frac{2}{5}, \frac{13}{32}$$

are the successive convergents. Testing denominator 5 recovers $2^5 \equiv 1 \pmod{31}$. Since 31 is prime, this is a demonstration of order recovery and does not yield a factorization.

Proposition 4.10 (Uniqueness of close rational approximations)**.** *If two distinct reduced fractions p/q and p'/q' both have positive denominators at most R, then $|p/q - p'/q'| \geq 1/R^2$. In particular a rational p/q with $q < R$ and $|x - p/q| < 1/(2R^2)$ is the unique reduced fraction with denominator below R in that interval.*

Proof. The integer $pq' - p'q$ is nonzero, hence has absolute value at least one, and

$$\left|\frac{p}{q} - \frac{p'}{q'}\right| = \frac{|pq' - p'q|}{qq'} \geq \frac{1}{R^2}.$$

If two candidates were both within $1/(2R^2)$ of x, the triangle inequality would contradict this separation. □

A classical continued-fraction theorem guarantees that p/q is a convergent whenever $|x - p/q| < 1/(2q^2)$. These observations explain why rational approximation can identify s/r when Q is sufficiently large. One must still validate candidate denominators by modular exponentiation because a reduced fraction s/r can have a smaller denominator when $\gcd(s, r) > 1$.

4.6.5 Grover iteration as an exact two-dimensional rotation

Assume the marked set contains M elements among N candidates. Define

$$|w\rangle = \frac{1}{\sqrt{M}} \sum_{f(x)=1} |x\rangle, \quad |b\rangle = \frac{1}{\sqrt{N - M}} \sum_{f(x)=0} |x\rangle.$$

They are orthonormal, and the uniform state is $|s\rangle = \sin\theta|w\rangle + \cos\theta|b\rangle$ with $\sin^2\theta = M/N$. In the ordered basis $(|w\rangle, |b\rangle)$, the oracle and diffusion reflections are

$$O_f = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad D = 2\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix}\begin{pmatrix} \sin\theta & \cos\theta \end{pmatrix} - I.$$

Multiplying gives

$$G = DO_f = \begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ -\sin(2\theta) & \cos(2\theta) \end{pmatrix}$$

under the corresponding component convention. Therefore

$$G^k|s\rangle = \sin((2k+1)\theta)|w\rangle + \cos((2k+1)\theta)|b\rangle, \quad P_k = \sin^2((2k+1)\theta).$$

To make the marked component close to one, choose the integer k nearest to $\pi/(4\theta) - 1/2$. For $M \ll N$, $\theta \sim \sqrt{M/N}$ and the number of oracle calls is $\Theta(\sqrt{N/M})$. The exact choice depends on the integer-rounding convention and the promised value of M.

Example 4.11 (Complete two-qubit amplitude calculation)**.** Set $N = 4$ and mark $|11\rangle$. Initially every amplitude is $1/2$. After the phase oracle the vector is $(1/2, 1/2, 1/2, -1/2)^T$. Its mean amplitude is $(1/2 + 1/2 + 1/2 - 1/2)/4 = 1/4$. The diffusion operation reflects each amplitude about the mean: $a_x \mapsto 2(1/4) - a_x$. The three unmarked amplitudes become zero, while the marked amplitude becomes 1. Hence exactly one Grover iteration prepares $|11\rangle$. The angle calculation agrees: $\sin\theta = 1/2$, so $\theta = \pi/6$ and $3\theta = \pi/2$.

4.6.6 Explicit oracle synthesis for a known marked bit string

When a particular target $w = w_1 \cdots w_n$ is known *for circuit construction*, a phase oracle for this target can be implemented by conjugating the multi-controlled Z that changes the sign of $|1^n\rangle$:

$$O_w = \left(\bigotimes_{j:w_j=0} X_j \right) C^{n-1}Z \left(\bigotimes_{j:w_j=0} X_j \right).$$

For instance, $w = 0010$ requires X on wires $1, 2, 4$ before and after the multi-controlled phase operation. This circuit is a useful concrete illustration, but it must not be misinterpreted as a free specification of the unknown answer in a black-box search problem: querying an oracle and constructing it from a known target are different computational models.

4.6.7 Amplitude amplification beyond uniform search

Let $A|0\rangle = \sqrt{p}|g\rangle + \sqrt{1-p}|b\rangle$ with orthogonal normalized good and bad states. A reflection about the good subspace followed by reflection about $A|0\rangle$ again produces a rotation in their span with angle $2\arcsin\sqrt{p}$. Repeating this rotation $O(1/\sqrt{p})$ times can raise success to a constant. This principle also underlies coherent amplification of low-probability block-encoding postselection. Its cost depends on coherent access to A, $A^\dagger$, and phase reflections. If only classical samples of the two outcome branches are available, the same square-root improvement cannot be inferred without those extra operations.

4.7 Arithmetic postprocessing, exact period states, and search variants

4.7.1 The modular order as a group-theoretic invariant

When $\gcd(a, N) = 1$, the residue class of a belongs to the finite multiplicative group of units modulo N. Its order r divides the group order $\varphi(N)$ by Lagrange's theorem. For a semiprime $N = pq$ with distinct odd primes, $\varphi(N) = (p-1)(q-1)$, but the algorithm does not require knowing p or q in advance. A candidate order obtained from quantum measurement can be verified by the classical test $a^r \equiv 1 \pmod{N}$. If the test succeeds, one should also determine whether a smaller divisor of r produces one, because the measured convergent denominator can be a multiple of the true order or, when the sampled Fourier numerator is not coprime to r, a proper divisor. The most direct test is modular exponentiation together with appropriate divisor checks.

A simple example shows why the numerator matters. If the true order is $r = 6$ and phase estimation identifies $s/r = 2/6$, the reduced rational is $1/3$ and continued fractions return denominator three. But a^3 need not equal one; for an element of order six, a^3 has order two. Trying a small multiple of the candidate denominator can recover the true order. Thus the continued-fraction algorithm produces *candidates*, not an unconditional exact order on every outcome.

4.7.2 An exact periodic state for factoring fifteen

For $N = 15$, $a = 7$, the order is four. With $Q = 16$ and modular-exponentiation output register measured as $7^2 \bmod 15 = 4$, the counting register becomes

$$|\psi\rangle = \frac{1}{2}(|2\rangle + |6\rangle + |10\rangle + |14\rangle).$$

Applying the inverse Fourier transform gives the amplitude at y,

$$\begin{aligned} A_y &= \frac{1}{8}\sum_{\ell=0}^{3} e^{-2\pi i(2+4\ell)y/16} \\ &= \frac{e^{-\pi i y/4}}{8}\sum_{\ell=0}^{3} e^{-\pi i \ell y/2}. \end{aligned}$$

The inner sum is four when $y \equiv 0 \pmod 4$ and zero otherwise. Thus the four outcomes $0, 4, 8, 12$ have equal probability $1/4$. In particular, outcome $y = 12$ gives $y/Q = 3/4$, which identifies order four. The subsequent classical computation uses $7^2 \equiv 4 \pmod{15}$, yielding factors $\gcd(4-1, 15) = 3$ and $\gcd(4+1, 15) = 5$. The entire two-stage calculation is exact because the order divides the Fourier register size in this toy example.

4.7.3 Continued-fraction convergents by recurrence

For a real number $\xi = [a_0; a_1, \ldots, a_m]$, define numerator and denominator sequences using

$$p_{-2} = 0,\ p_{-1} = 1, \quad q_{-2} = 1,\ q_{-1} = 0,$$

and

$$p_j = a_j p_{j-1} + p_{j-2}, \qquad q_j = a_j q_{j-1} + q_{j-2}.$$

These produce the convergents p_j/q_j. The determinant identity

$$p_j q_{j-1} - p_{j-1} q_j = (-1)^{j-1}$$

follows by induction: the recurrence replaces the determinant at stage j by the negative of its value at stage $j-1$. It implies that successive convergents are reduced and that their denominators form the appropriate sequence of best rational approximants. In Shor postprocessing, one filters these denominators by the bound $q_j < N$ and verifies the modular order condition.

Example 4.12 (Reconstructing 13/32 explicitly)**.** Perform the Euclidean algorithm $32 = 2 \cdot 13 + 6$, $13 = 2 \cdot 6 + 1$, and $6 = 6 \cdot 1$. The continued fraction is $13/32 = [0; 2, 2, 6]$. The recurrence gives $(p_0, q_0) = (0, 1)$, $(p_1, q_1) = (1, 2)$, $(p_2, q_2) = (2, 5)$, and $(p_3, q_3) = (13, 32)$. The convergent $2/5$ approximates $13/32$ with error $|13/32 - 2/5| = |65 - 64|/160 = 1/160$. When $N = 31$, $a = 2$, the candidate denominator five passes $2^5 \equiv 1 \pmod{31}$. This calculation displays both the rational approximation and its arithmetic validation.

4.7.4 Why even order alone is insufficient

If r is even, $z = a^{r/2} \bmod N$ satisfies $z^2 \equiv 1 \pmod{N}$, but it may equal $-1 \pmod{N}$. In that case $\gcd(z+1, N) = N$ and $\gcd(z-1, N)$ may be one, so the two gcd calls are uninformative. For example, modulo the odd prime 7, the element $a = 3$ has powers $3, 2, 6, 4, 5, 1$ and order six; its half-period residue is $3^3 \equiv 6 \equiv -1 \pmod{7}$. The order computation is correct, but no nontrivial factorization can result because the modulus is prime. The example isolates the separate logical steps: finding an order, finding a nontrivial square root of unity, and extracting a factor are not equivalent tasks.

4.7.5 Grover amplitude amplification with several marked items

For N total possibilities and M marked items, define the normalized marked and unmarked uniform superpositions $|w\rangle, |b\rangle$. The initial uniform state has marked probability M/N and can be expressed as $\sin\theta|w\rangle + \cos\theta|b\rangle$ with $\theta = \arcsin\sqrt{M/N}$. One Grover iteration rotates the state by 2θ, and after k iterations the success probability is

$$P_k = \sin^2((2k+1)\theta).$$

For example, if $N = 16$, $M = 4$, then $\sin\theta = 1/2$ and $\theta = \pi/6$. One iteration yields $P_1 = \sin^2(\pi/2) = 1$, meaning the algorithm finds *some* marked item with certainty. Conditional on success, measurement is uniform over the four marked possibilities in the symmetric oracle model, so it does not identify a particular one of the four unless the promise provides a unique target.

If instead $N = 16$, $M = 1$, then $\theta = \arcsin(1/4)$. A near-optimal integer number of iterations is the closest integer to $\pi/(4\theta) - 1/2$. Applying too many iterations decreases the success probability because the state rotates past the marked axis and continues around the two-dimensional plane. Grover iteration is oscillatory, not monotonically improving. For unknown M, adaptive or randomized schedules can avoid the need to know the exact optimal iteration count, but these require a different algorithmic analysis from the fixed-M formula.

4.7.6 Diffusion as reflection about the mean amplitude

Let $|\psi\rangle = \sum_{x=0}^{N-1} a_x|x\rangle$, and let $\overline{a} = N^{-1}\sum_x a_x$. Since $|s\rangle = N^{-1/2}\sum_x |x\rangle$, the component of $(2|s\rangle\langle s| - I)|\psi\rangle$ at index x is

$$2\langle x|s\rangle\langle s|\psi\rangle - a_x = 2\overline{a} - a_x.$$

This reflection identity holds for complex amplitudes as well as real amplitudes, although the common elementary picture of points reflected about a real horizontal mean is most intuitive in the real-amplitude case. The diffusion operator preserves the Euclidean norm because it is Hermitian and squares to the identity. Its standard decomposition is

$$D = H^{\otimes n}(2|0^n\rangle\langle 0^n| - I)H^{\otimes n}.$$

The middle reflection can be built from X gates and a multi-controlled phase flip, with an overall global sign choice depending on whether the implementation flips $|0^n\rangle$ or its orthogonal complement. Global signs do not change Grover success probabilities when applied uniformly to an iteration, but should be tracked if the iteration itself becomes a controlled subroutine.

4.7.7 Comparison of query and gate complexity

A Grover phase oracle count of $\Theta(\sqrt{N})$ presupposes access to a black-box predicate $f(x)$ at unit cost. If one query requires a reversible implementation of a classical subroutine with gate cost C_f, the gate cost contributed by oracle use is $\Theta(C_f\sqrt{N})$, plus diffusion and workspace overhead. Similarly, the polynomial complexity of modular exponentiation is an essential part of Shor's full resource analysis even though the conceptual quantum speedup is explained by period finding. These are not objections to the oracle model; they are reminders to specify whether the bound concerns black-box queries or a compiled quantum circuit.

CHAPTER 5

Stabilizer Quantum Error Correction and Fault Tolerance

5.1 Discrete error models and repetition codes

Under the binary symmetric channel, each classical bit flips independently with probability $p \in [0, 1/2]$. The three-bit repetition map $0 \mapsto 000$, $1 \mapsto 111$ corrects one flip by majority vote. The logical failure probability is

$$p_{\mathrm{L}} = \binom{3}{2} p^2(1-p) + p^3 = 3p^2 - 2p^3 = O(p^2).$$

For $0 < p < 1/2$, $p_{\mathrm{L}} < p$; at $p = 0.1$, $p_{\mathrm{L}} = 0.028$. This improvement depends on the noise assumptions. Quantum coding cannot be obtained by copying an unknown $|\psi\rangle$ three times, as cloning is forbidden.

The three-qubit **bit-flip code** encodes

$$\alpha|0\rangle + \beta|1\rangle \longmapsto \alpha|000\rangle + \beta|111\rangle.$$

Its commuting stabilizer generators are $S_1 = Z_1Z_2$ and $S_2 = Z_2Z_3$. Let syndrome bit $s_i = 0$ for eigenvalue $+1$ and $s_i = 1$ for eigenvalue -1. The syndrome table is

Error	I	X_1	X_2	X_3
(s_1, s_2)	00	10	11	01
Recovery	I	X_1	X_2	X_3

The **phase-flip code** is the conjugated encoding

$$|\psi_{\mathrm{L}}\rangle = \alpha|{+}{+}{+}\rangle + \beta|{-}{-}{-}\rangle.$$

Its generators are X_1X_2 and X_2X_3, and the Z_1, Z_2, Z_3 syndromes are respectively $10, 11, 01$. In general, Z-type checks reveal X errors, while X-type checks reveal Z errors.

Proposition 5.1 (Syndrome from anticommutation)**.** *Let $S|\psi_{\mathrm{L}}\rangle = |\psi_{\mathrm{L}}\rangle$ and let E be a Pauli error. If $SE = -ES$, then $S(E|\psi_{\mathrm{L}}\rangle) = -E|\psi_{\mathrm{L}}\rangle$; if $SE = ES$, then the eigenvalue remains $+1$.*

Measuring stabilizer parities therefore reveals the error syndrome without measuring the logical amplitudes α, β. For a Z_iZ_j check, an ancilla initialized in $|0\rangle$ can accumulate the parity through two data-to-ancilla CNOTs and then be measured in the computational basis. This elementary circuit by itself is not guaranteed to be fault tolerant: ancilla faults can propagate.

5.2 The nine-qubit Shor code

Define $|G_\pm\rangle = (|000\rangle \pm |111\rangle)/\sqrt{2}$. The Shor codewords are

$$|0_L\rangle = |G_+\rangle^{\otimes 3}, \qquad |1_L\rangle = |G_-\rangle^{\otimes 3}, \qquad |\psi_L\rangle = \alpha|0_L\rangle + \beta|1_L\rangle.$$

The six intra-block Z checks are

$$Z_1Z_2,\ Z_2Z_3,\ Z_4Z_5,\ Z_5Z_6,\ Z_7Z_8,\ Z_8Z_9,$$

and the inter-block X checks are

$$X_1X_2X_3X_4X_5X_6, \qquad X_4X_5X_6X_7X_8X_9.$$

For an error at position five, X_5 anticommutes with Z_4Z_5 and Z_5Z_6; Z_5 anticommutes with both inter-block checks; Y_5 activates both patterns. An undetected distinction among Z_4, Z_5, Z_6 within the middle block is a form of **degeneracy**: these physical errors have equivalent action on the encoded subspace. The Shor code has parameters $[[9, 1, 3]]$ and corrects arbitrary single-qubit errors, but not all two-qubit errors.

5.3 Pauli expansion and correction criteria

Any complex 2×2 operator admits the decomposition

$$E = a_0 I + a_1 X + a_2 Y + a_3 Z.$$

Equivalently, for $E = \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right)$,

$$E = \frac{a+d}{2} I + \frac{b+c}{2} X + \frac{a-d}{2} Z + \frac{b-c}{2} XZ.$$

The decomposition matters because quantum errors need not be classical random choices of a Pauli; they may be coherent superpositions of error operators.

Theorem 5.2 (Knill–Laflamme condition)**.** *Let P project onto a code subspace C and let $\{E_a\}$ be a collection of error operators. There exists a recovery channel correcting all errors in the linear span of $\{E_a\}$ exactly if and only if*

$$PE_a^\dagger E_b P = c_{ab} P$$

for some Hermitian matrix (c_{ab}). In an orthonormal logical basis this is

$$\langle i_L | E_a^\dagger E_b | j_L \rangle = c_{ab} \delta_{ij}.$$

A single-qubit depolarizing channel under the specified Pauli-error convention is

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z).$$

For nine independent qubits, the probability of at least two nonidentity Pauli faults is

$$1 - (1-p)^9 - 9p(1-p)^8 = 36p^2 + O(p^3).$$

It is a conservative upper bound on logical failure after ideal single-error correction, not an equality: some higher-weight errors may also be harmless or correctable. Correlated noise invalidates this independent-fault calculation.

5.4 Classical Hamming codes and binary parity checks

Work over $\mathbb{F}_2$ and order coordinates as (a, b, c, d, x, y, z). The systematic $(7, 4, 3)$ Hamming code uses

$$x = a \oplus b \oplus d, \qquad y = a \oplus c \oplus d, \qquad z = b \oplus c \oplus d.$$

Its generator and parity-check matrices are

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

Thus $c = mG$, $Hc^\top = 0$, and $HG^\top = 0$ over $\mathbb{F}_2$. For received $r = c \oplus e$, the syndrome is $s = Hr^\top = He^\top$. A one-bit error at position j has syndrome exactly the jth column of H:

Position	none	a	b	c	d	x	y	z
s	000	110	101	011	111	100	010	001

The seven nonzero three-bit vectors occur once each. The perfect-code counting identity $2^4(1 + 7) = 2^7$ accounts for the packing of radius-one Hamming balls around the 16 codewords.

Example 5.3 (A full classical syndrome calculation)**.** Message $(a, b, c, d) = (1, 1, 0, 1)$ gives parity bits $(x, y, z) = (1, 0, 0)$ and codeword $c = 1101100$. Flipping coordinate b gives $r = 1001100$. Computing $Hr^\top = (1, 0, 1)^\top$ identifies the second column and hence the flipped b coordinate. The bit string 101 is a column label under the specified H, not the binary integer five interpreted as a coordinate number.

5.5 The Steane $[[7, 1, 3]]$ CSS code

Let $C = \ker H \subseteq \mathbb{F}_2^7$ be the Hamming code, $D = C^\perp = \mathrm{rowspan}(H)$, and $v = 1111111 \in C \setminus D$. Here $\dim D = 3$, $|D| = 8$, and $HH^\top = 0$ over $\mathbb{F}_2$. The logical codewords are

$$|0_\mathrm{L}\rangle = \frac{1}{\sqrt{8}} \sum_{u \in D} |u\rangle, \qquad |1_\mathrm{L}\rangle = \frac{1}{\sqrt{8}} \sum_{u \in D} |u \oplus v\rangle.$$

The full dual code and complementary coset are

$$\begin{aligned} D = \{&0000000, 1101100, 1011010, 0110110, \\ &0111001, 1010101, 1100011, 0001111\}, \\ v + D = \{&1111111, 0010011, 0100101, 1001001, \\ &1000110, 0101010, 0011100, 1110000\}. \end{aligned}$$

The two disjoint cosets produce orthonormal logical vectors. For each row h of H, introduce $Z(h) = \bigotimes_{j=1}^7 Z_j^{h_j}$ and $X(h) = \bigotimes_{j=1}^7 X_j^{h_j}$. The six generators are

Binary support	Z-type generator	X-type generator
1101100	$Z_1Z_2Z_4Z_5$	$X_1X_2X_4X_5$
1011010	$Z_1Z_3Z_4Z_6$	$X_1X_3X_4X_6$
0111001	$Z_2Z_3Z_4Z_7$	$X_2X_3X_4X_7$

Since $HH^\top = 0$, all X and Z generators commute. Six independent stabilizer constraints reduce the 2^7-dimensional space to dimension $2^{7-6} = 2$, encoding one logical qubit. A valid pair of logical Pauli operators is $\overline{X} = X^{\otimes 7}$ and $\overline{Z} = Z^{\otimes 7}$.

Let s_Z denote outcomes of Z-type generators and s_X those of X-type generators. For a single-qubit Pauli at position j,

E_j	X_j	Z_j	Y_j
s_Z	$H_{\cdot,j}$	000	$H_{\cdot,j}$
s_X	000	$H_{\cdot,j}$	$H_{\cdot,j}$

.

For example, Y_4 has $s_Z = s_X = 111$ and is corrected by Y_4, up to global phase. The distance $d = 3$ permits $t = \lfloor (d-1)/2 \rfloor = 1$ arbitrary physical error to be corrected. A zero syndrome indicates commutation with the measured checks, not the absence of every conceivable logical error.

5.6 Fault-tolerant gates and concatenation

Conjugating faults through CNOT gives $X_c \mapsto X_cX_t$, $X_t \mapsto X_t$, $Z_c \mapsto Z_c$, $Z_t \mapsto Z_cZ_t$. This propagation law motivates **transversal** gates: corresponding physical qubits interact only across separate encoded blocks, preventing one local fault from creating multiple errors in a single block through that gate.

For the Steane code, transversal $H^{\otimes 7}$ acts as logical Hadamard, transversal $(S^\dagger)^{\otimes 7}$ acts as logical S, and seven parallel inter-block CNOTs implement logical CNOT. Indeed the weights in D are zero or four, whereas the weights in $v + D$ are three or seven; multiplying a basis word of weight w by $(-i)^w$ proves $(S^\dagger)^{\otimes 7}|0_\mathrm{L}\rangle = |0_\mathrm{L}\rangle$ and $(S^\dagger)^{\otimes 7}|1_\mathrm{L}\rangle = i|1_\mathrm{L}\rangle$. A logical T is not implemented merely by transversal physical T in this code; non-Clifford gate injection requires encoded ancilla resources, measurement, and feedforward. Clifford gates alone are not universal.

In the simplified independent-fault concatenation model

$$p_0 = p, \qquad p_{j+1} = cp_j^2, \qquad p_k = \frac{(cp)^{2^k}}{c}.$$

For $cp < 1$, the effective logical fault probability decreases doubly exponentially with concatenation depth k, while the number of physical data qubits grows as 7^k. For a computation with L relevant logical locations and union bound $p_\mathrm{fail} \le Lp_k$, a sufficient depth is

$$k \ge \log_2 \left(\frac{\log(L/(c\varepsilon))}{\log(1/(cp))} \right),$$

provided the logarithms are in the regime where this bound is meaningful and the integer depth is rounded upward. With $c = 10$ and $p = 0.01$, the sequence is $p_0 = 10^{-2}, p_1 = 10^{-3}, p_2 = 10^{-5}, p_3 = 10^{-9}$. Actual threshold theorems also require fault-tolerant preparation, memory, syndrome extraction, and measurement; these are not consequences of encoding alone.

5.7 Further syndrome and threshold calculations

Example 5.4 (Phase-flip error and codeword-independent recovery)**.** For $|\psi_{\mathrm{L}}\rangle = \alpha|{+}{+}{+}\rangle + \beta|{-}{-}{-}\rangle$, a physical Z_2 error anticommutes with both X_1X_2 and X_2X_3. Both stabilizer measurements return eigenvalue -1, so the syndrome is 11. Applying Z_2 again corrects the error because $Z_2^2 = I$. The argument makes no reference to the unknown amplitudes α and β.

Example 5.5 (Steane syndrome on the sixth physical qubit)**.** The sixth column of H is 010. A Y_6 error therefore produces $s_Z = s_X = 010$ and is corrected by Y_6. The same location under an X_6 error has $(s_Z, s_X) = (010, 000)$, while a Z_6 error has $(000, 010)$. Identifying the nonzero column and identifying the Pauli type are separate decisions.

Example 5.6 (Determining concatenation depth)**.** Suppose the simplified recursion has $c = 12$, physical fault probability $p = 0.02$, and a computation has $L = 10^6$ fault locations. For target $\varepsilon = 10^{-3}$, let $q = cp = 0.24$. The requirement $Lp_k \le \varepsilon$ is equivalent to

$$2^k \ge \frac{\log(L/(c\varepsilon))}{\log(1/q)} \approx 12.78.$$

The smallest integer depth is $k = 4$, corresponding to $7^4 = 2401$ physical data qubits per logical qubit in a simple Steane concatenation tree. This count does not include physical syndrome, routing, ancilla, and scheduling overhead of an actual architecture.

5.8 Syndrome decoding, code constructions, and fault propagation in detail

5.8.1 Classical redundancy and the exact logical failure probability

Under a binary symmetric channel, let each physical bit flip independently with probability p. A three-bit repetition code maps $0 \mapsto 000$ and $1 \mapsto 111$. Majority-vote decoding fails precisely when at least two of the three bits flip. With the error weight $W \sim \mathrm{Binomial}(3, p)$, the probability is

$$p_{\mathrm{L}} = \Pr(W \ge 2) = \binom{3}{2}p^2(1-p) + \binom{3}{3}p^3 = 3p^2 - 2p^3.$$

A direct subtraction shows

$$p_{\mathrm{L}} - p = -p(1-p)(1-2p).$$

Thus the repetition code improves the raw error probability when $0 < p < 1/2$; for example $p = 0.1$ gives $p_{\mathrm{L}} = 0.028$. At $p = 1/2$ there is no benefit; for $p > 1/2$, majority voting as stated is inferior to a channel-aware decoder that first reverses the more likely bit value. The quadratic suppression $O(p^2)$ critically relies on independent errors. A common-mode event that flips all three bits with probability p would instead generate a logical failure rate of order p.

5.8.2 The three-qubit bit-flip code as a stabilizer code

The encoding is $|0_{\mathrm{L}}\rangle = |000\rangle$, $|1_{\mathrm{L}}\rangle = |111\rangle$ and $\alpha|0\rangle + \beta|1\rangle \mapsto \alpha|000\rangle + \beta|111\rangle$. Define $S_1 = Z_1Z_2$, $S_2 = Z_2Z_3$. Since both logical basis states are $+1$ eigenvectors, every encoded superposition is stabilized by both checks. For any Pauli error E, if $S_jE = (-1)^{s_j}ES_j$, then

$$S_jE|\psi_{\mathrm{L}}\rangle = (-1)^{s_j}E|\psi_{\mathrm{L}}\rangle.$$

The pair s_1s_2 depends on the error, not on α, β. The syndrome table is

E	I	X_1	X_2	X_3
(s_1, s_2)	00	10	11	01

For instance X_2 anticommutes with both checks, so the measurement outputs are $-1, -1$ and the bit syndrome is 11. Recovery by X_2 restores the encoded state since $X_2^2 = I$. The code protects against a single X fault but cannot distinguish all single-qubit Pauli errors: for example Z_1 commutes with both checks and acts as a logical phase. Therefore its distance with respect to unrestricted Pauli errors is one, even though it detects one error in the restricted bit-flip model.

5.8.3 Measurement circuits and the limits of naive syndrome extraction

To measure Z_1Z_2, introduce an ancilla in $|0\rangle$, apply CNOT from data qubit 1 to ancilla and from data qubit 2 to ancilla, then measure ancilla in the computational basis. On a basis state $|x_1x_2\rangle$, the ancilla becomes $|x_1 \oplus x_2\rangle$. The measured bit is zero exactly when Z_1Z_2 has eigenvalue $+1$, and one exactly when the eigenvalue is -1. By linearity, the circuit performs a nondestructive parity measurement on superpositions within the eigenspaces, although it may project across eigenspaces.

A circuit that correctly measures a stabilizer in the absence of noise is not automatically fault tolerant. An X fault on a target ancilla may propagate differently from a Z fault; through conjugation by CNOT, Z on the target propagates backward to Z on both control and target. Multiple contacts with a common ancilla can therefore correlate data faults. Verified ancillae, flag circuits, cat-state methods, or other specially designed extraction schemes are necessary when a single fault must not create an uncorrectable correlated error.

5.8.4 The phase-flip code by Hadamard conjugation

Conjugating the bit-flip code by $H^{\otimes 3}$ gives $|0_L\rangle = |+++\rangle$ and $|1_L\rangle = |---\rangle$. Because $HXH = Z$ and $HZH = X$, the stabilizer checks become X_1X_2 and X_2X_3. A physical phase fault Z_j behaves, in the rotated basis, exactly like X_j in the original bit-flip code, producing the sequence $10, 11, 01$ for $j = 1, 2, 3$. The recovery applies the matching Z_j. This basis transformation is not merely a mnemonic; it is a unitary equivalence that carries stabilizers, logical operators, error sets, and measurement projectors into the new representation.

Example 5.7 (A coherent single-qubit fault)**.** Suppose the first physical qubit experiences a small rotation $E = e^{-i\theta X_1/2} = \cos(\theta/2)I - i\sin(\theta/2)X_1$. The corrupted codeword is a coherent superposition of the zero-syndrome and syndrome-10 subspaces. Stabilizer measurement returns syndrome 00 with probability $\cos^2(\theta/2)$ and syndrome 10 with probability $\sin^2(\theta/2)$. Conditional correction leaves the original logical state in either branch, up to a branch-dependent global phase. This calculation explains why discrete Pauli correction can correct a continuous family of coherent errors in the span of a correctable Pauli set.

5.8.5 Shor's nine-qubit code: bit and phase checks

Write $|G_\pm\rangle = (|000\rangle \pm |111\rangle)/\sqrt{2}$. The codewords are

$$|0_L\rangle = |G_+\rangle^{\otimes 3}, \qquad |1_L\rangle = |G_-\rangle^{\otimes 3}.$$

The six intra-block parity checks are Z_1Z_2, Z_2Z_3 and their translates to blocks $4, 5, 6$ and $7, 8, 9$. The two inter-block checks are $X_1X_2X_3X_4X_5X_6$ and $X_4X_5X_6X_7X_8X_9$. The first six checks localize bit flips within each block. The latter two localize phase changes to one of the three blocks; within a GHZ block the operators Z_1, Z_2, Z_3 act equivalently on the encoded subspace, since their pairwise products are stabilizers. This is stabilizer-code degeneracy: distinct physical errors can share a syndrome without creating ambiguity for the logical action.

Example 5.8 (A Y fault in the middle block)**.** Take $E = Y_5 = iX_5Z_5$ up to the usual Pauli phase convention. Because of the X_5 component, E anticommutes with both intra-block checks Z_4Z_5 and Z_5Z_6. Because of its Z_5 component, it anticommutes with both inter-block X checks, each containing X_5. Every other check commutes. Thus the syndrome specifies position 5 for the bit fault and the middle block for the phase fault; applying Y_5 corrects both components simultaneously.

The minimum weight of a nontrivial logical Pauli is three, so this is an $[[9, 1, 3]]$ code and any one-qubit Pauli error is correctable. The notation $[[n, k, d]]$ states that n physical qubits encode k logical qubits and that the minimum weight of a logical Pauli in the normalizer but outside the stabilizer is d. The quantity $t = \lfloor (d-1)/2 \rfloor$ is a sufficient correction radius for arbitrary unknown Pauli errors under ideal error extraction.

5.8.6 Pauli operator bases and the Knill–Laflamme criterion

Because I, X, Y, Z form a basis of $\mathbb{C}^{2\times 2}$, every single-qubit linear operator has a unique expansion

$$E = a_0 I + a_x X + a_y Y + a_z Z, \qquad a_P = \tfrac{1}{2}\operatorname{Tr}(P^\dagger E).$$

For example the amplitude damping Kraus operator $E_1 = \sqrt{\gamma}|0\rangle\langle 1|$ is $\sqrt{\gamma}(X + iY)/2$. It is not itself a Pauli operation. Nonetheless, if the code corrects every pair from the Pauli basis, linearity of the recovery and the algebraic correction conditions extends correction to their span. This does not imply that a code correcting all single-qubit faults also corrects arbitrary two-qubit interactions.

For a projector P onto the code, the exact error-correction condition is

$$PE_a^\dagger E_b P = c_{ab} P,$$

where $C = (c_{ab})$ is a positive-semidefinite matrix. Necessity follows because distinct logical input vectors must induce distinguishable, information-preserving recovery branches, and the environment must not learn which logical codeword was sent. Sufficiency can be seen by diagonalizing C: a unitary change of Kraus basis yields new errors F_a satisfying $PF_a^\dagger F_b P = \lambda_a \delta_{ab} P$. The nonzero error images $F_a C$ are orthogonal, and a recovery can identify the error subspace and invert its isometric action on the code. This argument also clarifies why degeneracy is allowed: C need not be diagonal in the original physical-error basis.

5.8.7 Hamming parity-check decoding by explicit finite-field algebra

Work over $\mathbb{F}_2$. A systematic $(7, 4)$ Hamming code with message $m = (a, b, c, d)$ may be constructed using parity bits $x = a \oplus b \oplus d$, $y = a \oplus c \oplus d$, and $z = b \oplus c \oplus d$. Then the codeword is $c = (a, b, c, d, x, y, z)$. Its parity-check matrix is

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

By construction $Hc^T = 0$. If a single bit at coordinate j flips, the received vector is $r = c + e_j$ and $Hr^T = He_j^T$, which is the j-th column of H. These columns are precisely the seven nonzero binary three-vectors, so the syndrome identifies the error position uniquely. The number of codewords is $2^4 = 16$. Each codeword has one radius-one Hamming ball of size $1 + \binom{7}{1} = 8$. Because the minimum distance is 3, these balls are disjoint, and their total size $16 \cdot 8 = 128 = 2^7$ covers the ambient space. This is the perfect-code property.

Example 5.9 (Encoding and correcting a data error)**.** For $m = (1, 1, 0, 1)$, the parity equations produce $x = 1$, $y = 0$, $z = 0$, so the codeword is 1101100. If the second coordinate flips, the received word is 1001100. Direct multiplication gives $Hr^T = 101$,

matching the second column of H; changing that bit again restores the codeword. The string 101 identifies the second column in this specified convention; it is not a universal formula converting a binary number directly into the one-based physical-qubit index.

5.8.8 The CSS construction underlying the Steane code

Let $C = \ker H \subseteq \mathbb{F}_2^7$ and $D = C^\perp = \operatorname{rowspan} H$. Since $HH^T = 0$, one has $D \subseteq C$. The dimensions are $\dim C = 4$ and $\dim D = 3$, giving $\dim(C/D) = 1$. Set $v = 1111111 \in C \setminus D$. The encoded states are the coset superpositions

$$|0_L\rangle = 2^{-3/2} \sum_{u \in D} |u\rangle, \qquad |1_L\rangle = 2^{-3/2} \sum_{u \in D} |u + v\rangle.$$

They are normalized because each coset has exactly eight elements, and orthogonal because D and $v + D$ are disjoint. For any $h \in D$, $X(h)$ permutes terms of each coset, so $X(h)|j_L\rangle = |j_L\rangle$. Furthermore, any $h \in D$ is orthogonal to C, and every $u, u + v$ lies in C, so $Z(h)$ multiplies every summand by $+1$. These are the six independent X- and Z-type stabilizer generators.

The full stabilizer has 2^6 elements, and the joint $+1$ eigenspace has dimension $2^7/2^6 = 2$, as required. The logical operators $\overline{X} = X^{\otimes 7}$ and $\overline{Z} = Z^{\otimes 7}$ commute with all stabilizers but anticommute with one another, because their supports overlap in seven (an odd number of) positions. They exchange or phase the logical basis states in the expected way.

5.8.9 Two independent three-bit syndromes in the Steane code

For a Pauli $E = X_j$, its effect on a Z-type check $Z(h)$ is anticommuting exactly when $h_j = 1$; all X-type checks commute. Therefore $(s_Z, s_X) = (H_{\cdot j}, 000)$. For $E = Z_j$ the two components swap, and for $E = Y_j$ both components equal $H_{\cdot j}$. For position 4 the fourth column is 111, giving

E	X_4	Z_4	Y_4
s_Z	111	000	111
s_X	000	111	111

.

A pair of three-bit syndromes therefore jointly determines both type and location for any single Pauli fault. The zero syndrome can still be produced by a nontrivial logical operator such as $\overline{Z}$; stabilizer checks detect deviations from the code space but do not reveal encoded information directly.

5.8.10 Transversal Clifford gates and propagation of faults

CNOT conjugates the single-qubit Paulis according to

$$X_c \mapsto X_c X_t, \quad Z_c \mapsto Z_c,$$
$$X_t \mapsto X_t, \qquad Z_t \mapsto Z_c Z_t.$$

These rules can be established by checking the action of CNOT on computational basis vectors. For instance, $\mathrm{CNOT}(X_c \otimes I)\,\mathrm{CNOT}\,|a,b\rangle = |a \oplus 1, b \oplus 1\rangle$, which is $X_c X_t |a,b\rangle$. A single uncontrolled CNOT spanning two physical qubits *within the same block* may spread one physical fault to two errors in that block. Applying CNOT transversally between corresponding qubits of two encoded blocks restricts each single fault to at most one affected qubit per block under the idealized location model.

The Steane code supports transversal H and a suitable transversal $S^\dagger$ representative. To verify the latter, note that $(S^\dagger)^{\otimes 7}|u\rangle = (-i)^{\mathrm{wt}(u)}|u\rangle$. Every element of D has weight 0 or 4, giving phase 1; every element of $v + D$ has weight 3 or 7, giving phase i. Thus the action on the logical basis is $\mathrm{diag}(1, i) = S$. The statement depends on the chosen logical basis and phase convention. The transversal Clifford group alone is not computationally universal: non-Clifford operations require an additional mechanism such as injection with a suitable encoded magic state.

5.8.11 Concatenation, scaling laws, and the threshold idealization

A simplified distance-three concatenation model obeys $p_{\ell+1} = c p_\ell^2$, where $c > 0$ summarizes malignant fault-pair locations. Induction gives

$$p_\ell = c^{2^\ell - 1} p_0^{2^\ell} = \frac{(c p_0)^{2^\ell}}{c}.$$

If $c p_0 < 1$, the effective failure probability decreases doubly exponentially as a function of concatenation level. If the algorithm has L logical locations and a target overall failure probability ε, the union bound asks for $L p_\ell \le \varepsilon$. A sufficient condition is

$$2^\ell \ge \frac{\ln(L/(c\varepsilon))}{\ln(1/(c p_0))}.$$

For $c = 12$, $p_0 = 0.02$, $L = 10^6$, and $\varepsilon = 10^{-3}$, the right side is about 12.78, so $\ell = 4$ is enough under this recursion. Four levels of a seven-qubit code require $7^4 = 2401$ physical *data* qubits per logical qubit, before accounting for syndrome measurement, routing, ancillae, and physical gate scheduling. A threshold theorem is a much stronger architectural statement: it requires a noise model, fault-tolerant implementations of all operations, and a bound on correlated malignant fault patterns. The scalar recursion is a useful illustration, not a stand-alone proof of an architectural threshold.

5.8.12 A comparison of fault probability models

With independent single-qubit depolarizing faults of total probability p, a nine-qubit code sees at least two nonidentity errors with probability $1-(1-p)^9-9p(1-p)^8 = 36p^2+O(p^3)$. This quantity is an upper bound on the failure probability of ideal single-error correction, because some weight-two errors may be equivalent to harmless stabilizers or may share correctable syndromes. It is not an equality without specifying the decoder. When physical faults exhibit temporal or spatial correlations, even the binomial formula is generally invalid; replacing p by a measured marginal rate does not restore independence.

5.9 Explicit binary codebooks and stabilizer consistency tests

5.9.1 The sixteen Hamming codewords in systematic form

For the systematic Hamming generator matrix used above, every message $m = (a,b,c,d) \in \mathbb{F}_2^4$ maps to a unique seven-bit word $mG = (a,b,c,d,a+b+d,a+c+d,b+c+d)$, with sums taken modulo two. The complete codebook can be arranged as follows:

message	codeword	message	codeword
0000	0000000	1000	1000110
0001	0001111	1001	1001001
0010	0010011	1010	1010101
0011	0011100	1011	1011010
0100	0100101	1100	1100011
0101	0101010	1101	1101100
0110	0110110	1110	1110000
0111	0111001	1111	1111111

Every listed word satisfies $Hc^T = 0$. Since G has rank four, these are all 2^4 codewords, and no duplicates occur. The nonzero minimum Hamming weight is three, establishing minimum distance three: for a linear code, the minimum distance between distinct codewords equals the minimum weight of a nonzero codeword. The rows with nonzero weight three, four, or seven are useful for checking both decoding and the Steane logical-coset decomposition.

The dual code $D = \operatorname{rowspan} H$ consists of eight codewords and is a linear subspace of C. In the coordinate convention above,

$$D = \{0000000, 1101100, 1011010, 0110110,\\ 0111001, 1010101, 1100011, 0001111\}.$$

Its nonzero elements all have weight four. Translating by $v = 1111111$ yields the

complementary coset

$$v + D = \{1111111, 0010011, 0100101, 1001001,\\ 1000110, 0101010, 0011100, 1110000\}.$$

The coset elements have weight three or seven. This explicitly displays the two logical Steane basis states as uniform sums over disjoint sets of eight computational basis vectors. Orthogonality is then immediate because distinct computational basis states are orthogonal.

5.9.2 The stabilizer syndrome as a symplectic inner product

Represent an n-qubit Pauli operator, up to global phase, by a binary pair $(x, z) \in \mathbb{F}_2^{2n}$, with X support x and Z support z. Two Paulis (x, z) and (x', z') commute exactly when

$$\langle (x, z), (x', z') \rangle_{\text{sp}} := x \cdot z' + z \cdot x' = 0 \pmod 2.$$

They anticommute when this symplectic inner product equals one. To verify the formula, move X factors past Z factors one qubit at a time; every overlapping XZ pair introduces a minus sign, and the total sign is $(-1)^{x \cdot z' + z \cdot x'}$. The stabilizer syndrome is therefore a matrix of binary symplectic inner products between the error vector and the stabilizer generators.

For the Steane code, the Z generators have support rows of H and no X support; the X generators have the same row supports and no Z support. Therefore the full syndrome of an error (x, z) is simply

$$\begin{pmatrix} s_Z \\ s_X \end{pmatrix} = \begin{pmatrix} Hx^T \\ Hz^T \end{pmatrix}.$$

This formula handles arbitrary multi-qubit Pauli patterns algebraically. For a single Y_j, both $x = e_j$ and $z = e_j$, so both syndrome halves equal the j-th column of H. For errors involving multiple positions, the syndrome is the XOR of the corresponding columns, which can cause different error patterns to share the same syndrome. Distance and decoder assumptions determine whether the ambiguity is correctable.

Example 5.10 (A two-error ambiguity)**.** Consider bit flips at Steane-code positions 1 and 2. Their Z-check syndrome is $H_{\cdot 1} + H_{\cdot 2} = 110 \oplus 101 = 011$, equal to the single-error syndrome of position 3. A decoder designed to correct only one error would apply X_3, leaving the residual error $X_1 X_2 X_3$. This residual operator has zero syndrome and is a nontrivial logical operation in the code, since $X_1 X_2 X_3$ has weight three and is not a stabilizer of weight four. The example demonstrates exactly why a distance-three code cannot guarantee correction of every weight-two error, even though every measured syndrome is well defined.

5.9.3 Logical operator verification by coset permutations

Let $\overline{X} = X^{\otimes 7}$. Its action on a computational basis vector is $\overline{X}|u\rangle = |u + v\rangle$, with $v = 1111111$. Therefore

$$\overline{X}|0_{\mathrm{L}}\rangle = \frac{1}{\sqrt{8}} \sum_{u \in D} |u + v\rangle = |1_{\mathrm{L}}\rangle, \quad \overline{X}|1_{\mathrm{L}}\rangle = |0_{\mathrm{L}}\rangle.$$

For $\overline{Z} = Z^{\otimes 7}$, the action is $\overline{Z}|u\rangle = (-1)^{\mathrm{wt}(u)}|u\rangle$. Every word in D has even weight, whereas every word in $v + D$ has odd weight; hence $\overline{Z}|0_{\mathrm{L}}\rangle = |0_{\mathrm{L}}\rangle$ and $\overline{Z}|1_{\mathrm{L}}\rangle = -|1_{\mathrm{L}}\rangle$. Finally, because $\overline{X}$ and $\overline{Z}$ overlap on an odd number of qubits, $\overline{X}\overline{Z} = -\overline{Z}\overline{X}$.

Logical operators are not unique physical matrices: multiplying a logical Pauli by any stabilizer produces a different physical Pauli with the same action on the codespace. For example, multiplying $\overline{X}$ by a weight-four X stabilizer can produce a weight-three representative. The minimum such weight among nontrivial logical representatives determines code distance, not the weight of an arbitrarily selected representative such as $X^{\otimes 7}$.

5.9.4 A Pauli-channel example with explicit subspace probabilities

Consider a single-qubit Pauli channel $\mathcal{E}(\rho) = (1 - p)\rho + (p/3)(X\rho X + Y\rho Y + Z\rho Z)$ applied independently on each of seven physical qubits. The probability of no nonidentity faults is $(1 - p)^7$ and of exactly one fault is $7p(1 - p)^6$. An ideal decoder guaranteed to correct every single-qubit fault thus has logical failure probability bounded above by

$$1 - (1 - p)^7 - 7p(1 - p)^6 = 21p^2 + O(p^3).$$

The coefficient 21 counts unordered pairs of faulty sites and is not an equality for the logical failure rate: some weight-two errors may also be corrected or may be equivalent to stabilizers depending on the code and decoder. Likewise, a coherent fault expansion cannot always be assigned a classical probability for each Pauli term before a syndrome measurement; probabilities arise from the relevant quantum instrument and orthogonal syndrome branches.

5.9.5 Fault propagation and why the ancilla arrangement matters

Take two CNOT gates that use a single ancilla as target, with controls data qubits 1 and 2. A Z error on the ancilla just before the second CNOT propagates backward under CNOT conjugation as $Z_2 Z_a$; if it occurred before both gates and is propagated through both, it can become $Z_1 Z_2 Z_a$. Thus a single ancillary fault can correlate multiple data errors in a backward-propagation analysis. The precise physical propagation direction depends on the location of the fault relative to the gates, but the algebraic possibility motivates verified and fault-tolerant syndrome-extraction circuits. A circuit that is correct in the noiseless model need not preserve the correctable-error set under single component failures.

5.10 Recovery theory, logical faults, and syndrome consistency

5.10.1 The Knill–Laflamme condition with a constructive recovery

Let $C \subseteq \mathcal{H}$ be a code space with orthogonal projector P, and suppose a noisy operation has Kraus operators $\{E_i\}_{i=1}^m$. The condition for an exact recovery that corrects the whole linear span of these errors is

$$PE_i^\dagger E_j P = c_{ij} P, \qquad 1 \le i, j \le m, \tag{5.1}$$

for a Hermitian positive-semidefinite matrix $C = (c_{ij})$. To see positivity without appealing to a general theorem, for any $z \in \mathbb{C}^m$ put $E(z) = \sum_j z_j E_j$; then

$$(z^\dagger C z) P = PE(z)^\dagger E(z) P \ge 0.$$

Consequently $C = V \operatorname{diag}(\lambda_1, \ldots, \lambda_m) V^\dagger$, with $\lambda_i \ge 0$. Define rotated errors $F_i = \sum_j E_j V_{ji}$. They satisfy

$$PF_i^\dagger F_j P = \lambda_i \delta_{ij} P.$$

For every positive λ_i, the map $W_i = F_i P / \sqrt{\lambda_i}$ is an isometry on the logical subspace, and

$$W_i^\dagger W_j = \delta_{ij} P, \qquad (W_i W_i^\dagger)(W_j W_j^\dagger) = 0 \quad (i \ne j).$$

Thus a recovery can project onto mutually orthogonal subspaces $\operatorname{im} W_i$, determine the syndrome index without measuring the logical amplitudes, and then apply $W_i^\dagger$ to return the state to C. For any input density operator $\rho = P\rho P$, linearity gives an output proportional to ρ on each nonzero syndrome branch. Summing the branches restores ρ because the original channel is trace preserving on the code. This proves the constructive sufficiency of (5.1); necessity follows because a reversible noise channel cannot imprint distinguishable logical input information on its inaccessible error environment.

5.10.2 Why correcting Pauli errors also corrects coherent errors

For one physical qubit, every complex 2×2 operator can be expanded as

$$E = a_I I + a_X X + a_Y Y + a_Z Z, \qquad a_P = \frac{1}{2} \operatorname{tr}(PE).$$

The four Pauli matrices form an orthogonal basis under the Hilbert–Schmidt inner product. If a code obeys the Knill–Laflamme condition for all the basis errors I, X_j, Y_j, Z_j acting on any fixed one-qubit location, then for general linear combinations $F = \sum_i a_i E_i$ and $G = \sum_j b_j E_j$,

$$PF^\dagger G P = \sum_{i,j} \overline{a_i} b_j c_{ij} P.$$

Therefore the code also corrects their coherent combinations, including small rotations such as $e^{-i\vartheta X_j/2} = \cos(\vartheta/2)I - i\sin(\vartheta/2)X_j$. This is the mathematical reason discrete Pauli syndrome extraction suffices to deal with a continuum of single-qubit operator errors, under the assumed locality model. It does not imply that an arbitrary correlated two-qubit error is correctable by a distance-three code.

5.10.3 A syndrome table that reveals a code's blind spot

For the three-qubit repetition code, use $S_1 = Z_1Z_2$ and $S_2 = Z_2Z_3$. Each Pauli error produces a two-bit outcome $s_i(E) = 0$ if $[E, S_i] = 0$ and $s_i(E) = 1$ if $\{E, S_i\} = 0$. The complete set of single-site Pauli syndromes is

E	I	X_1	X_2	X_3	Z_1	Z_2	Z_3	Y_1	Y_2	Y_3
s_1s_2	00	10	11	01	00	00	00	10	11	01

The code can locate an X_j error but cannot distinguish a Z_j error from no error. Indeed $Z_j(\alpha|000\rangle + \beta|111\rangle) = \alpha|000\rangle - \beta|111\rangle$, a nontrivial logical phase transformation with syndrome 00. If $Y_2 = iX_2Z_2$ occurs, the measured syndrome is 11. An X_2 recovery removes the bit flip but leaves, up to global phase, Z_2 acting as logical Z. The same syndrome therefore does not establish full arbitrary-Pauli correctability. The phase-flip repetition code exchanges X and Z by conjugating every physical qubit with H.

5.10.4 The stabilizer normalizer and the meaning of distance

An $[[n, k, d]]$ stabilizer code has an abelian subgroup $\mathcal{S}$ of the n-qubit Pauli group, containing no $-I$, with $n - k$ independent generators. The code is their simultaneous $+1$ eigenspace. A Pauli E that anticommutes with some stabilizer is detectable through a nonzero syndrome. A Pauli in the normalizer $N(\mathcal{S})$ commutes with every stabilizer and preserves the code space. The elements of $\mathcal{S}$ themselves act trivially on encoded information; elements of $N(\mathcal{S}) \setminus \mathcal{S}$ can act as nontrivial logical Paulis. Thus the distance is

$$d = \min\{\mathrm{wt}(E) : E \in N(\mathcal{S}) \setminus \mathcal{S}\}.$$

If the error set contains all Paulis of weight at most t, then products $E_i^\dagger E_j$ have weight at most $2t$. The sufficient condition $2t < d$ ensures every such product is either a harmless stabilizer or has a detectable syndrome. Consequently any distance-d code corrects arbitrary errors on at most $\lfloor (d-1)/2 \rfloor$ sites; code degeneracy may allow additional specially structured errors beyond this guaranteed radius.

5.10.5 Binary symplectic representation with a complete example

Ignoring overall phase, write a Pauli as X^aZ^b with $(a, b) \in \mathbb{F}_2^{2n}$. The commutation sign between X^aZ^b and $X^{a'}Z^{b'}$ is

$$(-1)^{a\cdot b' + b\cdot a'}.$$

This is the binary symplectic pairing $\langle (a,b),(a',b')\rangle_s = a\cdot b' + b\cdot a'$. For the three-qubit repetition stabilizers Z_1Z_2 and Z_2Z_3, the binary check vectors are (000 | 110) and (000 | 011). The error Y_2 corresponds, up to phase, to (010 | 010). Its symplectic pairings are $010\cdot 110 = 1$ and $010\cdot 011 = 1$; hence $s(Y_2) = 11$. The binary calculation agrees with the operator anticommutation table and scales naturally to CSS code parity-check matrices.

5.10.6 Second-order suppression and the boundary of a threshold argument

If n sites suffer independent errors of rate p and an ideal distance-three recovery corrects all single errors, the probability of at least two faulty sites is

$$\Pr\{W \geq 2\} = 1 - (1-p)^n - np(1-p)^{n-1} = \binom{n}{2}p^2 + O(p^3).$$

This gives a conservative upper bound on failure from physical data errors when every single-site fault is corrected perfectly. Actual logical failure may be strictly smaller because some multiple faults are harmless or still correctable. Fault-tolerant circuits require a different accounting: noisy syndrome extraction, gate propagation, correlated faults, and noisy classical-control interfaces must be included. If a gadget admits a bound $p_{\ell+1} \leq Cp_\ell^2$ with $p_0 = p$, then induction yields

$$p_\ell \leq \frac{1}{C}(Cp)^{2^\ell}.$$

The expression decreases doubly exponentially in the concatenation level when $Cp < 1$. The coefficient C depends on the precise gadget and fault model; the recurrence alone does not prove a universal numerical threshold for an unspecified architecture.

CHAPTER 6

Graph States and Measurement-Based Quantum Computation

6.1 Graph-state resources

Definition 6.1 (Graph state and stabilizer generators)**.** Let $G = (V, E)$ be a finite undirected simple graph. Its graph state is

$$|G\rangle = \left(\prod_{\{u,v\} \in E} CZ_{uv} \right) |+\rangle^{\otimes |V|}.$$

For a vertex v with neighborhood $N(v)$, define

$$K_v = X_v \prod_{u \in N(v)} Z_u.$$

Every K_v stabilizes $|G\rangle$, and the K_v form an independent commuting generating set of the graph-state stabilizer group.

The proof is direct: $|+\rangle^{\otimes |V|}$ is stabilized by each X_v, and conjugating X_v through CZ on every adjacent edge introduces exactly one Z on each neighbor, since $CZ_{uv} X_u CZ_{uv} = X_u Z_v$.

A two-vertex graph is

$$|G_2\rangle = CZ|++\rangle = \tfrac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle),$$

which is locally unitary equivalent to a Bell pair:

$$(I \otimes H)|G_2\rangle = |\Phi^+\rangle.$$

One-dimensional paths support logical wires, while suitably connected two-dimensional cluster graphs permit entangling operations between wires. Three-dimensional or otherwise structured resource states can additionally support topological fault-tolerant schemes, subject to the required noise assumptions and decoding.

For the path $1 - 2 - 3$, the generators are $K_1 = X_1 Z_2$, $K_2 = Z_1 X_2 Z_3$, and $K_3 = Z_2 X_3$. A Z_2 fault anticommutes only with K_2, giving syndrome $(0, 1, 0)$; X_2 anticommutes with K_1, K_3 and gives $(1, 0, 1)$. On a four-vertex star with center 2 and leaves $1, 3, 4$, the generators are $K_2 = Z_1 X_2 Z_3 Z_4$ and $K_j = X_j Z_2$ for each leaf j; this is a convenient local test for arbitrary Pauli-error syndromes.

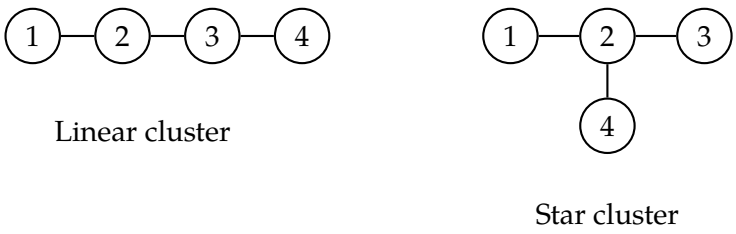


Figure 6.1: Two graph-state geometries. Edges denote controlled-Z entangling operations.

6.2 Graph reduction through local measurements

Proposition 6.2 (Computational-basis vertex deletion)**.** *For a vertex v, write $G - v$ for the graph with v and its incident edges removed. Then*

$$|G\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_v|G-v\rangle + |1\rangle_v \prod_{u\in N(v)} Z_u|G-v\rangle\right).$$

A Z_v measurement yielding eigenvalue $(-1)^s$ leaves the normalized state

$$\prod_{u\in N(v)} Z_u^s|G-v\rangle, \qquad s\in\{0,1\}.$$

The vertex and all adjacent edges disappear, but a -1 outcome creates Z byproducts on neighbors. X measurements can have a qualitatively different effect: measuring the center of a path $1-2-3$ in the X basis leaves $|\Phi^+\rangle_{13}$ for outcome $+$ and $|\Psi^+\rangle_{13}$ for outcome $-$ (under the graph-state phase conventions). Thus an intermediate measured qubit can mediate entanglement between distant endpoints.

6.3 One-bit teleportation and the identity wire

Attach an input $|\psi\rangle = a|0\rangle + b|1\rangle$ to an ancilla $|+\rangle$ through CZ. The state becomes $a|0\rangle|+\rangle + b|1\rangle|-\rangle$. Measuring the first qubit in the X basis with outcome eigenvalue $(-1)^s$ yields

$$|\psi_{\text{out}}\rangle = X^s H|\psi\rangle$$

after normalization. Each outcome has probability $1/2$ independent of $|\psi\rangle$. This primitive is a **half-teleportation** step: the logical information moves one graph edge and acquires a known unitary and a random Pauli byproduct.

For a three-vertex chain with two consecutive X measurements, the final qubit contains

$$X^{s_2}HX^{s_1}H|\psi\rangle = X^{s_2}Z^{s_1}|\psi\rangle,$$

with outcomes $(s_1, s_2) = 00, 01, 10, 11$ corresponding respectively to byproducts I, X, Z, XZ, up to a physically irrelevant phase. The identity wire is therefore the identity **modulo** a classically trackable Pauli frame.

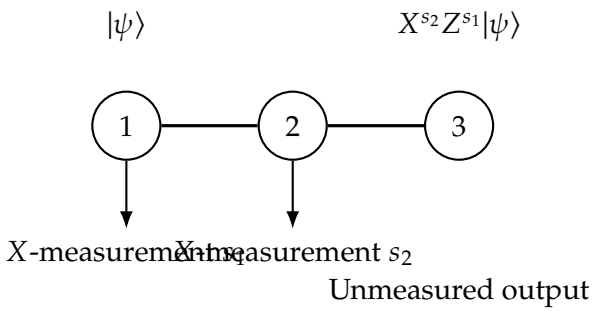


Figure 6.2: A three-site measurement-based identity wire; the output includes conditional Pauli byproducts.

6.4 Rotated measurements and single-qubit gates

For this chapter define

$$U_z(\alpha) = e^{i\alpha Z} = R_Z(-2\alpha), \qquad U_x(\beta) = e^{i\beta X} = HU_z(\beta)H.$$

These are **not** the same angular convention as $R_P(\theta) = e^{-i\theta P/2}$. Define the measurement eigenvectors

$$|\phi_s(\alpha)\rangle = \frac{|0\rangle + (-1)^s e^{2i\alpha}|1\rangle}{\sqrt{2}}, \qquad s \in \{0,1\}.$$

Projection of the entangled pair onto $\langle\phi_s(\alpha)|$ introduces the complex conjugate phase $e^{-2i\alpha}$. After normalization the output is

$$|\psi_{\text{out}}\rangle = X^s HU_z(\alpha)|\psi\rangle,$$

up to overall phase. Equivalently, write $J(\alpha) = HU_z(\alpha)$ and regard each measured hop as $X^s J(\alpha)$. Each measurement outcome is still unbiased.

For sequential idealized hops, the chronological operations multiply from **right to left**:

$$J(\gamma)J(\beta)J(\alpha) = HU_z(\gamma)U_x(\beta)U_z(\alpha).$$

The inner three rotations form a Z–X–Z Euler decomposition; a fixed leading H remains. Producing precisely $U_z(\gamma)U_x(\beta)U_z(\alpha)$ requires accounting for that deterministic H through an additional basis change or an adjoining pattern. Correcting random Paulis alone does not remove it.

6.5 Pauli frames, adaptive angles, and causal order

Track the current physical state as $X^a Z^b|\psi_{\text{logical}}\rangle$, where $a, b \in \mathbb{F}_2$. The Pauli commutation relations imply, modulo global phase,

$$J(\alpha')X^a Z^b = X^b Z^a J((-1)^a \alpha').$$

To implement logical $J(\alpha)$, select the physical angle

$$\alpha_{\text{phys}} = (-1)^a \alpha.$$

After obtaining measurement bit s, update the frame by

$$\boxed{(a', b') = (s \oplus b,\ a).}$$

For three successive measurements, beginning at $(0, 0)$, the frame evolves as

$$(0,0) \longrightarrow (s_1, 0) \longrightarrow (s_2, s_1) \longrightarrow (s_3 \oplus s_1, s_2).$$

Thus the third logical rotation angle γ requires physical angle $(-1)^{s_2}\gamma$. The measured outcomes are inherently random but **heralded**, so a classical controller can feed them forward. Fixed Pauli-basis measurements often require no change of physical basis when the sign flips, although the outcome labels and final Pauli frame still need tracking. Non-Pauli angles impose partial temporal ordering on measurements.

6.6 Two-qubit gates and network composition

The identity

$$\text{CNOT} = (I \otimes H)CZ(I \otimes H)$$

explains how controlled-Z links, logical Hadamards, and local measurements can assemble entangling CNOT operations. A graph edge is not automatically a CNOT on arbitrary preexisting logical inputs: a complete pattern must specify input and output vertices, measurements, byproducts, and dependencies. Computational-basis measurements can excise unused branches; subsequent patterns compose by propagating frames through their Clifford or non-Clifford operations.

6.7 Probabilistic entangling operations and cluster growth

Suppose a heralded entangling attempt succeeds with probability p. Requiring m independent successes without repair has probability p^m and naive restart expectation p^{-m}. In a particular local cluster-growth model, success adds one site and failure removes two:

$$\Delta L = \begin{cases} +1, & \text{with probability } p, \\ -2, & \text{with probability } 1 - p. \end{cases}$$

Therefore $\mathbb{E}[\Delta L] = 3p - 2$, so mean growth is positive precisely when $p > 2/3$. At $p = 0.8$ the average gain is 0.4 sites per attempt; 100 attempts give expected gain 40, but not a certainty of gain 40. With success gain $g > 0$ and failure loss $\ell > 0$, the correct threshold is $p > \ell/(g + \ell)$; $2/3$ is not a hardware-independent fundamental threshold.

6.8 Worked feedforward and stabilizer calculations

Example 6.3 (Center faults on a star graph)**.** For a star with center 2 and neighboring vertices $1, 3, 4$, order stabilizers as (K_1, K_2, K_3, K_4). The center fault X_2 anticommutes with the Z_2 factors in all three leaf checks, giving syndrome 1011. The fault Z_2 anticommutes only with the center generator K_2, giving 0100. Since $Y_2 = iX_2Z_2$, its syndrome is the bitwise XOR $1011 \oplus 0100 = 1111$. These are graph-state stabilizer signs; the graph is not thereby claimed to be an error-correcting code.

Example 6.4 (Two nonadaptive rotated teleportations)**.** Take $\alpha = \pi/4$, $\beta = -\pi/6$, and measurement outcomes $(s_1, s_2) = (1, 0)$. Without adapting the second angle, the chronological output is

$$J_{s_2}(\beta)J_{s_1}(\alpha)|\psi\rangle = HU_z(-\pi/6)XHU_z(\pi/4)|\psi\rangle = U_x(-\pi/6)ZU_z(\pi/4)|\psi\rangle,$$

where $J_s(\alpha) = X^s HU_z(\alpha)$. A Z correction cannot in general be commuted through $U_x(\beta)$ without changing the sign of the corresponding rotation angle.

Example 6.5 (Adapted triple-hop evolution)**.** Suppose the intended ideal sequence is $J(\gamma)J(\beta)J(\alpha)$ and $(s_1, s_2, s_3) = (1, 0, 1)$. The physical angles are $\alpha, -\beta, \gamma$. The Pauli frame evolves

$$(0, 0) \to (1, 0) \to (0, 1) \to (0, 0).$$

Consequently the random outcomes cancel in the final frame, leaving precisely the ideal triple-J transformation. If one chooses to describe only its Euler rotation part, one must still account for the deterministic outer Hadamard.

6.9 Graph-state calculus and executable measurement patterns

6.9.1 Graph states from commuting controlled-phase edges

Let $G = (V, E)$ be a simple undirected graph. The controlled-Z gate acts diagonally by $CZ|ab\rangle = (-1)^{ab}|ab\rangle$, so all edge gates commute, even when edges share a vertex. Starting from $|+\rangle^{\otimes|V|}$, the corresponding graph state admits the explicit expansion

$$|G\rangle = 2^{-|V|/2} \sum_{x\in\{0,1\}^V} (-1)^{\sum_{\{u,v\}\in E} x_u x_v} |x\rangle.$$

This formula is often more useful than a circuit picture: it gives the amplitude of any basis string immediately. For a two-vertex edge,

$$|G_2\rangle = \tfrac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle).$$

Applying $I \otimes H$ maps this state to $(|00\rangle + |11\rangle)/\sqrt{2}$. Thus a two-vertex graph state is entangled despite each computational-basis amplitude having equal magnitude.

Theorem 6.6 (Graph-state stabilizer theorem)**.** *For each $v \in V$, define $K_v = X_v \prod_{u \in N(v)} Z_u$. Then $K_v|G\rangle = |G\rangle$; the $|V|$ generators commute and are independent.*

Proof. Before entangling, $X_v|+\rangle^{\otimes|V|} = |+\rangle^{\otimes|V|}$. The identity $CZ_{uv}X_vCZ_{uv} = Z_uX_v$ shows that conjugating by the product of all edge gates replaces X_v with K_v. Since conjugation preserves products and commutators, the K_v commute. Independence follows because a product over a nonempty subset of generators contains X or Y on the selected vertices and cannot equal the identity; equivalently the X support vector of the product is the subset's indicator vector. Hence the common +1 eigenspace is one-dimensional and contains $|G\rangle$. □

6.9.2 An explicit four-vertex star and its Pauli signs

Consider the star with edges $\{1,2\},\{2,3\},\{2,4\}$, with vertex 2 central. Its stabilizers are

$$K_1 = X_1Z_2, \qquad K_2 = Z_1X_2Z_3Z_4, \qquad K_3 = Z_2X_3, \qquad K_4 = Z_2X_4.$$

An X_2 error anticommutes with K_1, K_3, K_4 but commutes with K_2, producing syndrome 1011 in generator order $(1,2,3,4)$. A Z_2 error anticommutes only with K_2, producing 0100. A Y_2 error anticommutes with all four and produces 1111, because its sign vector is the bitwise XOR of the signs for X_2 and Z_2. These signs may be used to characterize physical errors on the graph state, but a graph state's unique stabilizer eigenspace is not automatically an encoded logical code. In particular, graph-state fault diagnosis and error-correcting-code decoding are distinct tasks unless a code subspace and recovery map are defined.

6.9.3 The computational-basis measurement deletion rule

Let v be one graph vertex. The graph state may be decomposed by first preparing the rest of the graph and applying the CZ edges adjacent to v:

$$|G\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_v|G-v\rangle + |1\rangle_v \prod_{u\in N(v)} Z_u|G-v\rangle\right).$$

Projecting the measured vertex onto $|s\rangle$ and normalizing leaves $\prod_{u\in N(v)} Z_u^s|G-v\rangle$. The vertex is deleted, but its neighbors acquire a Pauli-Z byproduct when $s = 1$. Both outcomes have probability $1/2$ because the two values of s had equal norm. An often-made mistake is to delete the vertex and omit the conditional phases. Those phases remain relevant for subsequent measurements in noncommuting bases.

Example 6.7 (Pruning a four-site linear graph)**.** For the path $1-2-3-4$, measuring vertex 2 in the computational basis yields a graph with the isolated vertex 1 and the surviving edge $3-4$. If the outcome is $s_2 = 1$, the state acquires Z_1Z_3 on the survivors; if $s_2 = 0$, no Z correction occurs. The original edge $1-2$ and $2-3$ do not become a direct edge $1-3$ merely through a Z measurement. Entanglement swapping is instead associated with suitable X-type measurements and local Clifford corrections.

6.9.4 Entanglement swapping on a three-site path

The graph state of $1-2-3$ is $CZ_{12}CZ_{23}|+\,+\,+\rangle$. It has stabilizers $K_1 = X_1Z_2$, $K_2 = Z_1X_2Z_3$, and $K_3 = Z_2X_3$. Measuring the middle qubit in the X basis with outcome $(-1)^s$ projects onto an eigenspace satisfying $X_2 = (-1)^s$. The remaining endpoint operators obey $Z_1Z_3 = (-1)^s$ by $K_2 = +1$ and $X_1X_3 = +1$ by multiplying $K_1K_3 = +1$. Hence the endpoints are stabilized by

$$X_1X_3 = +1, \qquad Z_1Z_3 = (-1)^s.$$

For $s = 0$ the unique normalized endpoint state is $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$; for $s = 1$ it is $|\Psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}$. The two branches differ by an endpoint X correction. This stabilizer derivation is shorter and less error-prone than expanding all eight amplitudes, while being fully equivalent to that calculation.

6.9.5 One-bit teleportation by direct amplitude projection

Take an arbitrary input $|\psi\rangle = a|0\rangle + b|1\rangle$, an ancilla $|+\rangle$, and apply CZ. Before measurement,

$$|\Psi\rangle = a|0\rangle|+\rangle + b|1\rangle|-\rangle.$$

Projecting the first qubit onto $|+\rangle$ gives the unnormalized output $(a|+\rangle + b|-\rangle)/\sqrt{2} = H|\psi\rangle/\sqrt{2}$. Projecting onto $|-\rangle$ gives $(a|+\rangle - b|-\rangle)/\sqrt{2} = XH|\psi\rangle/\sqrt{2}$. Each branch has norm squared $1/2$. Thus the normalized output is $X^sH|\psi\rangle$. The H arises from teleportation itself, while X^s is a measurement-dependent byproduct. Omitting the deterministic H would incorrectly identify one hop as an identity channel.

Two consecutive hops multiply chronologically as

$$X^{s_2}HX^{s_1}H|\psi\rangle = X^{s_2}Z^{s_1}|\psi\rangle$$

because $HXH = Z$. The four possible outcomes produce the Pauli-frame correction classes I, X, Z, and XZ. Applying the inverse frame restores the logical state, or the frame may be stored classically and propagated through later gates.

6.9.6 Rotated equatorial measurements and the angle convention

Use the convention $U_z(\alpha) = e^{i\alpha Z}$ and $U_x(\beta) = e^{i\beta X}$; these are related to the standard rotations by $U_z(\alpha) = R_Z(-2\alpha)$ and $U_x(\beta) = R_X(-2\beta)$. An equatorial measurement with normalized basis vectors

$$|\phi_s(\alpha)\rangle = \frac{|0\rangle + (-1)^s e^{2i\alpha}|1\rangle}{\sqrt{2}}$$

produces, after a one-bit teleportation hop, the output

$$X^sHU_z(\alpha)|\psi\rangle,$$

up to a global phase. To see the sign, note that the bra contains $e^{-2i\alpha}$; the post-measurement branch is proportional to $a|+\rangle + (-1)^s e^{-2i\alpha} b|-\rangle$. Multiplication of $U_z(\alpha)|\psi\rangle$ gives the relative coefficient $e^{-2i\alpha}$ after discarding the common phase $e^{i\alpha}$. The conjugation of the measured ket is essential: using $e^{+2i\alpha}$ in the bra would reverse the intended rotation.

Define $J(\alpha) = HU_z(\alpha)$. In a noiseless branch with no byproduct, three hops implement

$$J(\gamma)J(\beta)J(\alpha) = HU_z(\gamma)U_x(\beta)U_z(\alpha),$$

where the Z–X–Z expression is an Euler rotation decomposition. The leading H must be included when comparing the pattern to a target unitary; it is not a random measurement error and cannot be removed by correcting only Pauli byproducts.

6.9.7 The Pauli frame as an algebraic automaton

Suppose the actual physical state is $X^a Z^b|\psi_{\text{logical}}\rangle$, with $a, b \in \mathbb{F}_2$. The identities

$$U_z(\theta)X = XU_z(-\theta), \qquad HX = ZH, \qquad HZ = XH$$

imply, modulo an irrelevant scalar phase,

$$J(\theta)X^a Z^b = X^b Z^a J((-1)^a \theta).$$

To realize the logical action $J(\alpha)$, choose the physical angle $\theta = (-1)^a \alpha$. The physical result after measuring bit s is

$$X^s J(\theta)X^a Z^b|\psi_{\text{logical}}\rangle = X^{s\oplus b} Z^a J(\alpha)|\psi_{\text{logical}}\rangle,$$

so the frame evolves by

$$(a, b) \longmapsto (s \oplus b, a).$$

The next measurement angle depends on the current X component but not directly on the Z component. This dependence is a causal constraint: a future measurement cannot be fixed until the earlier outcomes on which it depends are known.

Example 6.8 (Three correlated adaptive choices)**.** Let the desired logical sequence be $J(\gamma)J(\beta)J(\alpha)$ and the measurement outcomes be $(s_1, s_2, s_3) = (1, 0, 1)$. Begin with $(a, b) = (0, 0)$. The first physical angle is α, and the frame becomes $(1, 0)$. The second physical angle is $-\beta$, yielding frame $(0, 1)$. The third physical angle is γ, yielding $(s_3 \oplus b, a) = (0, 0)$. Thus the net operation is exactly the intended triple-J unitary with no remaining Pauli correction in this branch. A nonadaptive choice β at step two would generally implement a different unitary.

Example 6.9 (The same pattern with nontrivial final correction)**.** For $(s_1, s_2, s_3) = (1, 1, 0)$, the frames are $(0, 0) \to (1, 0) \to (1, 1) \to (1, 1)$. The physical angles must be $\alpha, -\beta, -\gamma$, and the output has byproduct XZ relative to the ideal logical state. The correction can be applied physically as ZX (equal up to global phase), or retained in software for downstream Clifford gates. This branch exhibits that adaptive basis choice and final Pauli correction solve different aspects of measurement randomness.

6.9.8 Graph composition, measurement order, and deterministic gates

The essential operations of a measurement pattern can be written in temporal order: prepare ancillary $|+\rangle$ states; apply the resource entangling CZ edges; measure appropriate vertices; update classical signals; perform dependent measurements; and interpret output qubits within the accumulated Pauli frame. Clifford measurements can often be moved in time because they do not require adaptive non-Pauli basis changes. Generic equatorial measurement angles introduce dependencies, often requiring column-by-column or layer-by-layer scheduling. A valid measurement pattern specifies which qubits are input and output, which observables are measured, and which outcome bits control later bases or corrections. A drawing of edges alone does not fully specify the implemented logical circuit.

The identity CNOT $= (I \otimes H)CZ(I \otimes H)$ can be checked on basis states:

$$(I \otimes H)CZ(I \otimes H)|c, t\rangle = |c, t \oplus c\rangle.$$

When $c = 0$, CZ acts trivially and the two target Hadamards cancel; when $c = 1$, the middle CZ acts as target Z and the surrounding H gates convert it to X. This construction explains how entangling graph edges and single-qubit patterns can realize a controlled-NOT, but in MBQC the required H transformations and accompanying byproducts must be incorporated into a complete measurement pattern.

6.9.9 A general randomized cluster-growth calculation

Suppose a heralded fusion attempt succeeds with probability p and increases chain length by g sites; if it fails it removes ℓ existing sites. The increment Δ satisfies

$$\mathbb{E}[\Delta] = pg - (1-p)\ell, \quad \mathrm{Var}(\Delta) = p(1-p)(g+\ell)^2.$$

Positive mean growth occurs precisely when $p > \ell/(g+\ell)$. For the example $g = 1$, $\ell = 2$, this condition is $p > 2/3$. When $p = 0.8$, the expected increment is 0.4 and the variance is $0.8 \cdot 0.2 \cdot 9 = 1.44$ per attempt. For $m = 100$ independent, identically distributed attempts that can be performed without boundary effects, the expected net gain is 40 and the standard deviation is $\sqrt{144} = 12$. These are expectation and variance statements rather than deterministic growth guarantees. Actual photonic cluster protocols must also account for available bonding sites, storage losses, adaptive reattempts, and whether a failed fusion destroys additional entanglement.

6.9.10 A compact graph-state circuit in editable TikZ

The following graph illustrates a path of four vertices with Z-basis deletion at the second vertex. The dashed arrows denote classical conditional byproducts, not entangling edges.

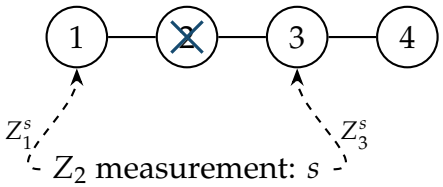


Figure 6.3: Computational-basis deletion of a graph vertex and its conditional neighbor corrections.

6.10 Measurement records, branch probabilities, and circuit identities

6.10.1 One-bit teleportation as a measurement instrument

The measurement calculus can be expressed as a pair of Kraus operators on the logical input. Prepare $|+\rangle$ on a second site, apply CZ, and measure the first site in the rotated basis $\{|\phi_0(\alpha)\rangle, |\phi_1(\alpha)\rangle\}$. The corresponding unnormalized maps are

$$M_s(\alpha) = \frac{1}{\sqrt{2}} X^s H U_z(\alpha), \qquad s \in \{0, 1\},$$

up to outcome-dependent global phase. Because $M_s^\dagger M_s = I/2$, each outcome occurs with probability exactly $1/2$, regardless of the input state. Moreover $\sum_s M_s^\dagger M_s = I$, so the two-branch instrument is trace preserving before conditioning. On each branch the measurement outputs a state related to the input by a known unitary; the random branch label is a classical resource that allows deterministic logical interpretation.

For two successive hops, the four composite Kraus operators are

$$M_{s_2}(\beta) M_{s_1}(\alpha) = \frac{1}{2} X^{s_2} H U_z(\beta) X^{s_1} H U_z(\alpha).$$

Every outcome pair has probability $1/4$ for any pure input when the second basis is a fixed orthonormal equatorial basis, even though the logical unitary represented by the branch can depend on the measurement record. In an adaptive implementation the basis β itself is replaced by a function of s_1, but each conditional branch still has probability $1/2$ for the next step. This provides a normalization check for every measured wire.

6.10.2 Circuit corrections as binary affine recurrences

Let the Pauli frame after the j-th measurement be $(a_j, b_j) \in \mathbb{F}_2^2$. With a chosen logical angle α_j, the physical angle is $(-1)^{a_{j-1}} \alpha_j$ and the update is

$$a_j = s_j \oplus b_{j-1}, \qquad b_j = a_{j-1}.$$

Eliminating b_{j-1} gives the second-order recurrence $a_j = s_j \oplus a_{j-2}$. For the initial frame $(a_0, b_0) = (0, 0)$,

measurement	a_j	b_j	physical angle
1	s_1	0	α_1
2	s_2	s_1	$(-1)^{s_1}\alpha_2$
3	$s_3 \oplus s_1$	s_2	$(-1)^{s_2}\alpha_3$
4	$s_4 \oplus s_2$	$s_3 \oplus s_1$	$(-1)^{s_3 \oplus s_1}\alpha_4$
5	$s_5 \oplus s_3 \oplus s_1$	$s_4 \oplus s_2$	$(-1)^{s_4 \oplus s_2}\alpha_5$

The recursion shows an alternating-parity memory of the measurement outcomes. It also makes the dependency graph explicit: measuring site four at a generic angle requires knowing s_1 and s_3, while a final Pauli correction can be determined from the complete record after all measurements. In a classical controller, this update requires only XOR operations and sign changes of angles.

6.10.3 An explicit angular example with noncommuting rotations

Let $J(\theta) = HU_z(\theta)$, and suppose the intended gate sequence contains two logical actions $J(\beta)J(\alpha)$. If the first measured outcome is $s_1 = 1$, applying the second physical angle β without adjustment yields

$$X^{s_2}J(\beta)XJ(\alpha) = X^{s_2}ZJ(-\beta)J(\alpha),$$

up to phase. This is not generally equivalent to $X^{s_2}ZJ(\beta)J(\alpha)$, because $J(\beta) \neq J(-\beta)$ for generic β. Replacing the second angle by $-\beta$ instead gives

$$X^{s_2}J(-\beta)XJ(\alpha) = X^{s_2}ZJ(\beta)J(\alpha),$$

which has the desired logical gate sequence with a trackable byproduct. The sign change is forced by Pauli conjugation, not by a convention that random outcomes should be ignored. At special Clifford angles $\beta \in \{0, \pi/2, \pi, \ldots\}$, some sign changes may be absorbed by relabeling outcomes or updating the Pauli frame, but the general algebra remains valid.

6.10.4 Partial trace as a warning against simply deleting measured vertices

If an intermediate vertex is measured and the measurement outcome is *discarded*, the correct operation on the unmeasured system is a mixture over measurement branches, not an arbitrarily chosen pure branch. For example, in the three-site path, an X measurement of the central site yields $|\Phi^+\rangle$ or $|\Psi^+\rangle$ on the endpoints, each with probability $1/2$. If the classical bit is forgotten, the endpoint state is

$$\rho_{13} = \tfrac{1}{2}|\Phi^+\rangle\langle\Phi^+| + \tfrac{1}{2}|\Psi^+\rangle\langle\Psi^+|.$$

This mixture is not the same as either conditional Bell state. In fact it is separable: expanding the Bell projectors yields $\rho_{13} = \frac{1}{2}(|++\rangle\langle++| + |--\rangle\langle--|)$. Thus heralded endpoint entanglement can be operationally lost if the measurement record is discarded. The calculation demonstrates why classical feedforward information is an integral component of measurement-based quantum computing.

6.10.5 A graph-state amplitude check for a four-site star

For a star with center 2 and leaves $1, 3, 4$, the graph-state basis amplitudes are

$$\langle x_1 x_2 x_3 x_4 | G \rangle = \frac{1}{4}(-1)^{x_2(x_1+x_3+x_4)}.$$

The state may be partitioned according to the central bit x_2:

$$|G\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_2|++\rangle_{134} + |1\rangle_2|--\rangle_{134}\right),$$

with the tensor factors reordered to place vertex two first. Measuring Z_2 thus yields either $|+++\rangle$ or $|---\rangle$ on the leaves, each with probability one half. Measuring X_2 instead produces, up to normalization, the three-qubit GHZ-type combinations $|+++\rangle \pm |---\rangle$. These distinct conditional states show that measurement basis choice changes the graph's entanglement pattern, not just the labels on its vertices.

6.10.6 Classical signaling versus quantum causal structure

Local measurements at spatially separated vertices do not themselves permit instantaneous transmission of unknown logical quantum data. A measurement result is a classical random bit; it can affect a later chosen basis only when the result has been communicated to the corresponding controller. The measurement pattern therefore imposes a partial order on operations. In a long chain, the initial creation of entanglement can be parallelized across edges because all CZ gates commute, whereas a sequence of adaptive non-Pauli measurements may need multiple rounds of classical communication. A complexity claim for measurement-based computation should distinguish preparation depth, quantum measurement depth, and classical feedforward depth.

CHAPTER 7

Variational Quantum Models and Quantum Neural Networks

7.1 Parametric learning objectives

Definition 7.1 (Parameterized quantum predictor)**.** For a classical input x, encoding unitary $U_{\text{in}}(x)$, variational circuit $V(\theta)$, and Hermitian readout O, define

$$|\psi(x;\theta)\rangle = V(\theta)U_{\text{in}}(x)|0^n\rangle, \qquad f(x;\theta) = \langle\psi(x;\theta)|O|\psi(x;\theta)\rangle.$$

For a Pauli readout $O = Z_0$ (Pauli Z on the first qubit), $f \in [-1, 1]$.

For data $\{(x^{(i)}, y^{(i)})\}_{i=1}^m$, a representative regression objective is

$$\mathcal{L}(\theta) = \frac{1}{m}\sum_{i=1}^{m}(f(x^{(i)};\theta) - y^{(i)})^2,$$

with gradient

$$\partial_{\theta_j}\mathcal{L} = \frac{2}{m}\sum_{i=1}^{m}(f_i - y^{(i)})\partial_{\theta_j} f_i.$$

The quantum subroutine estimates outputs and derivatives, while a classical optimizer constructs the loss and performs $\theta_j \leftarrow \theta_j - \eta\partial_{\theta_j}\mathcal{L}$. An observable expectation, a loss, and a loss gradient are distinct objects.

7.2 Data encoding models

Angle encoding assigns feature x_j to a rotation, for example

$$U_{\text{ang}}(x) = \bigotimes_{j=0}^{d-1} R_Y(x_j), \qquad R_Y(x)|0\rangle = \cos\frac{x}{2}|0\rangle + \sin\frac{x}{2}|1\rangle.$$

Since $R_Y(x + 2\pi) = -R_Y(x)$, the output density matrix is periodic with period 2π. Scaling features can prevent phase aliasing; rotations are not undefined outside a fixed interval.

Basis encoding of $x \in \mathbb{F}_2^d$ uses $\bigotimes_{j=0}^{d-1} X_j^{x_j}|0^d\rangle = |x\rangle$. For $x = 101101$, apply X to wires $0, 2, 3, 5$ under zero-based indexing; basis encoding requires d qubits.

Amplitude encoding of a nonzero vector $v \in \mathbb{C}^d$ uses

$$|v\rangle = \frac{1}{\|v\|_2} \sum_{j=0}^{d-1} v_j |j\rangle, \qquad n = \lceil \log_2 d \rceil,$$

with zero padding to length 2^n. The logarithmic qubit count is not a logarithmic guarantee for preparing an arbitrary state; gate count or data-access assumptions must be included in any efficiency statement.

Data re-uploading repeatedly interleaves data gates with trainable unitaries:

$$|\psi(x;\theta)\rangle = V_L(\theta_L)U(x)\cdots V_1(\theta_1)U(x)|0^n\rangle.$$

In one-dimensional periodic encodings, interference of the phase factors produces a trigonometric series $f(x;\theta) = \sum_{k\in\Omega_L} c_k(\theta)e^{ikx}$. If the assumed accessible frequencies are $\Omega_L = \{0, \pm 1, \ldots, \pm L\}$, there are $2L+1$ distinct frequencies; for $L = 3$, there are seven. For real-valued f, $c_{-k} = \overline{c_k}$. The precise frequency set depends on the encoding generators, gate repetitions, and architecture, not simply the number of parameters.

7.3 Ansatz design, trainability, and quantum convolution

A hardware-efficient ansatz typically alternates trainable local R_X, R_Y, R_Z rotations with entangling gates native to the processor. A quantum convolutional neural network (QCNN) instead reuses local variational convolution blocks and performs pooling operations that reduce the number of active qubits. If a particular hierarchical architecture halves the active width each layer and shares p independent weights per layer, it has $L \simeq \log_2 n$ scales and $P \simeq p \log_2 n$ learned parameters. This scaling requires the stated parameter-sharing assumptions and is not universal to every QCNN.

The **Fourier frequency spectrum**, the number of trainable parameters, and gradient concentration are different characteristics. A large parameter count does not by itself prove greater accessible Fourier bandwidth; a constant-gradient parameter can even be completely unidentifiable.

7.4 Exact parameter-shift differentiation

Theorem 7.2 (Pauli parameter-shift rule)**.** *Suppose a parameter appears exactly once through $R_P(\theta) = e^{-i\theta P/2}$, with $P = P^\dagger$ and $P^2 = I$, and all other operations are independent of θ. If*

$$f(\theta) = \langle\phi|R_P(\theta)^\dagger A R_P(\theta)|\phi\rangle, \qquad A = A^\dagger,$$

then

$$\boxed{\frac{df}{d\theta} = \frac{1}{2}\left[f\left(\theta + \frac{\pi}{2}\right) - f\left(\theta - \frac{\pi}{2}\right)\right].}$$

Proof. Write $R_P(\theta) = cI - isP$ with $c = \cos(\theta/2)$ and $s = \sin(\theta/2)$. Direct multiplication yields

$$R_P^\dagger A R_P = \frac{A + PAP}{2} + \frac{A - PAP}{2}\cos\theta + \frac{i[P, A]}{2}\sin\theta.$$

Consequently $f(\theta) = \alpha + \beta\cos\theta + \gamma\sin\theta$, with coefficients independent of θ. Shifting by $\pm\pi/2$ and subtracting gives the displayed exact derivative. □

For P independently parameterized single-occurrence Pauli rotations, a straightforward full gradient takes $2P$ **circuit settings** (for example, 24 settings for 12 parameters), before multiplying by the number of shots. Shared parameter occurrences require summing derivatives of each occurrence by the chain rule. The result is an analytic identity rather than a small-step finite-difference approximation.

For Z readout, let $\widehat{f}_\pm$ be independently measured sample means using $S_\pm$ shots. Then

$$\widehat{g} = \tfrac{1}{2}(\widehat{f}_+ - \widehat{f}_-), \qquad \mathrm{Var}(\widehat{g}) = \frac{1}{4}\left(\frac{1 - f_+^2}{S_+} + \frac{1 - f_-^2}{S_-}\right).$$

Thus the sampling standard deviation scales as $S^{-1/2}$. Circuit settings, shot count, and elementary gate complexity should be reported separately.

7.5 Heisenberg-picture parameter relevance

For a circuit $U = U_m \cdots U_1$ and observable O, propagate the effective observable backward from $O_m = O$ by $O_{j-1} = U_j^\dagger O_j U_j$. If a parameterized gate commutes with the effective observable at that position, its parameter has no effect on the output. This is a structural zero derivative, distinct from a small gradient caused by a particular initialization or a barren plateau.

Example 7.3 (A two-layer, three-qubit re-uploading circuit)**.** Consider two identical data-uploading layers, each applying $R_Y(x_j)$ on wires $j = 0, 1, 2$, then $R_Z(w_{\ell j})$ on each wire, followed by $CZ_{01}CZ_{12}$. Measure Z_0 at the end. The circuit contains six data rotations, six trainable R_Z gates, four CZ gates, and six independent declared weights.

All gates after the final R_Y block are diagonal in the computational basis and commute with Z_0. Hence every w_{1j} has identically zero influence. A more complete backward simplification for this **specific** circuit gives

$$f(x; w) = \cos^2 x_0 - \sin^2 x_0 \cos(w_{00}) \cos x_1.$$

Therefore w_{01} and w_{02} are also irrelevant: of the six declared weights, only w_{00} appears in this particular predictor. This stronger statement depends on the precise readout and CZ chain and does not extend automatically to other connectivity or gate orders.

When translating a programmatically generated quantum circuit, distinguish nested loop structure from physical parallelism. A loop over layers establishes chronology; an inner loop over wires establishes which gates are present. Data rotations do not become trainable just because they are repeated, and a CZ edge has no asymmetric control/target direction.

7.6 Heisenberg reduction and gradient composition

For the three-qubit circuit previously specified, start from the measured observable Z_0. Its backward conjugation by the final diagonal layer is still Z_0. The last $R_Y(x_0)$ changes it according to

$$R_Y(x_0)^\dagger Z_0 R_Y(x_0) = \cos x_0 \, Z_0 - \sin x_0 \, X_0.$$

Passing backward through CZ_{01} sends $X_0 \mapsto X_0 Z_1$ and leaves Z_0 invariant, so the effective observable becomes

$$O_{\text{eff}} = \cos x_0 \, Z_0 - \sin x_0 \, X_0 Z_1.$$

In the earlier product state before the first CZ layer,

$$\langle Z_0 \rangle = \cos x_0, \qquad \langle X_0 \rangle = \sin x_0 \cos w_{00}, \qquad \langle Z_1 \rangle = \cos x_1.$$

The expectation hence equals $\cos^2 x_0 - \sin^2 x_0 \cos w_{00} \cos x_1$, reproducing the exact closed expression. As boundary checks, $x_0 = 0$ forces output 1, and $x_1 = w_{00} = 0$ gives $\cos(2x_0)$.

A shifted **observable** gradient is not in general a shifted **nonlinear loss** gradient. For instance, if $f(\theta) = \cos\theta$ and $\mathcal{L}(\theta) = (f(\theta) - y)^2$, then $\mathcal{L}$ contains a $\cos 2\theta$ Fourier harmonic. The one-frequency parameter-shift rule must be applied to f and then combined with $\partial\mathcal{L}/\partial f = 2(f - y)$, unless a separate valid shift formula for $\mathcal{L}$ has been proved.

7.7 Detailed learning-model derivations and circuit reductions

7.7.1 A mathematical specification of variational quantum learning

Let $x \in \mathcal{X} \subseteq \mathbb{R}^d$ be a classical input, $U_{\text{enc}}(x)$ a data-dependent unitary, and $V(\theta)$ a parameterized trainable circuit with $\theta \in \mathbb{R}^P$. With a fixed Hermitian observable O, a simple variational prediction is

$$f_\theta(x) = \langle 0^n | U_{\text{enc}}(x)^\dagger V(\theta)^\dagger O V(\theta) U_{\text{enc}}(x) | 0^n \rangle.$$

A supervised training set $\{(x_i, y_i)\}_{i=1}^{m}$ can be fitted by minimizing $L(\theta) = m^{-1}\sum_i \ell(f_\theta(x_i), y_i)$, possibly with a regularizer. The outcome of a single projective measurement is not generally $f_\theta(x)$; it is a random eigenvalue of O. Repeated shots are needed to estimate an expectation, except in deterministic cases. One must distinguish the dimensions of input d, the number of qubits n, the number of trainable parameters P, circuit depth, and number of measurements. None is automatically equal to the others.

7.7.2 Angle, basis, amplitude, and repeated data encoding

Angle encoding maps coordinates to rotation angles, for example $U_{\mathrm{enc}}(x) = \bigotimes_{j=1}^{\min(d,n)} R_Y(x_j)$. On initial $|0\rangle$ this gives $\cos(x_j/2)|0\rangle + \sin(x_j/2)|1\rangle$ for each encoded qubit. Its dependence on x_j is periodic in output statistics, because $R_Y(x_j + 2\pi) = -R_Y(x_j)$ differs only by a global sign in an uncontrolled single-gate setting. This periodicity should not be confused with a bounded input domain: features may first be rescaled as part of a data-processing pipeline.

Basis encoding maps $b_1 \cdots b_n \in \{0,1\}^n$ to $|b_1 \cdots b_n\rangle$ by applying an X gate at each position with $b_j = 1$. For instance 101101 is prepared by X on wires $1, 3, 4, 6$ and identity on wires $2, 5$. This uses six qubits for six binary coordinates. The basis states are mutually orthogonal, so simple computational-basis readout can recover the input bits exactly, provided no subsequent transformation obscures them.

Amplitude encoding of a nonzero vector $u \in \mathbb{C}^D$ is $|u\rangle = \|u\|_2^{-1}\sum_{j=0}^{D-1} u_j|j\rangle$, typically with $n = \lceil \log_2 D \rceil$ qubits and zero-padding to dimension 2^n. For $u = (2, 1, 0, 0)^T$,

$$|u\rangle = \frac{2}{\sqrt{5}}|00\rangle + \frac{1}{\sqrt{5}}|01\rangle.$$

The representation uses logarithmically many qubits, but preparation of an arbitrary dense amplitude vector does not automatically take logarithmic time. The data-loading oracle or state-preparation circuit must be included in the computational model.

Data re-uploading interleaves encoding and trainable operations, for example

$$U(x, \theta) = W_L(\theta_L)U_L(x)\cdots W_2(\theta_2)U_2(x)W_1(\theta_1)U_1(x).$$

The same input may appear in multiple rotations. Even when the number of trainable parameters remains fixed, repeated data generators can enlarge the set of representable Fourier frequencies. Conversely, adding many parameters that commute with the measured observable may leave the function class unchanged.

7.7.3 Fourier frequencies created by data-dependent rotations

Suppose a scalar datum x enters through $R_Z(x) = e^{-ixZ/2}$ and the remaining gates are independent of x. Each computational amplitude is a linear combination of $e^{ix/2}$ and

$e^{-ix/2}$; an expectation formed by multiplying amplitudes with their conjugates has frequencies in $\{0, \pm 1\}$. With repeated appearances, products of such exponentials can create frequencies of the form sums of individual generator-eigenvalue differences. For L identical simple appearances, a common frequency envelope is $\{-L, \dots, L\}$, though coefficients may vanish because of circuit symmetries. A real-valued prediction has Fourier coefficients satisfying $c_{-k} = \overline{c_k}$.

Example 7.4 (Frequency counting)**.** Suppose the frequency set is $\{0, \pm 1, \pm 2, \pm 3\}$. It has seven distinct labels, not six. The zero-frequency term is counted once and each nonzero positive frequency has a negative counterpart. This count describes a frequency support upper bound; it does not assert that every coefficient is nonzero or independently trainable.

7.7.4 The parameter-shift identity from Pauli spectra

Consider a gate $R_P(\theta) = e^{-i\theta P/2}$ where $P^2 = I$ and an expectation $f(\theta) = \langle\psi|R_P(\theta)^\dagger O R_P(\theta)|\psi\rangle$ with all other gates absorbed into $|\psi\rangle$ and O. Since $R_P(\theta) = cI - isP$ for $c = \cos(\theta/2)$, $s = \sin(\theta/2)$,

$$\begin{aligned} f(\theta) &= c^2\langle O\rangle + s^2\langle POP\rangle + ics\langle[P, O]\rangle \\ &= a + b\cos\theta + c'\sin\theta, \end{aligned}$$

where the final coefficients are real because f is real. Differentiating gives $f'(\theta) = -b\sin\theta + c'\cos\theta$. Using $\cos(\theta \pm \pi/2) = \mp\sin\theta$ and $\sin(\theta \pm \pi/2) = \pm\cos\theta$, one obtains the exact identity

$$\boxed{\frac{df}{d\theta} = \frac{1}{2}\left[f(\theta + \pi/2) - f(\theta - \pi/2)\right].}$$

The rule is exact for a parameter that appears in a suitable single two-eigenvalue generator. If the same parameter appears in multiple locations, the derivative is the sum of separate contributions from each occurrence, not necessarily one pair of globally shifted circuits.

Example 7.5 (An elementary derivative check)**.** Set $f(\theta) = \langle 0|R_Y(\theta)^\dagger Z R_Y(\theta)|0\rangle = \cos\theta$. Differentiating gives $-\sin\theta$. The shifted measurements give $\frac{1}{2}[\cos(\theta + \pi/2) - \cos(\theta - \pi/2)] = \frac{1}{2}[-\sin\theta - \sin\theta] = -\sin\theta$. At $\theta = \pi/3$ the derivative equals $-\sqrt{3}/2$.

7.7.5 Shot noise and the variance of a shifted gradient

For a Pauli observable, each single shot $B \in \{-1, +1\}$ has expectation f and variance $1 - f^2$. The estimator $\widehat{f} = S^{-1}\sum_{j=1}^{S} B_j$ is unbiased with variance $(1 - f^2)/S$. For independent estimates of the two shifted expectation values with S_+ and S_- shots,

$$\widehat{\partial_\theta f} = \frac{\widehat{f}_+ - \widehat{f}_-}{2}, \quad \mathrm{Var}(\widehat{\partial_\theta f}) = \frac{1}{4}\left(\frac{1 - f_+^2}{S_+} + \frac{1 - f_-^2}{S_-}\right).$$

For equal shot budgets $S_+ = S_- = S$, the variance is at most $1/(2S)$. The resulting standard deviation decreases as $S^{-1/2}$, not S^{-1}. Evaluating a gradient with respect to P distinct parameters requires $2P$ circuit *settings* under the basic rule; if each setting uses S shots, the total measurement count is $2PS$. These are different resource units and must not be conflated.

7.7.6 Convolution, pooling, and parameter sharing

Quantum convolutional models commonly organize repeated local trainable blocks into layers and then reduce the number of active degrees of freedom through measurement or pooling. Consider a block $W(\eta)$ with q independently tuned parameters, applied to every adjacent two-qubit pair within a layer. If all copies share the same parameter vector η, applying the block at r distinct positions produces rq parameterized gate instances but only q independent parameters. This is the mathematical distinction between *circuit size* and *parameter count*. A balanced hierarchy reducing active qubits by a constant factor per layer has $O(\log n)$ layers, but no unconditional theorem says that every quantum convolutional model has $O(\log n)$ total gates or parameters; architecture and sharing assumptions must be stated.

A pooling step may measure or discard designated qubits and condition subsequent operations on the measurement result. Since the partial trace or measurement is not a unitary acting solely on the retained subsystem, it changes the channel structure. One cannot infer the effect of pooling just by deleting the measured wire from a unitary circuit: the measurement outcome and conditional update may contribute to the retained state.

7.7.7 A complete nested-loop circuit accounting

Consider the architecture obtained by executing, for each layer $\ell \in \{0, 1\}$, the sequence $R_Y(x_i)$ on all three wires $i = 0, 1, 2$, followed by $R_Z(w_{\ell i})$ on all three wires, followed by the two entanglers CZ_{01} and CZ_{12}. The final observable is Z_0. This architecture contains

$$\#R_Y = 2 \cdot 3 = 6, \quad \#R_Z = 2 \cdot 3 = 6, \quad \#CZ = 2 \cdot (3 - 1) = 4,$$

and six declared trainable weights $w_{\ell i}$. The repeated x_i are data, not newly independent trainable parameters. The time order is *outer layer first, inner wire second*, not all R_Y operations across both layers followed by all R_Z operations.

Let $V_0 = R_Y(x_0) \otimes R_Y(x_1) \otimes R_Y(x_2)$ and $D_\ell = CZ_{12}CZ_{01}(R_Z(w_{\ell 0}) \otimes R_Z(w_{\ell 1}) \otimes R_Z(w_{\ell 2}))$. The complete circuit is $U = D_1 V_0 D_0 V_0$. Since D_1 and Z_0 are both diagonal, $D_1^\dagger Z_0 D_1 = Z_0$. Therefore every final-layer weight has an identically zero derivative, for every input and earlier weight choice. This is an exact structural statement, not a small-gradient approximation.

7.7.8 A stronger Heisenberg-picture reduction

For this particular three-wire CZ-chain architecture and initial $|000\rangle$, a further reduction yields

$$\boxed{f(x;w) = \cos^2 x_0 - \sin^2 x_0 \cos(w_{00}) \cos x_1.}$$

A derivation begins by propagating Z_0 backwards through the terminal D_1, leaving Z_0. Through the second $R_Y(x_0)$ on wire zero, the observable becomes $\cos x_0 \, Z_0 - \sin x_0 \, X_0$ under the appropriate $U^\dagger O U$ convention. The first term contributes $\cos^2 x_0$ after the initial data rotation. The second acquires $\cos(w_{00})$ when propagated through the first $R_Z(w_{00})$ and a neighboring Z_1 from CZ_{01}; the initial product state gives the remaining $\sin x_0 \cos x_1$ factor. Terms containing Y_0 have vanishing expectation in the relevant real-rotation input state, while the second entangler CZ_{12} and diagonal rotations on other wires commute through or cancel in this observable's backward light cone.

Thus not only w_{10}, w_{11}, w_{12} but also w_{01} and w_{02} are irrelevant to f. Although six weights are declared, exactly one enters this measured prediction under the stated architecture. This cannot be transferred without proof to other observables, additional non-diagonal gates, or modified entanglement graphs.

Example 7.6 (A numerical evaluation and gradient)**.** Choose $x_0 = \pi/4$, $x_1 = \pi/3$, and $w_{00} = \pi/2$. The analytic expression gives $f = (1/2) - (1/2) \cdot 0 \cdot (1/2) = 1/2$. Differentiation with respect to the only relevant weight gives

$$\frac{\partial f}{\partial w_{00}} = \sin^2 x_0 \sin(w_{00}) \cos x_1 = \tfrac{1}{2} \cdot 1 \cdot \tfrac{1}{2} = \tfrac{1}{4}.$$

All five other declared weight derivatives vanish. If $x_1 = \pi/2$, even the remaining parameter disappears from the prediction for that particular input. Input-dependent vanishing of a derivative is different from structural vanishing for all inputs.

7.7.9 Gradient vanishing and trainability are logically distinct

Three phenomena should be distinguished. First, a *structural zero* follows from an algebraic symmetry such as commuting with the measured observable, so it holds at every parameter value. Second, a *stationary point* satisfies $\nabla L(\theta) = 0$ only at some particular configuration or data distribution. Third, a *barren plateau* is a scaling phenomenon in which the variance of gradients over a specified random ensemble may decay exponentially with system size. A single observed small gradient establishes none of these mechanisms by itself. Rigorous statements about trainability require specifying the random circuit ensemble, parameter initialization, observable locality, data encoding, and problem size.

7.8 Additional analytic variational-circuit case studies

7.8.1 A data-encoding circuit with an exactly computable decision function

Begin with $|0\rangle$, apply $R_Y(x)$ and then $R_Y(\theta)$, and measure Z. Because rotations about the same axis compose, $R_Y(\theta)R_Y(x) = R_Y(x+\theta)$, so

$$f_\theta(x) = \langle Z \rangle = \cos(x+\theta).$$

The model is a trigonometric decision function, not an arbitrary nonlinear function of x. Its derivative is $\partial_\theta f = -\sin(x+\theta)$ and $\partial_x f = -\sin(x+\theta)$, so this architecture cannot independently control the phase shift and the frequency of the input variable. Adding a second layer with only further Y rotations changes the net angle by addition but does not enlarge the function class. Noncommuting rotations or entangling operations are needed to create new structures in this particular setting.

Now interleave a Z rotation between two Y rotations:

$$U(x,\theta) = R_Y(x)R_Z(\theta)R_Y(x).$$

For the observable Z, direct Bloch-vector propagation gives

$$f_\theta(x) = \cos^2 x - \sin^2 x \cos\theta.$$

A derivation rotates the initial Bloch vector $(0,0,1)$ by x about Y to $(\sin x, 0, \cos x)$, by θ about Z to $(\sin x \cos\theta, \sin x \sin\theta, \cos x)$, and by x about Y again. The final z component is $-\sin x(\sin x \cos\theta) + \cos x \cos x$. The expression has input frequency two because $\cos^2 x = (1+\cos 2x)/2$ and $\sin^2 x = (1-\cos 2x)/2$. This illustrates how noncommuting generators can modify accessible Fourier frequencies.

7.8.2 Why a shared variable may need multiple parameter-shift terms

Consider $U(\theta) = R_Y(\theta)R_Z(\varphi)R_Y(\theta)$, in which the same mathematical variable θ occurs in two gates. The derivative of an expectation $f(\theta)$ obeys the product rule:

$$\frac{df}{d\theta} = \left.\frac{\partial f(\theta_1,\theta_2)}{\partial\theta_1}\right|_{\theta_1=\theta_2=\theta} + \left.\frac{\partial f(\theta_1,\theta_2)}{\partial\theta_2}\right|_{\theta_1=\theta_2=\theta}.$$

Each partial derivative may be computed by a two-shift rule because each occurrence is a single Pauli rotation. However, shifting both occurrences simultaneously by $\pm\pi/2$ is not in general equivalent to the sum of the two partial rules. In the case $\varphi = 0$, the circuit simplifies to $R_Y(2\theta)$ and $f(\theta) = \cos(2\theta)$. The naive simultaneous shift would give $[\cos(2\theta+\pi) - \cos(2\theta-\pi)]/2 = 0$, whereas the true derivative is $-2\sin(2\theta)$. This counterexample is decisive: the location of every parameter occurrence matters.

7.8.3 Estimating an expectation and its mean-squared error

If a Pauli observable has outcomes $B_s \in \{-1, +1\}$ and mean μ, the sample mean $\widehat{\mu}$ has mean-square error $\mathbb{E}[(\widehat{\mu} - \mu)^2] = (1 - \mu^2)/S$. The expression is maximized at $\mu = 0$ and vanishes when $|\mu| = 1$, reflecting the fact that deterministic measurements need no statistical averaging to estimate their own expectation in an ideal noiseless model. For two shifted expectations with independent samples, the parameter-shift gradient mean-square error equals its variance and can be optimized by allocating different shot budgets to the two settings. Given a fixed total $S_+ + S_- = S_{\text{tot}}$, minimizing

$$\frac{1}{4}\left(\frac{\sigma_+^2}{S_+} + \frac{\sigma_-^2}{S_-}\right), \qquad \sigma_\pm^2 = 1 - f_\pm^2,$$

by Lagrange multipliers yields $S_+ : S_- = \sigma_+ : \sigma_-$ when both variances are nonzero. Equal allocation is simple and often reasonable, but it is not universally variance-optimal.

7.8.4 A local light-cone criterion for parameter relevance

Write a circuit in chronological order as $U = G_m \cdots G_1$ and let O be the measured observable. Propagate O backward: set $O_m = O$ and $O_{j-1} = G_j^\dagger O_j G_j$. If a parametrized gate $G_j(\theta)$ commutes with O_j for all θ, then $G_j^\dagger O_j G_j = O_j$ and the parameter has no effect on the final expectation. This criterion is stronger than reasoning from the gate's location alone. A final R_Z commutes with a final Z measurement and is irrelevant, but the same R_Z followed by H can influence the result since $H^\dagger Z H = X$ does not commute with R_Z.

For instance, consider $|0\rangle \xrightarrow{H} R_Z(\theta) \xrightarrow{H}$ measure Z. The output is

$$f(\theta) = \langle 0|H R_Z(\theta)^\dagger H Z H R_Z(\theta) H|0\rangle = \cos\theta.$$

The R_Z lies between Hadamards, so its relative phase is converted into a population difference. By contrast, the circuit $|0\rangle \xrightarrow{H}\xrightarrow{H} R_Z(\theta) \rightarrow$ measure Z has prediction 1 independently of θ, because the terminal diagonal rotation is invisible to the measured Z observable.

7.8.5 Parameter identifiability versus declared circuit dimension

A parameter vector $\theta \in \mathbb{R}^P$ is *locally identifiable* through the model outputs only if the Jacobian with respect to relevant observations has sufficient rank. Let $F(\theta) = (f_\theta(x_1), \ldots, f_\theta(x_m))$. Even when P distinct gate parameters are declared, the Jacobian $J_{ij} = \partial f_\theta(x_i)/\partial\theta_j$ can have rank much smaller than P. In the three-wire two-layer example above, the output depends on a single weight w_{00} for all inputs, so the Jacobian rank is at most one. The existence of six stored parameters does not imply a

six-dimensional statistical model. This is distinct from finite-shot gradient variance: increasing shots cannot recover derivatives that are exactly zero by symmetry.

A useful numerical consistency test for any symbolic derivative is a centered finite difference:

$$\partial_j f(\theta) \approx \frac{f(\theta + he_j) - f(\theta - he_j)}{2h},$$

with truncation error $O(h^2)$ for smooth f. The finite difference is not a replacement for an exact hardware parameter-shift identity; it is a validation technique for algebra or code at moderate precision. Excessively small h may aggravate numerical cancellation or shot-noise amplification, so its error must be considered separately from the exact shift rule.

CHAPTER 8

Quantum Feature Maps and Kernel Methods

8.1 Classical positive-semidefinite kernels

A feature map $\varphi : \mathcal{X} \to \mathcal{F}$ into an inner-product space induces a kernel $k(x, z) = \langle \varphi(x), \varphi(z) \rangle$. Kernel methods compute the resulting inner products without constructing every feature coordinate explicitly. For polynomials in d variables, the number of monomials of total degree exactly r is $\binom{d+r-1}{r}$ and the number of all monomials of degree at most p is

$$D(d, p) = \sum_{r=0}^{p} \binom{d + r - 1}{r} = \binom{d + p}{p}.$$

For three variables up to quadratic degree, there is one constant, three linear, and six quadratic terms, giving $D(3, 2) = 10$.

8.2 Fidelity kernels as Hilbert–Schmidt kernels

Definition 8.1 (Quantum fidelity feature kernel)**.** Let $U_\varphi(x)$ prepare $|\varphi(x)\rangle = U_\varphi(x)|0^n\rangle$. The pure-state fidelity kernel is

$$K_Q(x, z) = |\langle \varphi(x) | \varphi(z) \rangle|^2 = \mathrm{Tr}[\rho(x)\rho(z)], \quad \rho(x) = |\varphi(x)\rangle\langle\varphi(x)|.$$

The kernel satisfies $0 \le K_Q(x, z) \le 1$, $K_Q(x, x) = 1$, and $K_Q(x, z) = 1$ if and only if the two prepared pure states differ by a global phase. Different classical inputs can therefore have fidelity one when the feature map is noninjective. Fidelity zero means orthogonal quantum states, not a failed circuit.

Proposition 8.2 (Positive semidefiniteness of the ideal Gram matrix)**.** *For any finite data set $x_1, \ldots, x_m$, the matrix $K_{ij} = K_Q(x_i, x_j)$ is Hermitian positive semidefinite.*

Proof. For arbitrary $c_1, \ldots, c_m \in \mathbb{C}$,

$$\sum_{i,j} \overline{c_i} c_j K_{ij} = \mathrm{Tr}\left[\left(\sum_i c_i \rho(x_i) \right)^\dagger \left(\sum_j c_j \rho(x_j) \right) \right] \ge 0.$$

The trace is the square of a Hilbert–Schmidt norm. □

8.3 Gram matrices, regression, and target alignment

Because the ideal kernel is symmetric and has known unit diagonal, an $m \times m$ Gram matrix requires $\binom{m}{2} = m(m-1)/2$ distinct off-diagonal fidelity values; when $m = 20$, the count is 190. This counts distinct pairs, not measurement shots.

Kernel ridge regression solves

$$\alpha = (K + \lambda I_m)^{-1} y, \qquad \widehat{y}(x) = \sum_{i=1}^{m} \alpha_i K_Q(x_i, x),$$

with regularization $\lambda > 0$. Measuring kernel entries is quantum computation; solving the dense $m \times m$ regularized linear system in this form is classical computation.

For label kernel $(K_y)_{ij} = y_i y_j$, the kernel-target alignment is

$$\mathcal{A}(K, K_y) = \frac{\mathrm{Tr}(K K_y)}{\sqrt{\mathrm{Tr}(K^2)\,\mathrm{Tr}(K_y^2)}}$$

for real symmetric matrices and nonzero denominators. A trainable feature map may optimize this quantity, differentiating individual kernel entries using parameter-shift identities and then applying the ordinary multivariable chain rule. Alignment concerns correspondence with labels; it is not a criterion for quantum computational advantage.

8.4 Compute–uncompute fidelity estimation

Prepare $U_\varphi(x)|0^n\rangle$ and then apply $U_\varphi(z)^\dagger$. The amplitude of $|0^n\rangle$ is

$$\langle 0^n | U_\varphi(z)^\dagger U_\varphi(x) | 0^n \rangle = \langle \varphi(z) | \varphi(x) \rangle.$$

Therefore

$$\Pr(0^n) = K_Q(x, z).$$

The scheme requires n qubits and an inverse state-preparation circuit. With S independent shots, its empirical Bernoulli estimator has variance $K_Q(1 - K_Q)/S$.

8.5 The SWAP test

Let $|a\rangle, |b\rangle$ be normalized n-qubit states. Initialize ancilla $|0\rangle$, apply H, controlled-SWAP between the two system registers, and another H. The final joint state is

$$\frac{1}{2}\Big[|0\rangle(|a\rangle|b\rangle + |b\rangle|a\rangle) + |1\rangle(|a\rangle|b\rangle - |b\rangle|a\rangle)\Big].$$

Taking the squared norm of the zero-ancilla component gives

$$p_0 = \frac{1 + |\langle a|b\rangle|^2}{2}, \qquad K_Q = 2p_0 - 1.$$

For S trials and n_0 observed zero outcomes, $\widehat{K}_Q = 2n_0/S - 1$. Its variance is

$$\mathrm{Var}(\widehat{K}_Q) = \frac{4p_0(1-p_0)}{S} = \frac{1 - K_Q^2}{S}.$$

The test uses $2n + 1$ qubits. For $n_0 = 900$ out of 1000 shots, $\widehat{K}_Q = 0.80$. For $n_0 = 875$, the result is 0.75, with estimated standard error $2\sqrt{0.875(0.125)/1000} \approx 0.0209$. A normal-approximation 95% interval is approximately $[0.709, 0.791]$, with the usual limitations of asymptotic interval estimation.

For the same ideal fidelity $K \in [0, 1]$ and shot budget, the single-shot variance $1 - K^2$ of the SWAP-test estimator is no smaller than $K(1 - K)$ for compute–uncompute. This ideal statistical comparison does not automatically determine the better practical circuit under hardware noise, connectivity, and inversion costs.

8.6 Circuit cancellation and exact classical reduction

Example 8.3 (A separable fidelity kernel hidden by controlled-phase gates)**.** Take $n = 2$. Define the feature unitary, in execution order, by $H \otimes H$, $R_Z(x_0) \otimes R_Z(x_1)$, CZ_{01}, and $R_Z(x_0x_1) \otimes R_Z(x_0x_1)$. To estimate overlap, execute $U_\varphi(x)$ followed by $U_\varphi(z)^\dagger$. The two CZ factors cancel because all intermediate R_Z gates are diagonal, so the overall compute–uncompute circuit reduces to

$$\big(HR_Z(\Delta_0)H\big) \otimes \big(HR_Z(\Delta_1)H\big),$$

where $\Delta_j = x_j - z_j + x_0x_1 - z_0z_1$. Since $\langle 0|HR_Z(\Delta)H|0\rangle = \cos(\Delta/2)$, the exact kernel is

$$\boxed{K_Q(x, z) = \cos^2(\Delta_0/2)\cos^2(\Delta_1/2).}$$

The original state-preparation circuit contains an entangling CZ, but this full overlap is easily calculated classically. A visible entangling gate alone is not a demonstration of computational separation.

The unreduced two-qubit compute–uncompute circuit contains four Hadamards, eight R_Z gates, and two CZ gates. Inverting a circuit requires reversing gate order as well as taking adjoints: $(U_k \cdots U_1)^\dagger = U_1^\dagger \cdots U_k^\dagger$.

8.7 Kernel concentration and sampling resolution

Suppose for many different inputs $K_{ii} = 1$ and $K_{ij} \approx c \ll 1$ with $i \neq j$. Then

$$K \approx (1 - c)I + c\mathbf{1}\mathbf{1}^\top,$$

and for $c \to 0$ the Gram matrix approaches the identity. The kernel remains PSD, but its off-diagonal entries contain little discriminative information about relations between points. If useful differences have size δ, additive shot noise of order $S^{-1/2}$ demands $S = \Omega(\delta^{-2})$ measurements to resolve them at fixed confidence under standard sampling assumptions. An exponentially small δ can therefore incur exponential sampling cost.

8.8 Kernel construction, estimation, and explicit circuits

8.8.1 The monomial feature space and a counting derivation

A degree-p polynomial feature map in d variables can include all monomials $x_1^{\alpha_1} \cdots x_d^{\alpha_d}$ with nonnegative exponents satisfying $\sum_j \alpha_j \le p$. To count monomials of exactly degree r, introduce a slack variable or use the stars-and-bars correspondence between weak compositions and the placement of $d-1$ separators among $r+d-1$ positions. The count is $\binom{d+r-1}{r}$. Summing over all degrees up to p and using the hockey-stick identity gives

$$D(d,p) = \sum_{r=0}^{p} \binom{d+r-1}{r} = \binom{d+p}{p}.$$

For $d = 3$, $p = 2$, one obtains $D(3,2) = \binom{5}{2} = 10$ features: 1, three degree-one monomials x_1, x_2, x_3, three pure squares, and three pairwise products. If the constant term is excluded, the dimension becomes nine. If only *homogeneous* degree-two monomials are used, the dimension becomes $\binom{4}{2} = 6$. All three counts are valid for different feature-map definitions; none should be substituted for another without checking the degree convention.

8.8.2 Positive semidefinite kernels as Gram matrices

Let $\phi : \mathcal{X} \to \mathcal{H}$ be a feature map into a real or complex Hilbert space and let $k(x,z) = \langle \phi(x), \phi(z) \rangle$. For any finite set $x_1, \ldots, x_m$, its Gram matrix $K_{ij} = k(x_i, x_j)$ is Hermitian and positive semidefinite. Indeed, for arbitrary $c \in \mathbb{C}^m$,

$$c^\dagger K c = \sum_{ij} \overline{c_i} c_j \langle \phi(x_i), \phi(x_j) \rangle = \left\| \sum_i c_i \phi(x_i) \right\|^2 \ge 0.$$

This one-line calculation justifies kernel-based linear models in implicit feature spaces. A symmetric table of numbers is not automatically a valid kernel matrix; the positive-semidefinite condition must hold for every finite set of inputs. A useful classical example is the polynomial kernel $(c + x \cdot z)^p$ with nonnegative c and integer $p \ge 0$, whose positive-semidefinite nature follows from its finite monomial-feature expansion.

8.8.3 Quantum fidelity as a classical Hilbert–Schmidt kernel

Suppose $U_\phi(x)$ prepares the normalized pure state $|\phi(x)\rangle = U_\phi(x)|0^n\rangle$. Define

$$K_Q(x,z) = |\langle\phi(x)|\phi(z)\rangle|^2.$$

Writing $\rho(x) = |\phi(x)\rangle\langle\phi(x)|$, the same quantity is $\mathrm{Tr}[\rho(x)\rho(z)]$, which is an inner product in the Hilbert–Schmidt operator space. Therefore the fidelity kernel is positive semidefinite even though the ordinary inner product $\langle\phi(x)|\phi(z)\rangle$ may have a complex phase. Moreover $0 \le K_Q(x,z) \le 1$ and $K_Q(x,x) = 1$. This construction identifies the implicit feature map as $x \mapsto \rho(x)$ rather than necessarily $x \mapsto |\phi(x)\rangle$ itself.

Example 8.4 (A one-qubit feature map)**.** Let $|\phi(x)\rangle = R_Y(x)|0\rangle$. Then

$$\langle\phi(x)|\phi(z)\rangle = \langle 0|R_Y(z)^\dagger R_Y(x)|0\rangle = \cos\frac{x-z}{2},$$

so $K_Q(x,z) = \cos^2((x-z)/2)$. The identity $\cos^2(t/2) = (1+\cos t)/2$ reveals that this simple quantum kernel has a low-dimensional explicit classical trigonometric feature representation. Quantum state preparation by itself is not evidence of a difficult-to-evaluate kernel.

8.8.4 Compute–uncompute estimation and finite-shot complexity

The compute–uncompute circuit is $U_\phi(z)^\dagger U_\phi(x)|0^n\rangle$. Its all-zero amplitude is

$$\langle 0^n|U_\phi(z)^\dagger U_\phi(x)|0^n\rangle = \langle\phi(z)|\phi(x)\rangle.$$

Hence the probability of measuring 0^n is $K_Q(x,z)$. If S independent shots return indicators $B_s = \mathbf{1}_{\{0^n\}}$, the sample average $\widehat{K} = S^{-1}\sum_s B_s$ is unbiased with variance

$$\mathrm{Var}(\widehat{K}) = \frac{K(1-K)}{S} \le \frac{1}{4S}.$$

The standard error is therefore at most $1/(2\sqrt{S})$. By Hoeffding's inequality, $\Pr(|\widehat{K} - K| \ge \varepsilon) \le 2e^{-2S\varepsilon^2}$, so $S \ge \ln(2/\delta)/(2\varepsilon^2)$ suffices for additive accuracy ε with failure probability at most δ. If many distinct kernel entries must be estimated simultaneously, the confidence budget must account for a union bound across entries.

For m training points, the full Gram matrix has m^2 entries but symmetry leaves only $m(m+1)/2$ distinct positions, and the pure-state diagonal entries are known exactly to be one. Thus at most $m(m-1)/2$ off-diagonal kernel estimations are needed when every pair is evaluated independently. Counting both K_{ij} and K_{ji} as two unrelated quantum experiments double-counts information.

8.8.5 The SWAP test by complete state evolution

Let $|a\rangle, |b\rangle$ be normalized and let an ancilla start in $|0\rangle$. After a Hadamard, controlled-SWAP, and another Hadamard, the total state is

$$\frac{1}{2}\big[|0\rangle(|a\rangle|b\rangle + |b\rangle|a\rangle) + |1\rangle(|a\rangle|b\rangle - |b\rangle|a\rangle)\big].$$

The probability of ancilla outcome zero is one quarter of the squared norm of the symmetric branch. Since

$$\| |a\rangle|b\rangle + |b\rangle|a\rangle\|^2 = 2 + 2|\langle a|b\rangle|^2,$$

one obtains

$$\boxed{p_0 = \frac{1 + |\langle a|b\rangle|^2}{2}, \qquad K = 2p_0 - 1.}$$

For identical states $K = 1$ and the ancilla is always zero. For orthogonal states $K = 0$ and the ancilla is unbiased. An intermediate overlap $|\langle a|b\rangle|^2 = 1/4$ gives $p_0 = 5/8$. The SWAP test and compute–uncompute estimate the same fidelity in this pure-state setting, but require different circuit primitives and may have different practical depths and noise sensitivities.

If S independent ancilla measurements estimate p_0, the induced estimate $\widehat{K} = 2\widehat{p}_0 - 1$ has variance $4p_0(1 - p_0)/S \le 1/S$. The mapping from probability to fidelity doubles the standard deviation; omitting this factor can understate the shot budget.

8.8.6 Exact simplification of an apparently entangling feature-map circuit

Consider two qubits and the feature map whose chronological gates are: Hadamard on both qubits; $R_Z(x_0)$ and $R_Z(x_1)$; a CZ edge; and $R_Z(x_0x_1)$ on each qubit. Let its total diagonal part after the initial Hadamards be $D(x)$. The compute–uncompute circuit is

$$U_\phi(z)^\dagger U_\phi(x) = H^{\otimes 2} D(z)^\dagger D(x) H^{\otimes 2}.$$

All of the gates inside $D(x)$ are diagonal in the same basis and therefore commute. In the product $D(z)^\dagger D(x)$, the two CZ factors cancel because $CZ^2 = I$. What remains is

$$D(z)^\dagger D(x) = R_Z(\Delta_0) \otimes R_Z(\Delta_1), \quad \Delta_i = (x_i - z_i) + (x_0x_1 - z_0z_1).$$

The all-zero output amplitude factorizes:

$$\begin{aligned}\langle 00|U_\phi(z)^\dagger U_\phi(x)|00\rangle &= \prod_{i=0}^{1} \langle 0|HR_Z(\Delta_i)H|0\rangle \\ &= \cos(\Delta_0/2)\cos(\Delta_1/2).\end{aligned}$$

Consequently the kernel has the closed form

$$\boxed{K_Q(x,z) = \cos^2(\Delta_0/2)\cos^2(\Delta_1/2).}$$

This is a concrete example in which entangling gates appear in each feature-map preparation but cancel from the particular overlap computation. It refutes the inference that the presence of CZ in a circuit automatically certifies a quantum computational advantage for the resulting kernel.

8.8.7 Kernel matrices, ridge regression, and alignment

For training data (x_i, y_i), kernel ridge regression with regularization $\lambda > 0$ solves $(K + \lambda I)\alpha = y$ and predicts

$$\widehat{y}(x) = k_x^T \alpha, \quad k_x = (k(x, x_1), \ldots, k(x, x_m))^T.$$

The regularization term improves invertibility even when K is singular. However, the numerical condition number of $K + \lambda I$ and the statistical quality of the kernel are different quantities. Kernel-target alignment is often defined as

$$\mathcal{A}(K, yy^T) = \frac{\langle K, yy^T\rangle_F}{\|K\|_F \|yy^T\|_F},$$

for nonzero target and kernel matrices. The numerator $y^T K y$ measures the compatibility between the kernel's geometry and the labels. Maximizing an alignment statistic is not identical to minimizing a supervised prediction loss; the objectives can respond differently to regularization and class imbalance.

8.8.8 Concentration of fidelity and statistical resolution

In some feature-map ensembles, distinct data states become nearly orthogonal as Hilbert-space dimension grows. The associated Gram matrix may approach a matrix of the form

$$K \approx (1 - c)I + c\mathbf{1}\mathbf{1}^T$$

with $c \ll 1$, so diagonal entries stay one while off-diagonal values cluster near c. This matrix is positive semidefinite for $0 \le c \le 1$: its eigenvalues are $1 - c$ with multiplicity $m - 1$ and $1 + (m - 1)c$ along $\mathbf{1}$. Thus concentration near identity does not imply loss of positive semidefiniteness. The concern is that similarities between *different* samples may become difficult to resolve or uninformative for generalization.

Suppose two meaningful similarities differ by $\delta > 0$ while each Bernoulli estimate has standard deviation of order $S^{-1/2}$. Resolving the difference with fixed signal-to-noise ratio requires $S = \Omega(\delta^{-2})$ in the generic bounded-variance regime. If δ decreases exponentially with the number of qubits, an exponential shot requirement may arise. Precise claims require a specified feature-map ensemble and data distribution: neither the existence of a quantum feature map nor a large number of qubits implies concentration in every instance.

8.9 Exact Gram matrices and statistical kernel calculations

8.9.1 A three-point fidelity Gram matrix with known spectrum

Use the one-qubit feature state $|\phi(x)\rangle = R_Y(x)|0\rangle$, whose fidelity kernel is $K(x,z) = \cos^2((x-z)/2)$. Choose training inputs $x_1 = 0$, $x_2 = \pi$, and $x_3 = \pi/2$. The matrix is

$$K = \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{pmatrix}.$$

The vector $(1,-1,0)^T/\sqrt{2}$ is an eigenvector with eigenvalue one. The orthogonal subspace spanned by $(1,1,0)^T/\sqrt{2}$ and $(0,0,1)^T$ has restricted matrix $\begin{pmatrix} 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1 \end{pmatrix}$, with eigenvalues $1+1/\sqrt{2}$ and $1-1/\sqrt{2}$. All are positive, agreeing with the theoretical PSD guarantee. In particular, the fact that the first and second training states are orthogonal does not force the third sample to be orthogonal to either of them.

8.9.2 A complete kernel ridge regression computation

Let the targets for these three inputs be $y = (1,-1,0)^T$. Kernel ridge regression with $\lambda = 1$ solves $(K+I)\alpha = y$. Since $Ky = y$, one has $(K+I)y = 2y$, so the coefficient vector is simply $\alpha = y/2 = (1/2,-1/2,0)^T$. For a test point $x = 0$, its kernel vector is $(1,0,1/2)^T$, giving $\widehat{y}(0) = 1/2$. At $x = \pi$, the vector is $(0,1,1/2)^T$ and prediction is $-1/2$. These are regularized predictions rather than interpolation of the original labels. For $\lambda \to 0$ in this positive-definite example, the kernel solution approaches exact training interpolation.

The centered label-target alignment numerator is $y^T K y = \|y\|^2 = 2$. The Frobenius norm of K is

$$\|K\|_F^2 = 3 + 4(1/2)^2 = 4, \qquad \|yy^T\|_F = \|y\|_2^2 = 2.$$

Thus $\mathcal{A}(K, yy^T) = 2/(2\cdot 2) = 1/2$. This calculation checks the dimensions of the target matrix and the fact that the Frobenius norm of yy^T equals $\|y\|^2$, not $\|y\|$.

8.9.3 The difference between estimating one entry and the full Gram matrix

If m points are given and each off-diagonal fidelity is estimated independently to additive error ε with individual failure probability δ', then at most $m(m-1)/2$ distinct experiments are required, excluding the exactly known pure-state diagonal. To guarantee simultaneous accuracy with total failure probability at most δ, a union bound permits $\delta' = 2\delta/[m(m-1)]$ for $m \geq 2$. Using the Hoeffding bound for Bernoulli measurements, it suffices to take

$$S \geq \frac{\ln\bigl(m(m-1)/\delta\bigr)}{2\varepsilon^2}$$

shots per off-diagonal entry. The corresponding total shot count is of order $m^2\varepsilon^{-2}\log(m/\delta)$, excluding the cost of implementing each feature-state circuit. This is a worst-case uniform-accuracy guarantee, not necessarily an optimal estimator for every learning objective.

The distinction between entrywise accuracy and spectral-matrix accuracy matters. If every entry of an $m \times m$ error matrix E satisfies $|E_{ij}| \le \varepsilon$, then $\|E\|_F \le m\varepsilon$ and hence $\|E\|_2 \le m\varepsilon$. Therefore obtaining operator-norm error at most η by this crude route requires entrywise error at most η/m. Stronger matrix-concentration methods can improve some estimates, but the naive conversion already illustrates why a small error on every individual kernel value does not automatically imply a comparably small error in the whole Gram matrix's spectrum.

8.9.4 SWAP test and compute–uncompute with a common toy pair

Take $|a\rangle = |0\rangle$ and $|b\rangle = |+\rangle$. Their overlap is $1/\sqrt{2}$, so the fidelity is $K = 1/2$. A compute–uncompute procedure that prepares $|a\rangle$ and then applies the inverse circuit for $|b\rangle$ measures all zeros with probability $1/2$. In a SWAP test, the ancilla-zero probability is $(1 + K)/2 = 3/4$. Thus the two measurements have different raw probabilities even though they estimate the same K. For $S = 100$ ideal shots, the theoretical variance of the compute–uncompute sample mean is $(1/2)(1/2)/100 = 0.0025$, while the variance of the SWAP-test-derived estimate $2\widehat{p}_0 - 1$ is $4(3/4)(1/4)/100 = 0.0075$.

8.9.5 A parameterized two-feature trigonometric kernel

For the two-qubit diagonal feature map reduced earlier, choose $x = (0, 0)$ and $z = (\pi, 0)$. Then $x_0x_1 - z_0z_1 = 0$, so $\Delta_0 = -\pi$ and $\Delta_1 = 0$. The fidelity kernel is $\cos^2(-\pi/2)\cos^2(0) = 0$. For $x = (0, 0)$ and $z = (\pi/2, 0)$, the result is $\cos^2(-\pi/4)\cos^2(0) = 1/2$. If instead $z = (\pi/2, \pi/2)$, the nonlinear product term contributes $-\pi^2/4$ to both phase differences; the kernel becomes a product of two nontrivial cosine squares. These examples demonstrate exactly where the pairwise feature interaction enters the simplified formula even though the CZ entanglers cancel from the compute–uncompute overlap circuit.

The final simplified expression is a function of both x and z through the differences $x_i - z_i$ and $x_0x_1 - z_0z_1$. It remains a valid kernel because it came from fidelity of normalized states, not because arbitrary products of trigonometric functions are automatically PSD in every pair of arguments. Its efficient classical evaluation, however, means that this particular overlap task provides no evidence of an exponential quantum advantage.

CHAPTER 9

Quantum Generative Models and Distributional Objectives

9.1 Probability-generating models

Generative learning seeks a model distribution P_θ approximating an unknown data distribution Q. An ordinary autoencoder minimizes reconstruction error $\|x - d(e(x))\|_2^2$ but does not by itself specify a conveniently sampled prior over its latent codes. A variational autoencoder regularizes a stochastic latent representation, for example through

$$\mathcal{L}_{\mathrm{VAE}} = \mathbb{E}_{z\sim q_\phi(z|x)}[-\log p_\theta(x \mid z)] + D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z)).$$

Adversarial generative models train a generator and discriminator with distinct parameter vectors. Quantum generative models can place a quantum circuit on either side or can define their output distribution directly through Born measurements.

9.2 Quantum circuit Born machines

Definition 9.1 (Quantum circuit Born machine (QCBM))**.** Let $U(\theta)$ act on n qubits initialized in $|0^n\rangle$. Its output probability distribution on bit strings is

$$P_\theta(x) = |\langle x|U(\theta)|0^n\rangle|^2, \quad x \in \mathbb{F}_2^n, \qquad \sum_x P_\theta(x) = 1.$$

A single computational-basis measurement yields one random sample $X \sim P_\theta$, not a table of 2^n exact probabilities. After S independent shots, the histogram estimator is $\widehat{P}_\theta(x) = S^{-1}\sum_{j=1}^{S} \mathbf{1}\{X_j = x\}$, with binomial standard deviation estimated by $\sqrt{\widehat{P}_\theta(x)(1-\widehat{P}_\theta(x))/S}$. Source data may be classical bit strings; no quantum state representation of each training observation is intrinsically required.

9.3 Total variation distance

Definition 9.2 (Total variation distance)**.** For distributions P, Q over a finite set Ω,

$$\mathrm{TV}(P,Q) = \frac{1}{2}\sum_{x\in\Omega}|P(x) - Q(x)| = \max_{A\subseteq\Omega}|P(A) - Q(A)|, \quad 0 \le \mathrm{TV} \le 1.$$

For $P = (0.40, 0.10, 0.30, 0.20)$ and $Q = (0.25, 0.25, 0.25, 0.25)$, the four absolute differences sum to 0.40, and $\mathrm{TV}(P, Q) = 0.20$. Where $P_\theta(x) \neq Q(x)$ for each x, its parameter derivative is

$$\partial_{\theta_j} \mathrm{TV}(P_\theta, Q) = \frac{1}{2} \sum_x \mathrm{sgn}(P_\theta(x) - Q(x)) \partial_{\theta_j} P_\theta(x).$$

At equality points the absolute value can be nondifferentiable; a subgradient or alternative smooth objective may be needed.

9.4 Maximum mean discrepancy

Definition 9.3 (Squared maximum mean discrepancy)**.** Given a positive-semidefinite kernel k with feature map ϕ into an RKHS $\mathcal{H}_k$, define the mean embedding $\mu_P = \mathbb{E}_{X \sim P}[\phi(X)]$. The squared maximum mean discrepancy is

$$\mathrm{MMD}_k^2(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}_k}^2.$$

For finite support, the objective has the expansion

$$\begin{aligned} \mathcal{L}_{\mathrm{MMD}}(\theta) = & \sum_{x,x'} P_\theta(x) P_\theta(x') k(x, x') + \sum_{y,y'} Q(y) Q(y') k(y, y') \\ & - 2 \sum_{x,y} P_\theta(x) Q(y) k(x, y). \end{aligned}$$

The three components are model–model similarity, data–data similarity, and twice model–data similarity with a minus sign. If the kernel is characteristic, MMD zero implies $P = Q$; the claim is not true for every degenerate kernel.

Given model samples $X_1, \ldots, X_n$ and data samples $Y_1, \ldots, Y_m$, the biased empirical estimator with diagonal terms is

$$\widehat{\mathrm{MMD}}^2 = \frac{1}{n^2} \sum_{i,i'} k(X_i, X_{i'}) + \frac{1}{m^2} \sum_{j,j'} k(Y_j, Y_{j'}) - \frac{2}{nm} \sum_{i,j} k(X_i, Y_j).$$

If k is symmetric and $k(x, x)$ is known without evaluation, the number of potentially distinct off-diagonal kernel evaluations is

$$\binom{n}{2} + \binom{m}{2} + nm.$$

For $n = m = 40$, this equals $780 + 780 + 1600 = 3160$. The nm cross-sample terms cannot be halved just because the kernel is symmetric; there is only one ordered cross block in the formula.

9.5 Differentiating MMD via sampled quantum probabilities

Proposition 9.4 (MMD gradient for a fixed symmetric kernel)**.** *For a differentiable model distribution P_θ and a parameter-independent symmetric kernel k, the MMD gradient is*

$$\partial_{\theta_j}\mathcal{L}_{\mathrm{MMD}} = 2\sum_x \partial_{\theta_j}P_\theta(x)\left[\sum_{x'} P_\theta(x')k(x,x') - \sum_y Q(y)k(x,y)\right].$$

Proof. Apply the product rule to $P_\theta(x)P_\theta(x')$ in the model–model term. Swapping the dummy indices x, x' and invoking $k(x,x') = k(x',x)$ makes the two contributions equal. The data–data term has zero derivative, and the mixed term contributes $-2\sum_{x,y}\partial_{\theta_j}P_\theta(x)Q(y)k(x,y)$. □

Let $\Pi_x = |x\rangle\langle x|$. Since $P_\theta(x) = \langle\psi_\theta|\Pi_x|\psi_\theta\rangle$, when θ_j occurs in one eligible Pauli rotation,

$$\partial_{\theta_j}P_\theta(x) = \frac{1}{2}\left(P_{\theta+(\pi/2)e_j}(x) - P_{\theta-(\pi/2)e_j}(x)\right).$$

One pair of shifted **circuit settings** can provide empirical histograms for all x simultaneously, but learning rare-event probabilities or an exponentially large distribution to a strong norm may still require very many measurement shots. The number of settings is not the number of independent, sufficiently precise probability estimates.

9.6 Expressivity and correlations

Example 9.5 (Circuit size and distribution dimension)**.** Suppose $n = 3$ and $L = 2$. In every layer, apply $R_X(a_i^{(\ell)})$, then $R_Z(b_i^{(\ell)})$ on each qubit, followed by CNOT $0 \to 1$ and CNOT $1 \to 2$. Finish with $R_Z(c_i)$ and $R_X(d_i)$ on each wire, and measure all three qubits. There are $Ln + n = 9$ R_X gates, nine R_Z gates, $L(n-1) = 4$ CNOTs, and $2Ln + 2n = 18$ declared trainable rotation parameters. The Born distribution has eight outcomes.

Removing all CNOT gates makes the final unitary a tensor product $U = \bigotimes_{i=1}^{3} U_i$ and forces $P(x_1x_2x_3) = \prod_i p_i(x_i)$. The model family has only three independent Bernoulli probabilities, although 18 rotation angles are written in the code. A general three-bit distribution has $2^3 - 1 = 7$ independent probabilities. In particular, the correlated target $P(000) = P(111) = 1/2$ with all other probabilities zero cannot be represented by an independent product distribution.

9.7 Quantum adversarial models

For a generator $G_\theta(z)$, discriminator D_ϕ, and latent variable $z \sim p_z$, the classical minimax objective is

$$\min_\theta \max_\phi V(\theta, \phi), \quad V = \mathbb{E}_{x\sim Q} \log D_\phi(x) + \mathbb{E}_{z\sim p_z} \log(1 - D_\phi(G_\theta(z))).$$

During discriminator updates θ is fixed; during generator updates ϕ is fixed. The saturating single-sample generator loss is $\ell_G = \log(1 - D_\phi(G_\theta(z)))$; a frequently used alternative is the nonsaturating loss $-\log D_\phi(G_\theta(z))$, which has a different derivative. A discriminator's labeled-sample binary cross entropy is $-[y \log D(s) + (1-y)\log(1 - D(s))]$.

In patch quantum generators, z is freshly sampled latent noise, potentially encoded by $R_Y(z_i)$, whereas θ denotes optimized angles, potentially in later $R_Y(\theta_i)$ and CZ layers. The two kinds of angles have different statistical roles despite using the same physical gate notation. If a patch circuit has n qubits of which a are ancillary, then a computational-basis distribution on the remaining $n - a$ wires contains 2^{n-a} categories. Under a one-category-per-pixel construction, the number of patches required for $W \times H$ pixels is

$$T = \left\lceil \frac{WH}{2^{n-a}} \right\rceil.$$

For $n = 5$, $a = 1$, and an 8×8 image, there are 16 pixels per patch and $T = 4$ patches, assuming the specified encoding/decoding convention.

Finally, if $g = G_\theta(z) \in \mathbb{R}^d$ enters a differentiable discriminator $D_\phi(g)$, the chain rule gives

$$\frac{\partial \ell_G}{\partial \theta_j} = -\frac{1}{1 - D_\phi(g)} \sum_{k=1}^{d} \frac{\partial D_\phi(g)}{\partial g_k} \frac{\partial G_{\theta,k}(z)}{\partial \theta_j}.$$

If g_k enters the discriminator exactly once through an eligible Pauli rotation, its input derivative may be obtained by shifting the **input angle** g_k, while holding ϕ fixed:

$$\partial_{g_k} D_\phi(g) = \tfrac{1}{2}[D_\phi(g + \tfrac{\pi}{2} e_k) - D_\phi(g - \tfrac{\pi}{2} e_k)].$$

Repeated input re-uploading requires a sum over gate occurrences, and hard discrete sampling may invalidate a naive pathwise derivative of G_θ.

9.8 Generative distributions, statistical losses, and gradient derivations

9.8.1 Born machines and their sampling semantics

A quantum circuit Born machine specifies a parameterized unitary $U(\theta)$ acting on n qubits and a distribution over all 2^n computational basis strings:

$$P_\theta(x) = |\langle x|U(\theta)|0^n\rangle|^2, \qquad x \in \{0,1\}^n.$$

Normalization follows from unitarity:

$$\sum_x P_\theta(x) = \sum_x \langle 0^n|U^\dagger|x\rangle\langle x|U|0^n\rangle = \langle 0^n|U^\dagger I U|0^n\rangle = 1.$$

One computational-basis measurement of the prepared state yields one sample $x \sim P_\theta$, not the entire numerical probability vector. An estimate of all probabilities is obtained by collecting a histogram, and accurate estimation of every low-probability entry can require many shots. The classification of a generative model as "explicit" or "implicit" depends on the formal definition being used. The operational fact about circuit sampling is unambiguous, but it does not by itself imply that an efficient classical likelihood oracle is available; that additional interface must be supplied separately.

9.8.2 Total variation and optimal distinguishing tests

For distributions P, Q on a finite set Ω, total variation is

$$\mathrm{TV}(P, Q) = \frac{1}{2}\sum_{x\in\Omega} |P(x) - Q(x)|.$$

The factor $1/2$ is essential: the sum of positive deviations equals minus the sum of negative deviations because both distributions normalize to one. If $A = \{x : P(x) \geq Q(x)\}$, then

$$P(A) - Q(A) = \sum_{x\in A}(P(x) - Q(x)) = \mathrm{TV}(P, Q).$$

It follows that $\mathrm{TV}(P, Q) = \max_{A\subseteq\Omega} |P(A) - Q(A)|$. Total variation therefore equals the largest possible difference in acceptance probabilities of a deterministic binary test on one draw. Its range is $[0, 1]$: it vanishes precisely when $P = Q$ and equals one for disjoint supports.

Example 9.6 (Four-outcome total variation)**.** Take the four-outcome distributions

$$P = (0.40, 0.10, 0.30, 0.20), \qquad Q = (0.25, 0.25, 0.25, 0.25).$$

The entrywise absolute differences are $0.15, 0.15, 0.05, 0.05$, whose sum is 0.40. Thus $\mathrm{TV}(P, Q) = 0.20$. The maximizing event is the set of first and third outcomes, for which $P(A) = 0.70$ and $Q(A) = 0.50$, again producing 0.20.

When $P = P_\theta$ depends on continuous parameters and $P_\theta(x) \neq Q(x)$ for every x, differentiation gives

$$\partial_j \mathrm{TV}(P_\theta, Q) = \frac{1}{2}\sum_x \mathrm{sgn}(P_\theta(x) - Q(x))\partial_j P_\theta(x).$$

If an entry satisfies $P_\theta(x) = Q(x)$, the absolute-value function is not differentiable there in the ordinary sense; subgradients or one-sided derivatives may be used. Consequently the displayed gradient is a piecewise formula, not an everywhere-smooth identity.

9.8.3 Maximum mean discrepancy as a squared feature-space distance

Let $k : \Omega \times \Omega \to \mathbb{R}$ be a positive-semidefinite kernel with feature map $\phi : \Omega \to \mathcal{H}$. Define the mean embedding $\mu_P = \mathbb{E}_{X\sim P}\phi(X)$, assuming the expectation exists. The squared maximum mean discrepancy is

$$\mathrm{MMD}_k^2(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}}^2.$$

Expanding the squared norm gives the standard three-term formula

$$\mathrm{MMD}_k^2(P, Q) = \mathbb{E}_{X,X'\sim P}k(X, X') + \mathbb{E}_{Y,Y'\sim Q}k(Y, Y') - 2\mathbb{E}_{X\sim P,Y\sim Q}k(X, Y),$$

where X, X' and Y, Y' are independent pairs. The kernel must be chosen before evaluating this loss. A characteristic kernel can distinguish distributions through their mean embeddings; an arbitrary PSD kernel may fail to distinguish some different distributions.

For a finite alphabet the same expression is

$$\mathrm{MMD}_k^2(P, Q) = \sum_{x,y\in\Omega} [P(x) - Q(x)]\, k(x, y)\, [P(y) - Q(y)].$$

Writing the vector of probabilities as p, q and the matrix $K_{xy} = k(x, y)$ gives the compact quadratic form $(p - q)^T K(p - q)$. Positive semidefiniteness ensures the result is nonnegative, even though the expanded three-term expression contains a negative cross term.

9.8.4 Finite-sample estimators and exact pair-counting

Given independent samples $X_1, \ldots, X_n \sim P$ and $Y_1, \ldots, Y_m \sim Q$, a biased but nonnegative sample estimate of the squared discrepancy is

$$\widehat{\mathrm{MMD}}_{\mathrm{b}}^2 = \frac{1}{n^2}\sum_{i,j=1}^{n} k(X_i, X_j) + \frac{1}{m^2}\sum_{i,j=1}^{m} k(Y_i, Y_j) - \frac{2}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} k(X_i, Y_j).$$

When kernel symmetry is exploited, the n^2 terms within the first sample include n diagonal evaluations and $n(n-1)/2$ unordered off-diagonal pairs; the cross term includes nm mixed pairs. For $n = m = 40$, the full *ordered-term arithmetic* involves $1600 + 1600 + 1600 = 4800$ kernel terms, whereas symmetry and any known diagonal formula can reduce the number of *distinct evaluations*. Which count is relevant depends on whether one counts algebraic summands or executed quantum kernel circuits.

An unbiased U-statistic version excludes diagonal within-sample terms:

$$\widehat{\mathrm{MMD}}_{\mathrm{u}}^2 = \frac{1}{n(n-1)}\sum_{i\neq j} k(X_i, X_j) + \frac{1}{m(m-1)}\sum_{i\neq j} k(Y_i, Y_j) - \frac{2}{nm}\sum_{i,j} k(X_i, Y_j).$$

This estimator is unbiased for the population squared discrepancy under independent sampling, but its numerical value on a finite sample need not be nonnegative. The biased and unbiased estimators should not be conflated.

9.8.5 Differentiating MMD without an unnecessary constant term

Let P_θ be a differentiable distribution and Q be fixed. Define

$$L(\theta) = \sum_{x,y} [P_\theta(x) - Q(x)]k(x,y)[P_\theta(y) - Q(y)].$$

Differentiate using the product rule. Symmetry $k(x,y) = k(y,x)$ allows the two resulting terms to be combined:

$$\begin{aligned}\partial_j L &= \sum_{x,y} \partial_j P_\theta(x) k(x,y)[P_\theta(y) - Q(y)] \\ &\quad + \sum_{x,y} [P_\theta(x) - Q(x)]k(x,y)\partial_j P_\theta(y) \\ &= 2\sum_{x,y} \partial_j P_\theta(x) k(x,y)[P_\theta(y) - Q(y)].\end{aligned}$$

The Q–Q term is constant in θ, so its derivative is zero. In matrix notation $\nabla_\theta L = 2J_P(\theta)^T K(p_\theta - q)$, where J_P is the Jacobian of the probability vector. If a textbook defines the unsquared MMD $\sqrt{L}$, differentiation introduces an additional factor $1/(2\sqrt{L})$ away from $L = 0$. Naming the exact loss is therefore essential.

9.8.6 Probability derivatives via shifted circuit distributions

Suppose θ_j occurs in one gate generated by a Pauli involution. For every outcome string x, the projector $|x\rangle\langle x|$ is an observable, so the parameter-shift rule applies to its expectation:

$$\partial_j P_\theta(x) = \frac{1}{2}\left[P_{\theta+\frac{\pi}{2}e_j}(x) - P_{\theta-\frac{\pi}{2}e_j}(x)\right].$$

Substitution into the MMD gradient yields

$$\partial_j L = \sum_{x,y} \left[P_{\theta+\frac{\pi}{2}e_j}(x) - P_{\theta-\frac{\pi}{2}e_j}(x)\right] k(x,y)[P_\theta(y) - Q(y)].$$

Two shifted *settings* are sufficient to define all outcome-probability derivatives for this one circuit occurrence: each setting generates a whole output distribution. However, recovering every probability to a small additive error on an alphabet of size 2^n may still be expensive. Two distinct settings do not imply two measurement shots or dimension-independent reconstruction of all 2^n probabilities.

9.8.7 A correlated two-qubit distribution and an expressivity obstruction

For an output state that is a product $|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$, computational-basis probabilities factorize:

$$P(x_1, \ldots, x_n) = \prod_{j=1}^{n} P_j(x_j).$$

Suppose the target distribution has $P(00) = P(11) = 1/2$ and $P(01) = P(10) = 0$. The marginals are both unbiased, hence any product distribution matching these marginals is uniform on all four strings, and cannot reproduce the desired correlations. More directly, if a product distribution assigns positive mass to both 00 and 11, then each single-qubit marginal assigns positive mass to both 0 and 1, so 01 and 10 must also have positive mass. Thus an architecture containing only local single-qubit gates acting on a product input cannot represent this target exactly. Entangling gates are one route to correlated output distributions, though their presence alone does not guarantee sufficient expressivity for every distribution.

9.8.8 Gate and parameter counts in a layered Born machine

Consider $n = 3$ qubits and $L = 2$ repeated layers. In each layer, every qubit receives an R_X and then an R_Z rotation, followed by a chain of CNOT gates $0 \to 1$ and $1 \to 2$. After the two layers, each qubit receives a final R_Z and then an R_X, and all qubits are measured. The counts are

$$\#R_X = Ln + n = 9, \qquad \#R_Z = Ln + n = 9, \qquad \#\mathrm{CNOT} = L(n-1) = 4.$$

With independent trainable angles for every rotation, the parameter count is $2Ln + 2n = 18$. The probability output has $2^3 = 8$ entries, indexed by $000, 001, \ldots, 111$. A shot budget of $S = 1000$ returns a multinomial histogram; it does not turn the eight probabilities into exact real numbers.

9.8.9 Adversarial objectives and their alternating optimization

A generative adversarial system consists of a generator distribution P_θ and a discriminator $D_\phi(x) \in (0, 1)$. A classical minimax objective has the form

$$\min_\theta \max_\phi V(\theta, \phi), \quad V = \mathbb{E}_{x \sim P_{\text{data}}} \log D_\phi(x) + \mathbb{E}_{z \sim P_z} \log(1 - D_\phi(G_\theta(z))).$$

The discriminator step updates ϕ with θ fixed; the generator step updates θ with ϕ fixed. If either component is implemented by quantum circuits, its input-output interface and differentiability assumptions must be specified: a circuit returning a differentiable expectation is not mathematically the same as a circuit returning an unprocessed discrete sample.

Let $g = G_\theta(z)$ be a differentiable generator output vector and let the discriminator receive g through circuit rotation angles. For the single-sample generator loss $\ell = \log(1 - D_\phi(g))$, the chain rule gives

$$\frac{\partial \ell}{\partial \theta_j} = -\frac{1}{1 - D_\phi(g)} \sum_k \frac{\partial D_\phi(g)}{\partial g_k} \frac{\partial G_{\theta,k}(z)}{\partial \theta_j}.$$

If g_k occurs once as a Pauli rotation parameter, then

$$\frac{\partial D_\phi(g)}{\partial g_k} = \tfrac{1}{2}[D_\phi(g + \tfrac{\pi}{2} e_k) - D_\phi(g - \tfrac{\pi}{2} e_k)].$$

The shift is applied to the *input angle* g_k rather than to a discriminator weight ϕ_k. Multiple uses of the same g_k require accounting for each gate occurrence. If g is formed only by hard discrete sampling, the displayed pathwise chain rule generally does not apply without an additional gradient-estimation construction.

9.8.10 Patch generators and output dimension arithmetic

Suppose a quantum sub-generator acts on n qubits, of which a are ancillae removed or conditioned on before the remaining measurement output is interpreted as pixel probabilities. The accessible computational-basis distribution of $n - a$ output qubits has 2^{n-a} coordinates. Under the model in which one basis-state probability produces one pixel intensity, a single sub-generator supplies 2^{n-a} pixel entries. To fill a $W \times H$ image requires

$$T = \left\lceil \frac{WH}{2^{n-a}} \right\rceil$$

sub-generators when partial final patches are allowed. With $n = 5$, $a = 1$, and $W = H = 8$, one patch contains $2^4 = 16$ entries and $T = \lceil 64/16 \rceil = 4$ patches suffice. This is a dimensional accounting identity for that image-encoding scheme, not a general lower bound on all possible quantum image generators.

9.9 Numerical losses and explicitly differentiable Born distributions

9.9.1 Total variation and MMD can assign different scales

For distributions on four outcomes, take

$$P = (0.40, 0.10, 0.30, 0.20), \quad Q = (0.25, 0.25, 0.25, 0.25).$$

Their total variation distance equals 0.20. If the MMD kernel is the Kronecker delta $k(x, y) = \mathbf{1}_{x=y}$, then its Gram matrix on the alphabet is the identity and

$$\mathrm{MMD}_k^2(P, Q) = \sum_x (P(x) - Q(x))^2 = 0.15^2 + (-0.15)^2 + 0.05^2 + (-0.05)^2 = 0.05.$$

These two losses are not numerically interchangeable: TV is an ℓ_1 discrepancy with factor $1/2$, while delta-kernel MMD squared is an ℓ_2^2 discrepancy. In particular, the probability difference 0.15 contributes linearly to TV and quadratically to the delta-kernel MMD. Changing the kernel changes the MMD even when P, Q are fixed, whereas TV is uniquely determined by the distributions.

9.9.2 A one-qubit Born machine with an exact loss gradient

Let $U(\theta) = R_Y(\theta)$ on initial $|0\rangle$. Its outcome probabilities are

$$P_\theta(0) = \cos^2(\theta/2) = \frac{1+\cos\theta}{2}, \qquad P_\theta(1) = \sin^2(\theta/2) = \frac{1-\cos\theta}{2}.$$

Fix $Q = (1/2, 1/2)$ and take the delta kernel. Then

$$L(\theta) = \mathrm{MMD}_\delta^2(P_\theta, Q) = \left(\frac{\cos\theta}{2}\right)^2 + \left(-\frac{\cos\theta}{2}\right)^2 = \frac{1}{2}\cos^2\theta.$$

Ordinary differentiation yields $L'(\theta) = -\sin\theta\cos\theta = -\frac{1}{2}\sin 2\theta$. The probability derivatives obtained by parameter shift are

$$P'_\theta(0) = -\frac{1}{2}\sin\theta, \qquad P'_\theta(1) = \frac{1}{2}\sin\theta.$$

Substituting them in $L' = 2\sum_x P'_\theta(x)[P_\theta(x) - Q(x)]$ reproduces the exact formula. This checks the factor two in the MMD gradient and confirms that probability normalization implies $\sum_x P'_\theta(x) = 0$.

At $\theta = \pi/2$, $P_\theta = Q$ and the squared MMD vanishes; its gradient also vanishes. At $\theta = 0$, $P_\theta = (1, 0)$ and $L = 1/2$, but the derivative is again zero because $\theta = 0$ is a stationary point of the chosen periodic parametrization. A zero gradient at a point therefore does not by itself mean that the distribution has converged to the target.

9.9.3 Distributions with matching marginals but incompatible correlations

Consider $P(00) = P(11) = 1/2$ and $P(01) = P(10) = 0$. Both marginal distributions are unbiased, so the product of the marginals is uniform with probability $1/4$ for all four outcomes. The total variation between the correlated target and that product distribution is

$$\frac{1}{2}\big(|1/2 - 1/4| + |0 - 1/4| + |0 - 1/4| + |1/2 - 1/4|\big) = \frac{1}{2}.$$

The target has mutual information one bit: $I(X : Y) = H(X) + H(Y) - H(X, Y) = 1+1-1 = 1$. The product distribution has mutual information zero. This information-theoretic discrepancy makes the expressive limitation of a purely product-state Born machine concrete. Entangling operations are needed to produce this target from an initially product state using only local unitaries before computational-basis measurement.

A two-qubit Bell state $|\Phi^+\rangle$ realizes the target directly: its computational-basis measurement returns 00 or 11, each with probability one half. The circuit H on the first qubit followed by CNOT with first-qubit control prepares this state. If the CNOT is removed, the resulting state is $|+0\rangle$, with outcomes 00 and 10 instead. This is a specific expressivity example; it does not imply that one entangling gate suffices for arbitrary high-dimensional correlated distributions.

9.9.4 A finite-sample MMD pair-count audit

For n samples from P and m samples from Q, the biased MMD statistic contains $n^2 + m^2 + nm$ written kernel summands when the factor two is kept as a scalar rather than duplicate cross terms. If one expands the -2 coefficient into two identical mixed sums, one obtains $n^2 + m^2 + 2nm = (n + m)^2$ algebraic terms. For $n = m = 40$, these count conventions give 4800 and 6400 terms respectively; they describe the same arithmetic expression, not different statistical estimators. The actual number of distinct kernel evaluations is smaller when symmetry is exploited, and the diagonal may be known analytically. In a computational resource estimate one must explicitly state whether the object being counted is written summands, ordered pairs, unique pairs, or executed circuit settings.

9.9.5 A patch-construction example beyond divisible dimensions

Suppose $n - a = 4$ output qubits are used per patch, giving 16 values. For an 8×8 image one needs exactly four patches. For a 10×10 image, $100/16 = 6.25$, so the ceiling formula gives $T = 7$ patches and the seventh has only four useful values under the same direct coordinate assignment. If every patch instead outputs a spatially structured subset with overlap or uses postprocessing to combine amplitudes into pixel intensities, the arithmetic changes. The formula $\lceil WH/2^{n-a} \rceil$ is therefore conditional on the one-basis-state-to-one-output-coordinate model, not a universal property of generative quantum networks.

9.10 Gradient estimators, computational costs, and objective geometry

9.10.1 The MMD gradient as a kernel-weighted signed measure

Let Ω be a finite set and $p_\theta, q \in \mathbb{R}^{|\Omega|}$ be probability vectors. If K is a symmetric positive-semidefinite kernel matrix, then

$$L(\theta) = (p_\theta - q)^T K (p_\theta - q).$$

Writing $J(\theta) = \partial p_\theta / \partial \theta \in \mathbb{R}^{|\Omega| \times P}$, differentiation gives

$$\nabla_\theta L(\theta) = 2 J(\theta)^T K (p_\theta - q).$$

This exhibits a useful two-stage interpretation: first compute the discrepancy vector $r = K(p_\theta - q)$, then project the discrepancy onto the tangent directions $\partial_j p_\theta$ that the model can realize. Even when $p_\theta \neq q$, the gradient can vanish if every allowed tangent direction is orthogonal to r. Thus stationarity of a parameterized MMD objective does not imply equality of distributions, especially when the model's expressivity is restricted.

Because every probability vector sums to one, differentiating $\mathbf{1}^T p_\theta = 1$ gives $\mathbf{1}^T \partial_j p_\theta = 0$. Consequently adding a constant vector to the discrepancy potential r does not change the gradient. Indeed $\sum_x \partial_j p_\theta(x) c = 0$ for any scalar c. This is a mathematical reason that kernels differing by certain constant-offset contributions can produce related parameter-gradient expressions, even though their raw MMD values may differ under more general formulations.

9.10.2 An unbiased two-shift estimate for a linear functional of the Born distribution

Fix any bounded function $h : \Omega \to [-1, 1]$ and define $F(\theta) = \sum_x h(x) P_\theta(x) = \mathbb{E}_{X \sim P_\theta}[h(X)]$. If parameter θ_j occurs in one Pauli-generated rotation, the shift rule gives

$$\partial_j F = \tfrac{1}{2} \left(F(\theta + \pi e_j/2) - F(\theta - \pi e_j/2) \right).$$

A shot-based unbiased estimator uses independent samples $X_1^+, \ldots, X_S^+$ from the plus-shift distribution and $X_1^-, \ldots, X_S^-$ from the minus-shift distribution:

$$\widehat{\partial_j F} = \frac{1}{2S} \sum_{s=1}^{S} [h(X_s^+) - h(X_s^-)].$$

Its variance is $[\mathrm{Var}_{P_+}(h) + \mathrm{Var}_{P_-}(h)]/(4S) \le 1/(2S)$. The estimator does not reconstruct either complete 2^n-component probability vector. This observation can be used when an MMD gradient is expressed as an expectation of a kernel-weighted potential, although dependence of that potential on p_θ introduces additional estimation and sample-splitting considerations.

9.10.3 Pair sampling for MMD without forming a full probability vector

If P_θ is available through sampling, the population MMD squared can be estimated by drawing independent pairs $(X, X') \sim P_\theta \times P_\theta$, $(Y, Y') \sim Q \times Q$, and $(X, Y) \sim P_\theta \times Q$. A Monte Carlo estimator of the three expectations avoids enumerating all $|\Omega|^2$ outcome pairs. For a bounded kernel $|k(x, y)| \le 1$, each sampled kernel value has variance at most one. Under independent sampling of the three terms, a budget of order ε^{-2} samples per term gives constant-probability additive accuracy $O(\varepsilon)$ by a basic variance or concentration argument. This does not remove the circuit cost of generating each quantum sample or computing each kernel value, and it does not automatically yield a gradient estimator with the same variance. Nevertheless it explains why writing MMD as a double sum over all 2^n outcomes is a mathematical identity rather than a mandate to construct an exponentially large classical array.

9.10.4 The total-variation subgradient at matching probabilities

For a scalar difference $d(\theta) = P_\theta(x) - Q(x)$, the derivative of $|d(\theta)|$ is $\operatorname{sgn}(d)d'$ when $d \neq 0$. At a point where $d = 0$, the absolute-value function has subdifferential $[-1, 1]$, and the subgradient contribution can be selected as $\xi d'$ with $\xi \in [-1, 1]$ under the appropriate chain-rule conditions. Thus a symbolic expression involving $\operatorname{sgn}(0)$ requires an explicit convention or subgradient treatment. Using $\operatorname{sgn}(0) = 0$ selects one possible subgradient, but it does not turn the original objective into a differentiable function at that point.

Compare the one-qubit distribution $P_\theta = (\frac{1+\cos\theta}{2}, \frac{1-\cos\theta}{2})$ to the unbiased target $Q = (1/2, 1/2)$. Then

$$\mathrm{TV}(P_\theta, Q) = \frac{1}{2}|\cos\theta|.$$

For $\cos\theta \neq 0$, its derivative is $-\frac{1}{2}\sin\theta\operatorname{sgn}(\cos\theta)$. At $\theta = \pi/2$, left and right derivatives disagree, whereas the delta-kernel MMD squared from the same pair equals $\frac{1}{2}\cos^2\theta$ and has a smooth zero derivative. This example shows that the choice of distributional loss affects optimization geometry even when the minimizing distribution is the same.

9.10.5 Discriminator gradient propagation through classical preprocessing

Suppose a generator produces a differentiable feature vector $g = G_\theta(z)$, which is mapped to discriminator angles $a(g)$ before entering a quantum circuit. The discriminator expectation is $D_\phi(a(g))$. The full chain rule is

$$\frac{\partial D_\phi}{\partial \theta_j} = \sum_{k,r} \frac{\partial D_\phi}{\partial a_k}\frac{\partial a_k}{\partial g_r}\frac{\partial G_{\theta,r}}{\partial \theta_j}.$$

A parameter-shift rule can evaluate $\partial D_\phi/\partial a_k$ when a_k is a qualifying single gate angle. The classical Jacobian $\partial a_k/\partial g_r$ must still be included; it is not automatically the identity when the generator output is normalized, rescaled, or passed through a nonlinear embedding. Finally the generator Jacobian $\partial G_{\theta,r}/\partial\theta_j$ may itself require shifted quantum evaluations if the generator output is an expectation. Thus an end-to-end QGAN gradient can involve multiple nested differentiation interfaces, even when each individual quantum derivative admits a simple two-shift representation.

9.10.6 A parameter count distinct from the dimension of the output simplex

An n-qubit Born machine produces 2^n probabilities subject to one normalization constraint, so the ambient probability simplex has real dimension $2^n - 1$. An architecture with P smoothly varying real parameters can locally trace at most a P-dimensional

manifold in this simplex, and possibly less when parameters are redundant. If $P < 2^n - 1$, a generic full-dimensional family of distributions cannot be covered by this smooth model without further structural mechanisms. This dimension comparison is only a local necessary observation; it does not by itself quantify expressivity on a restricted target family or prove that an architecture with $P \geq 2^n - 1$ can represent every distribution.

CHAPTER 10

Quantum Numerical Linear Algebra: Access Models and Block-Encodings

10.1 Quantum input and output models

Let $N = 2^n$. A nonzero vector $v \in \mathbb{C}^N$ is represented by the normalized amplitude state

$$|v\rangle = \frac{1}{\|v\|_2} \sum_{j=0}^{N-1} v_j |j\rangle.$$

For example, $v = (2, 1, 0, 0)^\top$ is represented by $(2|00\rangle + |01\rangle)/\sqrt{5}$. An input-access assumption may supply a state-preparation unitary O_v such that $O_v|0^n\rangle = |v\rangle$. Efficient access is not automatic for arbitrary dense vectors. In particular, using $\log_2 N$ physical qubits does not imply that loading N classical complex numbers takes $\mathrm{poly}(\log N)$ time.

Quantum numerical linear algebra must specify at least three resources: the cost of preparing $|v\rangle$, the cost of querying a matrix access oracle, and the cost of extracting the requested classical output information. Query complexity, fault-tolerant gate complexity, and measurement-readout complexity are not interchangeable.

For a normalized state $|\psi\rangle = \sum_j c_j |j\rangle$, computational-basis measurement yields label j with probability $|c_j|^2$; for a Hermitian observable O, $\langle O\rangle_\psi = \langle\psi|O|\psi\rangle$. A quantum algorithm preparing the state proportional to $A^{-1}b$ does not thereby print the full classical vector $A^{-1}b$.

10.2 Matrix block-encoding

Definition 10.1 (Exact (α, a) block-encoding)**.** For $A \in \mathbb{C}^{N\times N}$, an (α, a) block-encoding is a unitary U_A acting on a ancilla qubits and the n-qubit system such that

$$(\langle 0^a| \otimes I_N) U_A (|0^a\rangle \otimes I_N) = A/\alpha, \qquad \alpha > 0.$$

An approximate block-encoding permits the left-hand side to differ from A/α in operator norm by a specified error.

In ancilla-block form, $U_A = \begin{pmatrix} A/\alpha & B \\ C & D \end{pmatrix}$. The other blocks are constrained by unitarity and cannot usually be omitted.

Proposition 10.2 (Necessary normalization)**.** *Every exact block-encoding satisfies* $\alpha \geq \|A\|$*, where* $\|\cdot\|$ *is the spectral operator norm.*

Proof. For every normalized $|u\rangle$, the quantity $(A/\alpha)|u\rangle$ is the projection of the normalized state $U_A|0^a\rangle|u\rangle$ onto the ancilla-zero subspace. Orthogonal projection cannot increase norm, so $\|(A/\alpha)|u\rangle\| \leq 1$. Taking the supremum over $|u\rangle$ gives $\|A\|/\alpha \leq 1$. □

10.3 Postselection and coherent success amplification

Applying U_A to the prepared input yields an orthogonal decomposition

$$U_A|0^a\rangle|u\rangle = |0^a\rangle\frac{A|u\rangle}{\alpha} + |\text{bad}\rangle, \qquad (\langle 0^a| \otimes I)|\text{bad}\rangle = 0.$$

The probability of measuring all-zero ancillas is

$$p_{\text{succ}} = \frac{\|A|u\rangle\|_2^2}{\alpha^2},$$

and on success the conditional system state is $A|u\rangle/\|A|u\rangle\|$. Independent direct repetition requires expected $1/p_{\text{succ}}$ attempts, whereas suitable amplitude amplification (when the requisite reflections are available) requires $O(p_{\text{succ}}^{-1/2})$ coherent invocations. If $A|u\rangle = 0$, success probability is zero and no normalized output exists. For $\alpha = 4$ and $\|A|u\rangle\| = 1$, the success probability is $1/16$, direct expectation is 16, and coherent amplification scale is $O(4)$.

For a Hermitian contraction $A = A^\dagger$ with $\|A\| \leq 1$, set $B = \sqrt{I - A^2}$. The explicit matrix

$$U_A^{\text{dil}} = \begin{pmatrix} A & B \\ B & -A \end{pmatrix}$$

is both Hermitian and unitary, since A and B commute and $A^2 + B^2 = I$. This exhibits an exact block-encoding in principle, but it is not a proof that a generic dense A has an efficient gate implementation.

10.4 Linear combination of unitaries

For accessible U_0, U_1, prepare $|0\rangle|u\rangle$, apply Hadamard to the ancilla, controlled-U_0/U_1 (SELECT), and a final Hadamard. The state is

$$\frac{|0\rangle}{2}(U_0 + U_1)|u\rangle + \frac{|1\rangle}{2}(U_0 - U_1)|u\rangle.$$

The success probability for outcome zero is $\|(U_0 + U_1)|u\rangle\|^2/4$, and the successful state is its normalized sum. In the special case $U_0 = I$, $U_1 = Z$, and $|u\rangle = |+\rangle$,

$$\frac{I + Z}{2}|+\rangle = \frac{|0\rangle}{\sqrt{2}},$$

so success probability is $1/2$ and the postselected state is $|0\rangle$.

Proposition 10.3 (General LCU block-encoding)**.** *Let* $S = \sum_{j=0}^{m-1} c_j U_j$ *with* $c_j \geq 0$ *and* $C = \sum_j c_j > 0$*. Suppose a unitary PREPARE maps* $|0\rangle$ *to* $|\chi\rangle = \sum_j \sqrt{c_j/C}|j\rangle$*, and SELECT is* $\sum_j |j\rangle\langle j| \otimes U_j$*. Then*

$$U_S = (\text{PREPARE}^\dagger \otimes I)\,\text{SELECT}\,(\text{PREPARE} \otimes I)$$

block-encodes S with normalization C.

Proof. The all-zero ancilla block is $\langle\chi|\text{SELECT}|\chi\rangle = \sum_j (c_j/C)U_j = S/C$. □

Complex scalar phases can be absorbed into the unitaries. The postselection probability is $\|S|u\rangle\|^2/C^2$, and suitable amplitude amplification incurs $O(C/\|S|u\rangle\|)$ coherent attempts. The costs of PREPARE, SELECT, and their inverses remain part of the full implementation.

10.5 Products and ancilla reuse

Suppose $U_0 = \begin{pmatrix} A_0/\alpha_0 & B_0 \\ C_0 & D_0 \end{pmatrix}$ and $U_1 = \begin{pmatrix} A_1/\alpha_1 & B_1 \\ C_1 & D_1 \end{pmatrix}$ act on the **same** ancilla. The upper-left block of U_1U_0 equals

$$\frac{A_1A_0}{\alpha_1\alpha_0} + B_1C_0,$$

not necessarily $A_1A_0/(\alpha_1\alpha_0)$. The cross-term represents amplitude returning from the failure subspace. Using independent ancilla registers and projecting each onto zero avoids this interference and produces a product block-encoding with normalization $\alpha_1\alpha_0$, subject to the indicated circuit construction.

10.6 Interference, normalization, and hybrid LCU estimators

For normalized $|a\rangle = U_0|u\rangle$ and $|b\rangle = U_1|u\rangle$, the balanced two-term LCU success probability is

$$p_0 = \left\| \frac{|a\rangle + |b\rangle}{2} \right\|^2 = \frac{1 + \text{Re}\langle a|b\rangle}{2}.$$

Orthogonal outputs give $p_0 = 1/2$ and conditional state $(|a\rangle + |b\rangle)/\sqrt{2}$; equal outputs give certainty; opposite outputs produce destructive cancellation and zero success probability. If the same matrix A is block-encoded with a larger normalization $r\alpha$ $(r > 1)$, its postselection success probability decreases by a factor r^2, its direct repetition cost grows by r^2, and amplitude-amplification calls grow linearly with r.

A related **hybrid LCU** method estimates expectation values without coherently preparing the entire linear combination. For $u = \sum_j c_j U_j |u_0\rangle$, the unnormalized quadratic form satisfies

$$\langle u|M|u\rangle = \sum_{j,k} \overline{c_j} c_k \langle u_0|U_j^\dagger M U_k|u_0\rangle.$$

When $c_j \geq 0$, let $C = \sum_j c_j$ and $q_j = c_j/C$. Then

$$\langle u|M|u\rangle = C^2 \, \mathbb{E}_{j,k\sim q}[\langle u_0|U_j^\dagger M U_k|u_0\rangle].$$

Sampling index pairs and estimating cross terms gives an alternative estimator. The normalized expectation $\langle u|M|u\rangle/\langle u|u\rangle$ also requires estimating its denominator. Avoiding coherent LCU preparation exchanges circuit requirements for statistical sampling costs rather than eliminating all costs.

10.7 Block-encoding as a computational interface: exact calculations

10.7.1 State access, matrix access, and what the algorithm actually returns

Let $N = 2^n$. Given a nonzero vector $b \in \mathbb{C}^N$, an amplitude-encoding state-preparation oracle O_b produces

$$O_b|0^n\rangle = |b\rangle := \frac{1}{\|b\|_2} \sum_{j=0}^{N-1} b_j |j\rangle.$$

Storage in $n = \log_2 N$ qubits does not guarantee that preparing an arbitrary specified N-component vector is efficient. In an oracle-complexity analysis, the cost of implementing O_b may be left symbolic. In a gate-complexity analysis, it must eventually be accounted for. Similarly, the ability to manipulate $|x\rangle$ does not mean that one can read all components x_j in a single experiment. A single measurement returns a random basis index, and expectation estimation typically requires repeated preparation of the state. These distinctions delimit the actual input/output promise of a quantum linear-algebra algorithm.

10.7.2 Block-encoding and its operator-norm constraint

Definition 10.4 (Exact block-encoding)**.** For $A \in \mathbb{C}^{N\times N}$, a unitary U_A on $a + n$ qubits is an $(\alpha, a, 0)$ block-encoding of A if

$$(\langle 0^a| \otimes I_N) U_A (|0^a\rangle \otimes I_N) = A/\alpha, \qquad \alpha > 0.$$

An (α, a, ε) approximate block-encoding allows operator-norm error at most ε in this identity.

The block-encoding property is not the statement $U_A = A/\alpha$ on the entire Hilbert space. In block notation one may write $U_A = \begin{pmatrix} A/\alpha & B \\ C & D \end{pmatrix}$, where the unknown blocks must collectively satisfy unitarity. A single upper-left block need not be unitary and can represent a nonunitary matrix transformation through postselection.

Proposition 10.5 (Necessary normalization factor)**.** *Every exact block-encoding satisfies* $\alpha \geq \|A\|_2$*, with* $\|A\|_2$ *the spectral/operator norm.*

Proof. For every normalized $|u\rangle$, the vector $|0^a\rangle \otimes |u\rangle$ has norm one. Unitarity preserves that norm, while projection onto $|0^a\rangle$ cannot increase it. Therefore $\|A|u\rangle/\alpha\|_2 \leq 1$. Taking the supremum over $\|u\|_2 = 1$ yields $\|A\|_2/\alpha \leq 1$. □

For an approximate block-encoding, $\|A\|_2/\alpha \leq 1 + \varepsilon$ rather than necessarily ≤ 1. The distinction matters when propagating approximation errors through multiple transformations.

10.7.3 An explicit unitary dilation for a Hermitian contraction

Let $A = A^\dagger$ and $\|A\|_2 \leq 1$. Functional calculus defines the positive operator $B = \sqrt{I - A^2}$. Since B is a function of A, $AB = BA$. Then

$$U = \begin{pmatrix} A & B \\ B & -A \end{pmatrix}$$

is Hermitian and unitary: squaring gives $U^2 = \begin{pmatrix} A^2 + B^2 & AB - BA \\ BA - AB & A^2 + B^2 \end{pmatrix} = I$. Its top-left block is exactly A, so U provides a one-ancilla existence proof of a $(1, 1, 0)$ block-encoding. But evaluating $\sqrt{I - A^2}$ and compiling this full unitary for a generic dense matrix may itself be expensive; an existence construction is not an efficient implementation theorem.

Example 10.6 (A scalar dilation)**.** Take $A = [3/5]$. Then $B = [4/5]$ and

$$U = \begin{pmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{pmatrix}.$$

Direct multiplication gives $U^2 = I$. Starting from ancilla state $|0\rangle$, the amplitude of returning to $|0\rangle$ is $3/5$ and the corresponding postselection probability is $9/25$. This simple matrix contains the essential distinction between a block amplitude and its squared probability.

10.7.4 The postselected output and its success probability

Suppose U_A is an exact $(\alpha, a, 0)$ block-encoding and $|u\rangle$ is a normalized input. The state after applying U_A to $|0^a\rangle|u\rangle$ decomposes orthogonally as

$$U_A|0^a\rangle|u\rangle = |0^a\rangle \frac{A|u\rangle}{\alpha} + |\text{bad}\rangle,$$

where $(\langle 0^a| \otimes I)|\text{bad}\rangle = 0$. Thus

$$p_{\text{succ}} = \frac{\||A|u\rangle\|_2^2}{\alpha^2}.$$

If $A|u\rangle \neq 0$, the normalized successful output is $A|u\rangle/\||A|u\rangle\|_2$. The inverse success probability is the expected number of independent repetitions until the first success. Under coherent amplitude-amplification assumptions, the cost can instead scale as $O(1/\sqrt{p_{\text{succ}}})$ uses of the relevant state-preparation and reflection operations. Neither statement includes the internal gate cost of a block-encoding call unless explicitly multiplied in.

Example 10.7 (Postselection with normalization factor four)**.** Suppose $\alpha = 4$ and $\||A|u\rangle\|_2 = 1$. The good branch has norm $1/4$ and probability $1/16$. Direct repeated trials therefore require an expected 16 uses of the experiment for one successful output. Amplitude amplification may reduce the coherent repetition scaling to order 4. Writing "probability $1/4$" would confuse a branch amplitude with its squared norm.

10.7.5 Interference in the two-unitary LCU construction

Given unitaries U_0, U_1, prepare an ancilla $|0\rangle$, apply H, apply SELECT $= |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1$, and then apply H again. On input $|u\rangle$, the three states are

$$\begin{aligned} |0\rangle|u\rangle &\longmapsto \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|u\rangle \\ &\longmapsto \tfrac{1}{\sqrt{2}}(|0\rangle U_0|u\rangle + |1\rangle U_1|u\rangle) \\ &\longmapsto \tfrac{1}{2}|0\rangle(U_0 + U_1)|u\rangle + \tfrac{1}{2}|1\rangle(U_0 - U_1)|u\rangle. \end{aligned}$$

Therefore the good ancilla branch implements $(U_0 + U_1)/2$. Its probability is

$$p_0 = \frac{1}{4}\|(U_0 + U_1)|u\rangle\|_2^2 = \frac{1 + \text{Re}\langle u|U_0^\dagger U_1|u\rangle}{2}.$$

The interference term determines the success probability. If the two transformed states are equal, then $p_0 = 1$; if they are opposite, $p_0 = 0$; if orthogonal, then $p_0 = 1/2$.

Example 10.8 (Constructing a rank-one projector by LCU)**.** Take $U_0 = I$, $U_1 = Z$ on one qubit, and input $|u\rangle = |+\rangle$. The good branch is $(I + Z)|+\rangle/2 = |0\rangle/\sqrt{2}$, so $p_0 = 1/2$ and the normalized conditional state is $|0\rangle$. Since $(I + Z)/2 = |0\rangle\langle 0|$, the LCU gadget has implemented a nonunitary projection as a unitary subblock. The bad branch is $(I - Z)|+\rangle/2 = |1\rangle/\sqrt{2}$; its norm squared is the complementary $1/2$.

10.7.6 The general PREPARE–SELECT–UNPREPARE identity

Suppose $S = \sum_{j=0}^{m-1} c_j U_j$ with real $c_j \geq 0$ and $C = \sum_j c_j > 0$. Let PREPARE map the ancilla zero state to $|\chi\rangle = \sum_j \sqrt{c_j/C}|j\rangle$ and define SELECT $= \sum_j |j\rangle\langle j| \otimes U_j$. The

composite circuit $W = (\mathrm{PREPARE}^\dagger \otimes I)\,\mathrm{SELECT}(\mathrm{PREPARE} \otimes I)$ has top-left block

$$(\langle 0| \otimes I)W(|0\rangle \otimes I) = \langle \chi|\,\mathrm{SELECT}\,|\chi\rangle = \sum_j \frac{c_j}{C} U_j = \frac{S}{C}.$$

If a coefficient c_j is complex, write $c_j = |c_j|e^{i\phi_j}$ and absorb $e^{i\phi_j}$ into the unitary U_j. Hence the same construction covers complex linear combinations, with normalization $C = \sum_j |c_j|$. A successful postselection on input $|u\rangle$ has probability $\|S|u\rangle\|_2^2/C^2$. The costs of PREPARE and SELECT must be included when translating this query construction into an actual gate count.

10.7.7 Why reusing a product ancilla can corrupt a block-encoding

Let two block-encodings, written relative to the same ancilla decomposition, be

$$U_0 = \begin{pmatrix} A_0/\alpha_0 & B_0 \\ C_0 & D_0 \end{pmatrix}, \qquad U_1 = \begin{pmatrix} A_1/\alpha_1 & B_1 \\ C_1 & D_1 \end{pmatrix}.$$

Then the upper-left block of their product is

$$(U_1U_0)_{00} = \frac{A_1A_0}{\alpha_1\alpha_0} + B_1C_0.$$

The extra term represents an amplitude that leaves the selected ancilla subspace during the first operation and returns during the second. This is not the desired matrix product. With two *separate* ancilla registers and the appropriate pair of zero projections, a standard product construction isolates the intended $A_1A_0/(\alpha_1\alpha_0)$ block. The correct implementation therefore depends on tracking which ancillary subspaces are projected and which failure branches may interfere.

Example 10.9 (A scalar counterexample to naive multiplication)**.** Let $U_0 = U_1 = H$ and use a single ancilla with no system. Each has top-left scalar block $1/\sqrt{2}$. Their desired block product is $1/2$. But $H^2 = I$, whose upper-left element is 1. The difference $1 - 1/2 = 1/2$ is exactly the contribution from the off-diagonal excursion B_1C_0. Thus the equality "product of block-encodings encodes product of blocks" is false if the same ancilla is reused without an appropriate construction.

10.7.8 Estimating observables of an LCU state without preparing it coherently

Consider the unnormalized vector $|v\rangle = \sum_j c_jU_j|u_0\rangle$ and a Hermitian observable M. Its quadratic expectation numerator expands as

$$\langle v|M|v\rangle = \sum_{j,k} c_j^* c_k \langle u_0|U_j^\dagger M U_k|u_0\rangle.$$

For nonnegative real coefficients, set $C = \sum_j c_j$ and sample J, K independently with probabilities $q_j = c_j/C$. Then

$$\langle v|M|v\rangle = C^2\,\mathbb{E}_{J,K}\left[\langle u_0|U_J^\dagger M U_K|u_0\rangle\right].$$

The generally complex cross term can be estimated by a suitable interference circuit (such as a Hadamard test), treating the real and imaginary parts separately. Normalization requires another quadratic form, $\langle v|v\rangle$, and a ratio estimator may inherit substantial variance if the norm is small. This hybrid LCU identity replaces coherent preparation of the entire coefficient superposition by stochastic index selection for certain expectation-estimation tasks; it does not directly output the pure state $|v\rangle/\|v\|$.

10.7.9 Resource accounting and assumptions

A block-encoding-based linear-algebra algorithm should report at least four distinct quantities: the number of calls to U_A or $U_A^\dagger$; the cost of preparing the state $|b\rangle$; the amplitude-amplification or postselection overhead; and the number of output measurements required for the desired classical statistic. Oracle query counts can be asymptotically small even when the state-loading or readout cost dominates. Conversely, when the output is consumed coherently by another quantum routine, reconstructing a full classical vector may be neither necessary nor meaningful. The output specification is therefore part of the formal problem, not a matter of presentation.

10.8 Matrix-embedding and LCU examples

10.8.1 The singular-value obstruction to unscaled embeddings

Take a diagonal Hermitian matrix $A = \mathrm{diag}(3, 1/2)$. Its spectral norm is three. No unitary can have A as a top-left subblock at scale one because the input basis vector $|0\rangle$ would be mapped to a selected block component with norm three, contradicting norm preservation. Any exact block-encoding requires $\alpha \geq 3$. Choosing $\alpha = 3$ gives the contraction $B = A/3 = \mathrm{diag}(1, 1/6)$, for which the Hermitian dilation

$$U_B = \begin{pmatrix} B & \sqrt{I - B^2} \\ \sqrt{I - B^2} & -B \end{pmatrix}$$

exists. On input $|1\rangle$ with a zero ancilla, the success probability is $\|B|1\rangle\|^2 = 1/36$, while on input $|0\rangle$ it is one. The matrix and the normalization factor are the same in both experiments; the postselection probabilities differ because of the input direction.

10.8.2 Two-unitary interference with a tunable relative phase

Let $U_0 = I$ and $U_1 = e^{i\varphi} I$ in the balanced one-ancilla LCU circuit. Then the good branch is $\frac{1}{2}(1 + e^{i\varphi})|u\rangle$, which simplifies to $e^{i\varphi/2}\cos(\varphi/2)|u\rangle$. Its probability is

$$p_0 = \cos^2(\varphi/2).$$

At $\varphi = 0$, constructive interference gives success probability one; at $\varphi = \pi$, destructive interference gives zero. At $\varphi = \pi/2$, success is one half. This example emphasizes that LCU postselection is governed by coherent interference between operator branches. An estimate based only on the norms $\|U_0 u\|$ and $\|U_1 u\|$, both equal to one, misses the entire dependence on their relative phase.

10.8.3 A three-term LCU including its normalization

Consider the operator $S = I + 2X + Z$ acting on one qubit. It is not unitary; for example its determinant is $\det\left(\begin{smallmatrix} 2 & 2 \\ 2 & 0 \end{smallmatrix}\right) = -4$. The coefficient sum is $C = 1 + 2 + 1 = 4$. The normalized ancilla PREPARE state is

$$|\chi\rangle = \frac{1}{2}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle + \frac{1}{2}|2\rangle.$$

The controlled SELECT operator applies I on index zero, X on index one, and Z on index two. PREPARE–SELECT–UNPREPARE therefore has top-left block $S/4$, which equals

$$\frac{S}{4} = \frac{1}{4}\begin{pmatrix} 2 & 2 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 0 \end{pmatrix}.$$

On input $|0\rangle$, the good branch is $(|0\rangle + |1\rangle)/2$, with squared norm $1/2$. The successful normalized state is $|+\rangle$. The alternative state obtained by applying S directly is $S|0\rangle = 2|0\rangle + 2|1\rangle$ and has norm $2\sqrt{2}$; dividing by that norm gives the same $|+\rangle$. This checks all three quantities separately: LCU normalization C, branch probability $1/2$, and output normalization after postselection.

10.8.4 Finite precision of a block encoding

Suppose an approximate block-encoding has projected matrix $A/\alpha + E$ with $\|E\|_2 \le \varepsilon$. On unit input $|u\rangle$, its good-branch vector differs from the ideal one by at most ε. However, if $A|u\rangle/\alpha$ has norm $\eta \ll 1$, the normalized conditional states can differ by order ε/η. Indeed, writing $v = A|u\rangle/\alpha$ and $\widetilde{v} = v + Eu$, one has

$$\left\| \frac{\widetilde{v}}{\|\widetilde{v}\|} - \frac{v}{\|v\|} \right\| \le \frac{2\varepsilon}{\eta}$$

when both branches are nonzero. This is the same normalization instability encountered later in inverse-polynomial linear-system solvers. A small operator-norm error before postselection is not automatically a comparably small error in the conditioned quantum state when success is rare.

10.8.5 Separate ancillae for a two-matrix product

Let U_A use ancilla register a and U_B use an independent register b, and suppose their zero-projected blocks are A/α and B/β. Starting with both ancillae in zero states, apply U_A controlled only by its own ancilla register, then apply U_B using its separate ancilla. The amplitude in which both ancillae return to zero is precisely $(BA)|u\rangle/(\alpha\beta)$. The bad branch from the first operation carries an orthogonal nonzero state of ancilla a; subsequent operations on b do not reset this register, so that amplitude cannot contaminate the joint all-zero block. This is the operational mechanism that eliminates the unwanted B_1C_0 interference term found when a single shared ancilla is reused.

CHAPTER 11

Qubitization, QSP, and Singular-Value Transformation

11.1 Spectral calculus and singular values

For Hermitian $A = \sum_j \lambda_j |v_j\rangle\langle v_j|$ and a function f defined on its spectrum,

$$f(A) = \sum_j f(\lambda_j)|v_j\rangle\langle v_j|.$$

A non-Hermitian matrix instead admits an SVD $A = W\Sigma V^\dagger$ with nonnegative singular values σ_j. Its singular-value transforms are defined through the singular vectors and $f(\Sigma)$, not by substituting arbitrary complex eigenvalues into a Hermitian matrix-functional-calculus formula. The parity and matrix-block conventions in QSVT determine whether the resulting block resembles $Wf(\Sigma)V^\dagger$ or a same-side singular-vector operator.

11.2 Chebyshev polynomials from planar rotations

Definition 11.1 (Chebyshev polynomial of the first kind)**.** The polynomial T_k is uniquely determined by

$$T_k(\cos\theta) = \cos(k\theta), \qquad k \in \mathbb{Z}_{\geq 0}.$$

The first terms are $T_0(x) = 1$, $T_1(x) = x$, $T_2(x) = 2x^2 - 1$, and $T_3(x) = 4x^3 - 3x$.

The identity $\cos((k+1)\theta) + \cos((k-1)\theta) = 2\cos\theta\cos(k\theta)$ gives

$$T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x).$$

Consequently $\deg T_k = k$ for $k \geq 1$, $T_k(-x) = (-1)^k T_k(x)$, and $|T_k(x)| \leq 1$ for $x \in [-1, 1]$.

For $\lambda \in [-1, 1]$, define

$$U(\lambda) = \begin{pmatrix} \lambda & \sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & -\lambda \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then $U(\lambda)^2 = I$, while

$$O(\lambda) = U(\lambda)Z = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \quad \theta = \arccos\lambda.$$

Therefore $(O(\lambda)^k)_{00} = \cos(k\theta) = T_k(\lambda)$. In particular, with $\lambda = 1/\sqrt{2}$, $(O(\lambda)^3)_{00} = \cos(3\pi/4) = -1/\sqrt{2}$. The factor Z is crucial: $U(\lambda)$ alone is a reflection with $U(\lambda)^2 = I$.

11.3 Hermitian qubitization: an invariant-subspace theorem

Theorem 11.2 (Chebyshev block-encoding by qubitization)**.** *Suppose $A = A^\dagger$, $\|A\| \le 1$, and U_A is both Hermitian and unitary, with one designated zero-ancilla block equal to A. Let $\Pi = |0\rangle\langle 0| \otimes I$ and $Z_\Pi = 2\Pi - I$. Then, for every nonnegative integer k,*

$$\boxed{(\langle 0| \otimes I)(U_A Z_\Pi)^k (|0\rangle \otimes I) = T_k(A).}$$

Proof. Let $A|v_j\rangle = \lambda_j|v_j\rangle$, set $|e_j\rangle = |0\rangle|v_j\rangle$, and assume first $|\lambda_j| < 1$. Since the projected component of $U_A|e_j\rangle$ is $\lambda_j|e_j\rangle$, write

$$U_A|e_j\rangle = \lambda_j|e_j\rangle + s_j|e_j^\perp\rangle, \qquad s_j = \sqrt{1-\lambda_j^2}, \quad \Pi|e_j^\perp\rangle = 0.$$

Using $U_A^2 = I$ gives $U_A|e_j^\perp\rangle = s_j|e_j\rangle - \lambda_j|e_j^\perp\rangle$. Hence the two-dimensional span of $|e_j\rangle, |e_j^\perp\rangle$ is invariant, and the restrictions of U_A and Z_Π are exactly

$$\begin{pmatrix} \lambda_j & s_j \\ s_j & -\lambda_j \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Their product rotates by $\arccos \lambda_j$, so its kth power has projected amplitude $T_k(\lambda_j)$ on $|e_j\rangle$. At $|\lambda_j| = 1$, the same identity follows directly from $U_A|e_j\rangle = \lambda_j|e_j\rangle$ and $T_k(\pm 1) = (\pm 1)^k$. Applying the result to an eigenbasis proves the operator identity. □

For $A = \mathrm{diag}(1/2, 2/3)$,

$$T_3(A) = 4A^3 - 3A = \mathrm{diag}(-1, -22/27),$$

whose entries respect the required absolute bound. The procedure acts coherently on all eigencomponents without finding the eigenvectors explicitly. Its output is still a block-encoding rather than a deterministic physical application of a nonunitary matrix.

The reflection assumption on U_A is substantive. An arbitrary block-encoding of a Hermitian matrix need not itself be Hermitian. More general qubitization schemes use $U_A, U_A^\dagger$ and appropriate projectors or reflections; one cannot simply assert the previous formula for an arbitrary non-Hermitian U_A. For example, constructions involving $U_A^\dagger Z_\Pi U_A Z_\Pi$ produce related even-degree transformations after the correct invariant subspaces and block projections are identified.

11.4 Quantum signal processing

QSP interpolates controlled phase rotations among fixed signal operations. A representative one-variable sequence with d signal queries is

$$F_{\boldsymbol{\phi}}(\lambda) = e^{i\phi_0 Z} O(\lambda) e^{i\phi_1 Z} O(\lambda) \cdots O(\lambda) e^{i\phi_d Z},$$

where $\boldsymbol{\phi} = (\phi_0, \ldots, \phi_d)$ are tunable real phases. Depending on the signal convention and basis, the final unitary can be described by polynomials P, Q constrained by

$$\deg P \le d, \quad \deg Q \le d-1, \qquad P(-\lambda) = (-1)^d P(\lambda), \quad Q(-\lambda) = (-1)^{d-1} Q(\lambda),$$

$$|P(\lambda)|^2 + (1-\lambda^2)|Q(\lambda)|^2 = 1 \quad (\lambda \in [-1,1]).$$

Unitarity implies $|P(\lambda)| \le 1$. These are structural restrictions of a fixed-length signal sequence, not arbitrary cosmetic restrictions on candidate polynomials.

For a real target polynomial intended for a **single parity-constrained** QSP/QSVT block, the key screening conditions are: finite polynomial degree d, parity $p(-x) = (-1)^d p(x)$, and boundedness $|p(x)| \le 1$ on $[-1,1]$. Appropriate QSP completion and, where necessary, additional ancilla/LCU constructions select the desired real polynomial from the implemented complex matrix element.

Candidate	Degree/parity/boundedness assessment
$2x^2 - 1$	Degree two, even, bounded; it is T_2
$(1+x)/2$	Bounded but mixed parity; requires combination of parity sectors rather than this single-parity form
$2x$	Odd degree one but exceeds the unit bound at $x = 1$
$1/(2x)$	Not a polynomial and singular at zero; requires a bounded polynomial approximation away from zero

Mixed parity is not a general physical impossibility: additional combinations of QSP blocks can implement more general matrix functions with the appropriate normalization. A degree-d QSVT sequence uses $O(d)$ calls to the input block-encoding or its inverse, plus the required controlled phases/reflections. This is an oracle-query bound; its physical cost depends on implementing the input oracle, phase synthesis, ancilla preparation, postselection, and readout.

11.5 Functional approximation with explicit spectral domains

To implement a matrix function $f(A)$, the mathematically relevant steps are: specify the spectrum or singular-value interval where accuracy is required; select a scale making the transform compatible with a unitary subblock; construct a polynomial satisfying boundedness and parity under the chosen QSVT scheme; and quantify the approximation error, polynomial degree, and success amplification. The behavior outside the occupied spectral interval matters for QSP boundedness even when the polynomial need not approximate f there.

11.6 Polynomial checks and phase synthesis

The recurrence for Chebyshev polynomials also gives $T_4(x) = 8x^4 - 8x^2 + 1$, so for Hermitian A,

$$T_2(A) = 2A^2 - I, \qquad T_4(A) = 8A^4 - 8A^2 + I.$$

It is essential to interpret scalar constant terms as multiples of the identity matrix. For diagonal A, all entries can be calculated by applying the same scalar polynomial to each eigenvalue; for general Hermitian A, spectral decomposition gives the same operation.

A QSP implementation entails three distinct computational tasks. First, approximate the desired function on the relevant spectral intervals by a bounded polynomial. Second, synthesize a phase vector ϕ realizing that polynomial under the chosen signal convention; this can itself require nontrivial numerical computation and finite-precision control. Third, implement the resulting oracle sequence, including ancillas, inverses, and readout. An $O(d)$ signal-query statement refers to the third task's oracle count, not a universal $O(d)$ bound on all three tasks together.

11.7 Polynomial spectral transforms: detailed proofs and examples

11.7.1 Matrix functions and the role of the spectral theorem

If $A = A^\dagger$ has spectral decomposition $A = \sum_j \lambda_j |v_j\rangle\langle v_j|$, any polynomial $p(x) = \sum_{k=0}^{d} a_k x^k$ defines a matrix polynomial $p(A) = \sum_{k=0}^{d} a_k A^k$ and satisfies

$$p(A) = \sum_j p(\lambda_j)|v_j\rangle\langle v_j|.$$

This follows because $A^k|v_j\rangle = \lambda_j^k|v_j\rangle$ for every nonnegative integer k. In particular the eigenvectors are preserved, while eigenvalues are transformed according to $\lambda_j \mapsto p(\lambda_j)$. The operator norm obeys $\|p(A)\|_2 = \max_j |p(\lambda_j)|$. Therefore if $\mathrm{spec}(A) \subseteq [-1, 1]$ and $|p(x)| \le 1$ on that interval, then $p(A)$ is a contraction and can, in principle, be block-encoded with scale one.

For a non-Hermitian matrix $A = W\Sigma V^\dagger$, a singular-value transformation acts on singular values σ_j and the corresponding singular vectors according to the parity and orientation of the transformation. It is not generally the same as substituting A into an ordinary polynomial. Thus claims of the form "QSVT implements $p(A)$" are cleanest for Hermitian A with a specified block-encoding and require additional care in the general singular-value setting.

11.7.2 Chebyshev polynomials from trigonometry

Define the Chebyshev polynomial of the first kind by

$$T_k(\cos\theta) = \cos(k\theta), \qquad k \geq 0.$$

Using $\cos((k+1)\theta) + \cos((k-1)\theta) = 2\cos\theta\cos(k\theta)$ gives the recurrence

$$T_0(x) = 1, \qquad T_1(x) = x, \qquad T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x).$$

The first few cases are

$$\begin{aligned} T_2(x) &= 2x^2 - 1, \\ T_3(x) &= 4x^3 - 3x, \\ T_4(x) &= 8x^4 - 8x^2 + 1, \\ T_5(x) &= 16x^5 - 20x^3 + 5x. \end{aligned}$$

Since $|\cos(k\theta)| \leq 1$, $|T_k(x)| \leq 1$ for $x \in [-1, 1]$. The parity $T_k(-x) = (-1)^k T_k(x)$ follows from either the recurrence or the identity $\arccos(-x) = \pi - \arccos x$. It is often faster to use recurrence than to expand a high power of a matrix directly.

Example 11.3 (Two independent evaluations of a Chebyshev value)**.** Take $x = 1/\sqrt{2}$. Since $\arccos x = \pi/4$,

$$T_3(1/\sqrt{2}) = \cos(3\pi/4) = -1/\sqrt{2}.$$

The polynomial calculation gives $4(1/\sqrt{2})^3 - 3(1/\sqrt{2}) = \sqrt{2} - 3/\sqrt{2} = -1/\sqrt{2}$, agreeing exactly. This example detects a common error: replacing the required rotation $U(x)Z$ by the reflection $U(x)$ alone would yield $U(x)^2 = I$ and incorrectly predict the same result for all odd powers.

11.7.3 Two reflections create a plane rotation

For $\lambda \in [-1, 1]$, define $s = \sqrt{1 - \lambda^2}$ and

$$U(\lambda) = \begin{pmatrix} \lambda & s \\ s & -\lambda \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Direct multiplication yields $U(\lambda)^\dagger = U(\lambda)$ and $U(\lambda)^2 = I$, so $U(\lambda)$ is a reflection. But

$$O(\lambda) = U(\lambda)Z = \begin{pmatrix} \lambda & -s \\ s & \lambda \end{pmatrix}$$

is a proper rotation. Let $\theta = \arccos\lambda$ so that $s = \sin\theta \geq 0$. Then

$$O(\lambda)^k = \begin{pmatrix} \cos(k\theta) & -\sin(k\theta) \\ \sin(k\theta) & \cos(k\theta) \end{pmatrix}, \qquad \langle 0|O(\lambda)^k|0\rangle = T_k(\lambda).$$

The calculation explains how a repeated controlled reflection can encode a high-degree polynomial in a selected matrix element while the full transformation remains unitary.

11.7.4 The qubitization invariant-subspace argument

Theorem 11.4 (Hermitian involutory block-encoding model)**.** *Let $A = A^\dagger$ with $\|A\|_2 \le 1$. Assume a unitary U_A is also Hermitian, satisfies $U_A^2 = I$, and obeys $\Pi U_A \Pi = |0^a\rangle\langle 0^a| \otimes A$ on the selected subspace, where $\Pi = |0^a\rangle\langle 0^a| \otimes I$. Put $Z_\Pi = 2\Pi - I$ and $W = U_A Z_\Pi$. Then*

$$(\langle 0^a| \otimes I) W^k (|0^a\rangle \otimes I) = T_k(A)$$

for every nonnegative integer k.

Proof. Fix an eigenpair $A|v\rangle = \lambda|v\rangle$. Write $|g\rangle = |0^a\rangle|v\rangle$ and $|h\rangle = U_A|g\rangle$. The block property gives $\langle g|h\rangle = \lambda$. Since $U_A^2 = I$, $U_A|h\rangle = |g\rangle$. Furthermore $\Pi|g\rangle = |g\rangle$ and $\Pi|h\rangle = \lambda|g\rangle$. Therefore

$$W|g\rangle = |h\rangle, \qquad W|h\rangle = 2\lambda|h\rangle - |g\rangle.$$

The span of $|g\rangle, |h\rangle$ is invariant. For $|\lambda| < 1$, orthonormalize it by setting $|g_\perp\rangle = (|h\rangle - \lambda|g\rangle)/\sqrt{1-\lambda^2}$. In the ordered basis $(|g\rangle, |g_\perp\rangle)$, W has the rotation matrix $\begin{pmatrix} \lambda & -\sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & \lambda \end{pmatrix}$. Hence $\langle g|W^k|g\rangle = T_k(\lambda)$. The endpoint cases $\lambda = \pm 1$ follow by continuity or by observing that $|h\rangle = \pm|g\rangle$. Applying the identity separately to each eigenvector $|v\rangle$ establishes the operator identity for the projected block. □

The assumption $U_A = U_A^\dagger$ is substantive. For a general non-Hermitian block-encoding unitary, the simple equation $U_A|h\rangle = |g\rangle$ no longer follows. A correct general qubitization/QSVT construction uses reflections and controlled access to both U_A and $U_A^\dagger$ with carefully identified left/right invariant subspaces. One must not copy this two-reflection proof verbatim into a model lacking its involutory assumption.

Example 11.5 (A diagonal matrix transformed by T_3)**.** Let $A = \mathrm{diag}(1/2, 2/3)$. Then

$$T_3(A) = 4A^3 - 3A = \mathrm{diag}(4(1/8) - 3/2,\ 4(8/27) - 2) = \mathrm{diag}(-1, -22/27).$$

Both transformed eigenvalues remain in $[-1, 1]$. The same matrix can be evaluated via $T_2(A) = 2A^2 - I$ and $T_3(A) = 2AT_2(A) - A$, which provides a check independent of direct cubic substitution.

11.7.5 The QSP phase sequence and polynomial constraints

The scalar quantum signal-processing construction inserts selectable Z-axis phase rotations between copies of a signal rotation $O(x)$. In one convention,

$$F_\Phi(x) = e^{i\phi_0 Z} O(x) e^{i\phi_1 Z} O(x) \cdots O(x) e^{i\phi_d Z},$$

where the signal unitary is called d times and $d + 1$ phases are specified. The unitary's entries can be described by two polynomials P, Q, whose degree and parity constraints depend on d:

$$\deg P \le d, \quad \deg Q \le d - 1, \quad P(-x) = (-1)^d P(x), \quad Q(-x) = (-1)^{d-1} Q(x).$$

Unitarity imposes, for $x \in [-1, 1]$,

$$|P(x)|^2 + (1 - x^2)|Q(x)|^2 = 1.$$

In particular $|P(x)| \leq 1$ on the full signal interval. This is a *necessary* entrywise constraint. Real-polynomial implementation results can use suitable choices of complex P, Q or additional interference constructions to realize desired real functions, often with constant normalization overhead; consequently a mixed-parity target may still be implementable by combining separate even and odd components, even though it cannot be the single parity-constrained entry of one degree-d signal sequence.

11.7.6 The three direct polynomial checks, with counterexamples

Before claiming a direct parity-constrained QSP/QSVT implementation for a real polynomial p, perform three checks. First, p must actually be a polynomial in the variable to be transformed. Second, its parity must match the signal-sequence degree: $p(-x) = (-1)^d p(x)$. Third, it must be bounded in magnitude by one over the *entire* interval $[-1, 1]$, not just at the eigenvalues used in a particular small numerical example.

The polynomial $2x^2 - 1 = T_2(x)$ passes all three tests at degree two. The linear polynomial $(1 + x)/2$ mixes even and odd terms and therefore does not satisfy the single-sequence parity rule at degree one, though it can be decomposed into constant and linear components. The function $2x$ has odd parity but violates boundedness at $x = 1$. The function $1/(2x)$ is not a polynomial and is singular at the origin; its inverse-like behavior can be approximated only on an interval excluding zero after appropriate scaling and polynomial approximation. These distinctions separate the direct-implementation test from the broader question of whether a more elaborate circuit construction can realize a desired matrix function.

11.7.7 Polynomial approximation on a gapped spectral domain

For inverse spectral functions, the admissible domain is typically

$$\mathcal{D}_\kappa = [-1, -1/\kappa] \cup [1/\kappa, 1], \qquad \kappa \geq 1.$$

On this set $1/x$ is bounded by κ, even though it diverges at zero. Scaling by $1/\kappa$ gives $|1/(\kappa x)| \leq 1$ on $\mathcal{D}_\kappa$. However, a polynomial intended as an entry of a unitary signal-processing circuit still needs an appropriate global bound on $[-1, 1]$, not merely pointwise agreement with the target on $\mathcal{D}_\kappa$. The construction can use a bounded odd polynomial that approximates the scaled reciprocal only on the two separated intervals, while remaining controlled in the excluded gap. This is why spectral-gap assumptions appear explicitly in quantum linear-system solvers.

11.7.8 Chebyshev recurrence as a stable symbolic construction

The recurrence $T_{k+1}(A) = 2AT_k(A) - T_{k-1}(A)$ can be implemented symbolically without diagonalizing a small matrix. Starting from $T_0(A) = I$ and $T_1(A) = A$ gives

$$T_2(A) = 2A^2 - I, \quad T_3(A) = 4A^3 - 3A, \quad T_4(A) = 8A^4 - 8A^2 + I.$$

The constant term always multiplies the identity matrix; replacing I by a scalar 1 in a non-scalar expression is dimensionally incorrect. The recurrence is an algebraic fact independent of quantum circuits; qubitization is the operational mechanism that can turn the same polynomial into a projected quantum transformation. Its asymptotic query count is linear in polynomial degree, provided the block-encoding primitive is available at the assumed cost.

11.7.9 A formal workflow for spectral-function design

A matrix-function algorithm begins by specifying the spectral or singular-value domain and the target function. Next it chooses a normalization so that the desired matrix function can appear as a unitary subblock, and derives a polynomial approximation with a quantified uniform error over the relevant domain. The polynomial must also be compatible with parity and global boundedness requirements of the chosen QSP/QSVT construction, possibly using ancillary combination steps. The signal sequence then implements the polynomial transformation with $O(d)$ uses of the block-encoding for a degree-d polynomial, up to the precise convention's constant factors. Finally, the algorithm must account for success probability, amplification, state preparation, and any classical measurement needed to obtain the desired output. Omitting one of these stages can turn a valid algebraic approximation into an incorrect claimed end-to-end complexity.

11.8 Spectral-transform case studies and mathematical consistency checks

11.8.1 An off-diagonal Hermitian matrix whose Chebyshev image is simple

Consider $A = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} = X/2$. It is Hermitian with eigenvalues $\pm 1/2$ and satisfies $A^2 = I/4$. Therefore

$$T_2(A) = 2A^2 - I = -\frac{I}{2}, \qquad T_3(A) = 4A^3 - 3A = 4 \cdot \frac{A}{4} - 3A = -2A = -X.$$

In the eigenbasis, $T_2(1/2) = T_2(-1/2) = -1/2$, and $T_3(1/2) = -1$, $T_3(-1/2) = +1$. The matrix calculation and eigenvalue substitution agree. This example distinguishes even and odd Chebyshev polynomials: T_2 gives the same value at opposite eigenvalues, whereas T_3 flips sign.

11.8.2 A qubitization matrix with one prescribed eigenvalue

Let $\lambda = 3/5$, $s = 4/5$ and

$$U = \begin{pmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{pmatrix}, \qquad Z = \mathrm{diag}(1, -1).$$

Then $O = UZ = \left(\begin{smallmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{smallmatrix}\right)$. Squaring gives

$$O^2 = \begin{pmatrix} -7/25 & -24/25 \\ 24/25 & -7/25 \end{pmatrix},$$

so the top-left entry is $-7/25 = T_2(3/5) = 2(9/25) - 1$. Multiplying once more yields top-left entry $T_3(3/5) = 4(27/125) - 9/5 = -117/125$. A direct multiplication of O^3 gives the same number. The identity is thus a sequence of exact rational calculations, with no reliance on floating-point trigonometric approximations.

11.8.3 Oddness of reciprocal approximations and the spectral gap

The target function $f(x) = 1/x$ is odd, so for a symmetric spectrum containing positive and negative eigenvalues, an odd approximating polynomial preserves the sign structure. In contrast, an even polynomial satisfies $p(-x) = p(x)$ and cannot approximate both $1/x$ and $-1/x$ at opposite points with error smaller than the absolute value of either target there. At $x = 1$ and $x = -1$, for instance, an even polynomial gives the same value at both locations, whereas the reciprocal takes values 1 and -1. Hence its maximum approximation error over the symmetric two-point set is at least one. This proves that the parity restriction is not merely a convention for inversion: for such symmetric spectral domains, oddness is structurally appropriate.

The gap parameter κ determines the allowed intervals $[-1, -1/\kappa] \cup [1/\kappa, 1]$. If $\kappa = 4$, the nearest allowed spectral points to zero are $\pm 1/4$ and the unscaled reciprocal reaches magnitude four. A scale of $1/(2\kappa) = 1/8$ brings the target's largest magnitude down to $1/2$, leaving room for approximation errors while satisfying the contraction requirement. The polynomial still needs to be controlled between $-1/4$ and $+1/4$ so that the full QSP sequence remains unitary throughout $[-1, 1]$. Merely interpolating exact reciprocal values at a few eigenvalues would not establish this necessary boundedness.

11.8.4 Polynomial degree versus implementation accuracy

Let p_d be a polynomial intended to approximate f on a specified spectral set. Its uniform approximation error is $\delta = \sup_{x \in \mathcal{D}} |p_d(x) - f(x)|$. For Hermitian A with spectrum in $\mathcal{D}$, spectral calculus yields

$$\|p_d(A) - f(A)\|_2 = \max_{\lambda \in \mathrm{spec}(A)} |p_d(\lambda) - f(\lambda)| \leq \delta.$$

This converts a scalar approximation guarantee into an operator-norm guarantee without an additional matrix-dimension factor. However, if the implemented signal sequence has error η per oracle call or per synthesized phase rotation, the final circuit error may have a separate dependence on $d\eta$ under a worst-case telescoping argument. In particular, a degree bound alone is not a full fault-tolerant gate count: the precision and cost of implementing each call and each phase gate remain part of the resource model.

11.8.5 A general telescoping bound for imperfect gate sequences

Let $U_1, \dots, U_d$ be ideal unitaries and $\widetilde{U}_1, \dots, \widetilde{U}_d$ their approximations with $\|U_j - \widetilde{U}_j\|_2 \le \eta_j$. Then

$$\|U_d \cdots U_1 - \widetilde{U}_d \cdots \widetilde{U}_1\|_2 \le \sum_{j=1}^{d} \eta_j.$$

To prove it, add and subtract one intermediate product at a time; each difference is a product of unitary matrices with the single factor $U_j - \widetilde{U}_j$ inserted, so its norm is at most η_j. The triangle inequality finishes the bound. The result is general and conservative; coherent errors can sometimes cancel, but a rigorous worst-case estimate cannot assume such cancellation. If each of d oracle uses has error at most η, the aggregate bound is $d\eta$, explaining why high-degree QSVT may impose increasing precision demands on its primitives.

11.9 Singular-value transformations, parity, and polynomial accuracy

11.9.1 The projector–unitary–projector viewpoint

For a general matrix $A \in \mathbb{C}^{N\times N}$, a block-encoding is more safely expressed in terms of an enlarged unitary U, a right input projector Π_R, and a left output projector Π_L, with $\Pi_L U \Pi_R$ representing the desired scaled operator. With $A = W\Sigma V^\dagger$ and singular vectors $A|v_j\rangle = \sigma_j|w_j\rangle$, $A^\dagger|w_j\rangle = \sigma_j|v_j\rangle$, the normalized singular-value problem consists of transforming $\sigma_j \in [0,1]$ while tracking which singular-vector space is the input and which is the output. Even-degree transformations of a standard projected signal sequence act within the same singular-vector space; odd-degree transformations generally move between the right and left spaces. This distinction is invisible for a Hermitian matrix whose eigenvectors may be used on both sides, but matters for a genuinely nonsymmetric or nonnormal matrix.

For example, take the real matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. It has one nonzero singular value, equal to one: $|v_1\rangle = |1\rangle$, $|w_1\rangle = |0\rangle$, and $A|v_1\rangle = |w_1\rangle$. However $A^2 = 0$. Applying the odd singular-value function $p(\sigma) = \sigma$ reproduces A, while ordinary matrix squaring gives

$A^2 = 0$. At even degree, $p(\sigma) = \sigma^2$ on the *right* singular space gives $A^\dagger A = |1\rangle\langle 1|$, whereas on the *left* singular space it gives $AA^\dagger = |0\rangle\langle 0|$. Neither is the naive matrix power A^2. The example demonstrates concretely why singular-value transformation must not be summarized as $p(A)$ for arbitrary non-Hermitian A.

11.9.2 An explicit singular-vector factorization

Let

$$A = \begin{pmatrix} 0 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}, \qquad A^\dagger A = \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{4} \end{pmatrix}, \qquad AA^\dagger = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & 0 \end{pmatrix}.$$

The nonzero singular value is $1/2$. The polynomial $p_3(x) = T_3(x) = 4x^3 - 3x$ satisfies $p_3(1/2) = -1$. An odd singular-value transform oriented from right to left therefore acts as

$$p_3^{(\mathrm{SV})}(A) = -|0\rangle\langle 1| = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$$

on the supported singular-vector pair. But the ordinary polynomial is $T_3(A) = 4A^3 - 3A = -3A$, because $A^2 = 0$; these two outputs disagree. Such a comparison is a direct test of whether a derivation has silently replaced a singular-value operation with ordinary polynomial substitution.

11.9.3 Polynomial parity from a signal sequence

The standard degree-d QSP sequence uses d signal interactions and phase rotations. In a fixed convention the signal-dependent matrix entries have the form

$$\begin{pmatrix} P(x) & i\sqrt{1-x^2}\,Q(x) \\ i\sqrt{1-x^2}\,\overline{Q(x)} & \overline{P(x)} \end{pmatrix},$$

where one may absorb convention-dependent signs and conjugations into the choice of phases. The essential algebraic conditions are $\deg P \le d$, $\deg Q \le d-1$, the respective parities d and $d-1$, and the unitarity condition $|P(x)|^2 + (1-x^2)|Q(x)|^2 = 1$ on $[-1, 1]$. These are conditions on a *coherent matrix entry*, not a statement that every arbitrary bounded scalar function is directly implementable by the same finite sequence.

As a degree-three admissibility test, $p(x) = 4x^3 - 3x$ is odd and has magnitude at most one on $[-1, 1]$, so it passes the necessary parity and boundedness checks. By contrast $p(x) = x^2 + x/4$ has degree two but is neither even nor odd; a single degree-two parity-constrained sequence cannot have this as its selected polynomial entry. Furthermore $p(x) = 3x/2$ has odd parity but fails the global magnitude bound at $x = 1$. Restricting a particular input matrix to small singular values does not remove the global unitarity constraint on the implementing signal sequence.

11.9.4 From operator error to state error

Suppose $A = A^\dagger$, $\mathrm{spec}(A) \subseteq \mathcal{D}_\kappa$, and p approximates a target inverse filter f uniformly: $\max_{x\in\mathcal{D}_\kappa} |p(x) - f(x)| \le \delta$. Spectral decomposition yields the exact operator-norm bound

$$\|p(A) - f(A)\|_2 \le \delta.$$

For a normalized right-hand side $|b\rangle$, write $u = f(A)|b\rangle$, $v = p(A)|b\rangle$. Then $\|u - v\| \le \delta$. If $u \neq 0$ and $\delta < \|u\|$, elementary triangle inequalities imply

$$\left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\| \le \frac{2\delta}{\|u\|}.$$

For $f(x) = 1/(\kappa x)$ on a gapped Hermitian contraction whose eigenvalues have magnitudes in $[1/\kappa, 1]$, every singular value of $f(A)$ is between $1/\kappa$ and 1, so $\|f(A)b\| \ge 1/\kappa$. Hence an $O(\varepsilon)$ error in the normalized target can be ensured by a uniform inverse-polynomial approximation error $\delta = O(\varepsilon/\kappa)$. This requirement must be fed back into the polynomial-degree estimate; estimating only the unnormalized block error at precision ε can be insufficient when the desired postselection branch is small.

11.9.5 Spectral projectors and a two-point inverse comparison

Let $A = \mathrm{diag}(1, 1/4)$, $|b\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and $\kappa = 4$. The normalized quantum solution is

$$|x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|} = \frac{|0\rangle + 4|1\rangle}{\sqrt{17}}.$$

The scaled reciprocal $f_1(x) = 1/(4x)$ has filtered branch $f_1(A)|b\rangle = (|0\rangle/4 + |1\rangle)/\sqrt{2}$, with probability $17/32$. The further scaled reciprocal $f_2(x) = 1/(8x)$ has branch one half as large, with probability $17/128$, yet its *conditional normalized state* is exactly the same $|x\rangle$. A change in scaling therefore changes the postselection cost without changing the mathematical solution, provided the scaling is uniform over the spectral domain.

11.9.6 A query ledger for a composite spectral algorithm

If a degree-d signal sequence uses $\Theta(d)$ calls to a block-encoding and $\Theta(d)$ one- or two-qubit phase primitives, a successful postselected application with probability p requires $\Theta(d/p)$ signal calls by independent repetition, or $O(d/\sqrt{p})$ with suitable coherent amplitude amplification, ignoring additional logarithmic and constant factors. If implementing one block-encoding query itself costs G_A elementary gates and one call to the input preparation costs G_b, then the gate cost is not simply $O(d)$; it contains terms on the order of $G_A d$ and the repeated state-preparation/reflection overhead. Estimation of a bounded observable to additive precision η by independent shots adds $O(\eta^{-2})$ output copies, while amplitude-estimation-based variants impose their

own coherent-access assumptions. Explicitly listing these separate resources prevents an oracle-speedup statement from being incorrectly interpreted as a complete end-to-end runtime bound.

CHAPTER 12

Quantum Linear Systems and Inverse Spectral Transformations

12.1 Formal computational task

Definition 12.1 (Quantum linear systems problem (QLSP))**.** Given oracle access to an invertible matrix $A \in \mathbb{C}^{N\times N}$ and a normalized state $|b\rangle$, prepare a state ε-close to

$$|x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|}.$$

The access model must specify implementations of a block-encoding for A and a state-preparation oracle for $|b\rangle$, and the output error must be specified in a state metric such as Euclidean norm up to global phase.

The classical task of listing every entry of $A^{-1}b$ is stronger than producing the amplitude state $|x\rangle$. A quantum solution can be useful for estimating observables $\langle x|O|x\rangle$, sampling, or later coherent quantum processing; extracting all N amplitudes incurs additional costs and may erase an apparent dimensional speedup.

Assume for the basic spectral analysis that $A = A^\dagger$ and $\|A\| = 1$. Its condition number is

$$\kappa = \|A\|\,\|A^{-1}\| = \frac{1}{\min_j |\lambda_j|}.$$

For $|b\rangle = \sum_j \beta_j |v_j\rangle$, one has

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j}|v_j\rangle.$$

The inversion amplifies components on small-magnitude eigenvalues, and this dependence is reflected in algorithmic complexity.

12.2 Spectral gap and bounded inverse scaling

The promised Hermitian spectrum lies in

$$\sigma(A) \subseteq [-1, -1/\kappa] \cup [1/\kappa, 1],$$

or in $[1/\kappa, 1]$ for positive-definite A. The function $1/x$ has a pole at zero and attains magnitude κ on the occupied spectral interval; it cannot itself appear as a unit-normalization block of a unitary for $\kappa > 1$.

The direction of $A^{-1}|b\rangle$ does not depend on nonzero scalar rescaling. Two commonly useful target functions are

$$f_1(x) = \frac{1}{\kappa x}, \qquad f_{1/2}(x) = \frac{1}{2\kappa x}.$$

Both produce the same normalized target state, while the second leaves a boundedness margin for polynomial approximation. The functions have singularities at zero; they must be approximated by globally bounded **polynomials**, with the required accuracy only on the promised spectral intervals. An odd polynomial is natural because $1/x$ is odd.

12.3 QSVT inverse construction

Choose an odd polynomial p with

$$\sup_{1/\kappa \le |x| \le 1} \left| p(x) - \frac{1}{2\kappa x} \right| \le \delta, \qquad \sup_{|x|\le 1} |p(x)| \le 1.$$

Use QSVT to block-encode $p(A)$. Applied to $|0^a\rangle|b\rangle$, the good ancilla component is $|0^a\rangle p(A)|b\rangle$. Conditioning on zero ancillas yields

$$|\widetilde{x}\rangle = \frac{p(A)|b\rangle}{\|p(A)|b\rangle\|} \approx |x\rangle.$$

For the ideal unapproximated transforms, the good-branch success probabilities are

$$p_1 = \frac{\|A^{-1}|b\rangle\|^2}{\kappa^2}, \qquad p_{1/2} = \frac{\|A^{-1}|b\rangle\|^2}{4\kappa^2} = \frac{p_1}{4}.$$

The normalized conditional states agree, but the success probabilities differ by exactly a factor of four.

12.4 A complete two-dimensional inverse calculation

Example 12.2 (Diagonal system with tunable condition number)**.** Let $\kappa \ge 1$, $A = \mathrm{diag}(1, 1/\kappa)$, and $|b\rangle = |+\rangle$. Then

$$A^{-1}|b\rangle = \frac{|0\rangle + \kappa|1\rangle}{\sqrt{2}}, \qquad |x\rangle = \frac{|0\rangle + \kappa|1\rangle}{\sqrt{1+\kappa^2}}.$$

The scaled inverse $f_1(A) = \mathrm{diag}(1/\kappa, 1)$ gives good branch $(|0\rangle/\kappa + |1\rangle)/\sqrt{2}$, with

$$p_1 = \frac{1+\kappa^{-2}}{2} = \frac{\kappa^2+1}{2\kappa^2}.$$

The half-scaled inverse gives $p_{1/2} = (\kappa^2+1)/(8\kappa^2)$. With $\kappa = 4$, the target state is $(|0\rangle + 4|1\rangle)/\sqrt{17}$, and the two ideal probabilities are $17/32 \approx 0.53125$ and $17/128 \approx 0.1328125$ respectively.

12.5 Approximation error after normalizing a state

An operator-polynomial approximation error must be translated into an error of **normalized** output states. For nonzero vectors u, v,

$$\left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\| \leq \frac{2\|u - v\|}{\|v\|}.$$

To see this, add and subtract $u/\|v\|$ and use $|\|u\| - \|v\|| \leq \|u - v\|$. In the half-scaled inverse scheme,

$$\|f_{1/2}(A)|b\rangle\| = \frac{\|A^{-1}|b\rangle\|}{2\kappa} \geq \frac{1}{2\kappa}$$

for $\|A\| = \|b\| = 1$ and invertible Hermitian A. Thus an additive error $\|[p(A) - f_{1/2}(A)]|b\rangle\| \leq \delta$ can cause normalized-state error of order $O(\kappa\delta)$. Choosing $\delta = O(\varepsilon/\kappa)$ suffices for state accuracy $O(\varepsilon)$ in this basic bound.

12.6 Baseline query complexity

A standard bounded reciprocal approximation on the gapped spectrum can be taken with degree

$$d = O\left(\kappa \log \frac{1}{\delta}\right) = O\left(\kappa \log \frac{\kappa}{\varepsilon}\right)$$

under the conventional approximation assumptions. A single QSVT attempt calls the matrix block-encoding $O(d)$ times. With half-scaled inversion, $\|A^{-1}|b\rangle\| \geq 1$ implies success probability at least $1/(4\kappa^2)$ in the idealized bound, so amplitude amplification costs $O(\kappa)$ coherent repetitions. This yields the **baseline** matrix-oracle bound

$$\boxed{O\left(\kappa^2 \log \frac{\kappa}{\varepsilon}\right)}$$

for this straightforward QSVT-plus-amplification approach. More sophisticated algorithms can improve the condition-number dependence. All such oracle expressions remain conditional on efficient access, implementation and requested output observables.

12.7 Non-Hermitian systems

A general square complex matrix can be embedded into the Hermitian matrix

$$\widetilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix} = \widetilde{A}^\dagger.$$

If $Ax = b$, then

$$\widetilde{A} \begin{pmatrix} 0 \\ x \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}.$$

This reduction shows that non-Hermiticity is not, by itself, a prohibition against block-encoding-based quantum linear solvers. Singular values, invertibility, condition number, block-encoding implementation, and output extraction all remain essential.

12.8 Cross-cutting resource accounting

Across block-encoding, LCU, QSVT, and QLSP, five distinct quantities must be separated: the ancilla-zero **amplitude**; its squared **success probability**; the **conditional normalized state**; the **coherent repetitions** necessary for amplitude amplification; and the **measurement and classical reconstruction** cost of the requested output. A small ancilla-zero amplitude is not itself the probability, and an oracle-query improvement is not automatically an improvement in every physical resource.

12.9 Spectral support of the right-hand side

For the same diagonal system $A = \mathrm{diag}(1, 1/\kappa)$, take

$$|b_\gamma\rangle = \cos\gamma|0\rangle + \sin\gamma|1\rangle.$$

Then

$$A^{-1}|b_\gamma\rangle = \cos\gamma|0\rangle + \kappa\sin\gamma|1\rangle,$$
$$\|A^{-1}|b_\gamma\rangle\|^2 = \cos^2\gamma + \kappa^2\sin^2\gamma.$$

For unit-maximal inverse scaling $f_1(A)$, the success probability is

$$p_1(\gamma) = \frac{\cos^2\gamma}{\kappa^2} + \sin^2\gamma.$$

This interpolates between $1/\kappa^2$ for $|b\rangle = |0\rangle$ and 1 for $|b\rangle = |1\rangle$. Thus postselection is controlled not only by κ but also by the input state's spectral weight. The condition-number factor is a worst-case guarantee rather than an equality for every input.

12.10 Comparing algorithmic scaling under explicit access assumptions

Under illustrative unit-normalized block-encoding models, representative asymptotic query trends include original phase-estimation-based quantum linear solvers with polynomial dependence on $1/\varepsilon$, the baseline QSVT approach with $O(\kappa^2\log(\kappa/\varepsilon))$, and improved quantum linear solvers that can achieve near-linear dependence on κ and logarithmic dependence on $1/\varepsilon$ subject to their specific oracle and success criteria. These asymptotic comparisons are meaningful only when input models, matrix normalization, condition number, error metric, and target output are matched. Preconditioning may reduce the effective condition number, but must itself be efficiently implementable in the quantum access model.

12.11 Solving linear systems by inverse spectral transformations

12.11.1 Classical vector output versus a normalized quantum solution state

Let $A \in \mathbb{C}^{N\times N}$ be invertible and $b \neq 0$. The classical linear-system problem requests the vector $x = A^{-1}b$, including its N numerical coordinates. A quantum-state version instead requests preparation of

$$|x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|_2},$$

where $|b\rangle = b/\|b\|_2$. The two specifications are not interchangeable. In particular, $|x\rangle$ loses the norm $\|A^{-1}b\|_2$, and measuring it in a computational basis returns only one coordinate index sampled with probability proportional to $|x_j|^2$. Recovering all complex entries of x to fixed accuracy requires additional information and repeated measurements, potentially defeating the intended dimension dependence. By contrast, estimating a small set of expectation values $\langle x|M|x\rangle$, sampling from the coordinate distribution, or feeding $|x\rangle$ into a downstream quantum computation can be compatible with the state-output interface.

Assume first that $A = A^\dagger$ and $\|A\|_2 = 1$. If its nonzero eigenvalues obey $1/\kappa \leq |\lambda_j| \leq 1$, its spectral condition number is $\kappa = \|A\|_2\|A^{-1}\|_2 = 1/\min_j |\lambda_j|$. For a general invertible matrix, the condition number is defined through singular values and the same eigenvalue interval cannot be invoked unless Hermiticity is established.

12.11.2 Why the inverse must be scaled

A unitary's subblock is a contraction, so it cannot directly contain A^{-1} with scale one whenever $\|A^{-1}\|_2 = \kappa > 1$. The inverse target must be scaled. Two mathematically useful choices on the gapped spectral interval are

$$f_1(x) = \frac{1}{\kappa x}, \qquad f_2(x) = \frac{1}{2\kappa x}.$$

Both are odd functions. For $|x| \in [1/\kappa, 1]$, their magnitudes are bounded by one and one half, respectively. The second choice provides additional margin for designing a polynomial approximation that remains bounded over the entire signal interval. Because the resulting vectors $f_1(A)|b\rangle$ and $f_2(A)|b\rangle$ differ only by a scalar factor, their *normalized* quantum states are equal, but their postselection probabilities differ by a factor of four. Mixing the two scales within a probability calculation is therefore a substantive error.

The inverse $1/x$ cannot be approximated uniformly by a bounded polynomial over the full interval $[-1, 1]$ because it has a singularity at zero. The promise $\mathrm{spec}(A) \subseteq$

$[-1, -1/\kappa] \cup [1/\kappa, 1]$ restricts approximation to a domain where the reciprocal is finite. A QSP polynomial must nonetheless satisfy the necessary global unitary bounds, so the desired polynomial is not simply the truncated Taylor expansion of $1/x$ at the origin.

12.11.3 A two-dimensional inverse worked from first principles

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1/\kappa \end{pmatrix}, \qquad |b\rangle = |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad \kappa \geq 1.$$

The inverse matrix is $A^{-1} = \mathrm{diag}(1, \kappa)$, hence

$$A^{-1}|b\rangle = \frac{|0\rangle + \kappa|1\rangle}{\sqrt{2}}, \qquad \|A^{-1}|b\rangle\|_2 = \sqrt{\frac{1+\kappa^2}{2}}.$$

The normalized exact solution is

$$\boxed{|x\rangle = \frac{|0\rangle + \kappa|1\rangle}{\sqrt{1+\kappa^2}}.}$$

Using $f_1(x) = 1/(\kappa x)$, the ideal transformed matrix is $f_1(A) = \mathrm{diag}(1/\kappa, 1)$. Its unnormalized good branch is

$$f_1(A)|b\rangle = \frac{\kappa^{-1}|0\rangle + |1\rangle}{\sqrt{2}}.$$

The probability of selecting the good branch is

$$\boxed{p_1 = \frac{1+\kappa^{-2}}{2} = \frac{\kappa^2+1}{2\kappa^2}.}$$

Dividing the good branch by $\sqrt{p_1}$ yields $(|0\rangle + \kappa|1\rangle)/\sqrt{1+\kappa^2}$, exactly the target solution state. The scalar rescaling disappears after normalization, as required.

With $f_2(x) = 1/(2\kappa x)$, the good branch is half as large, hence its probability is $p_2 = p_1/4 = (\kappa^2+1)/(8\kappa^2)$, but its normalized state is unchanged. This pair of formulas provides a direct arithmetic test for the intended reciprocal normalization.

Example 12.3 (Numerical comparison at condition number four)**.** Take $\kappa = 4$. Then $|x\rangle = (|0\rangle + 4|1\rangle)/\sqrt{17}$. Under f_1, the success probability is $p_1 = (1+1/16)/2 = 17/32 \approx 0.53125$. Under f_2, it is $p_2 = 17/128 \approx 0.1328125$. Both conventions prepare the same normalized solution when postselection succeeds. The success probabilities differ, so a complexity analysis must say which convention is used.

12.11.4 Dependence on the input's spectral support

For the same matrix, take a general normalized input $|b_\gamma\rangle = \cos\gamma|0\rangle + \sin\gamma|1\rangle$. Then

$$A^{-1}|b_\gamma\rangle = \cos\gamma|0\rangle + \kappa\sin\gamma|1\rangle, \quad \|A^{-1}|b_\gamma\rangle\|^2 = \cos^2\gamma + \kappa^2\sin^2\gamma.$$

Under f_1, the postselection probability is

$$p_{1,\gamma} = \frac{\cos^2\gamma}{\kappa^2} + \sin^2\gamma.$$

If $|b\rangle = |0\rangle$ (the largest-eigenvalue eigenspace), then $p = 1/\kappa^2$. If $|b\rangle = |1\rangle$ (the smallest-eigenvalue eigenspace), then $p = 1$. This direction dependence explains why worst-case amplitude amplification is proportional to κ even though particular input states can have constant postselection probability.

More generally, for $A = \sum_j \lambda_j|v_j\rangle\langle v_j|$ and $|b\rangle = \sum_j \beta_j|v_j\rangle$, the inverse-transformed good branch has squared norm

$$\|f_1(A)|b\rangle\|^2 = \sum_j \frac{|\beta_j|^2}{\kappa^2\lambda_j^2}.$$

For $\lambda_j \in [-1, -1/\kappa] \cup [1/\kappa, 1]$ and normalized b, this lies between $1/\kappa^2$ and 1. The bounds are saturated when the input occupies the extreme spectral directions. A condition number alone thus provides only a worst-case guarantee; the actual success probability depends on the right-hand side.

12.11.5 Approximation error before and after normalization

Suppose $p(A)$ approximates a scaled inverse $f(A)$ in operator norm with $\|p(A) - f(A)\|_2 \le \delta$. Since $\|b\| = 1$,

$$\|p(A)|b\rangle - f(A)|b\rangle\| \le \delta.$$

This is an *unnormalized branch* error. The desired output, however, is a normalized state. For nonzero vectors u, v,

$$\left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\| \le \frac{\|u - v\|}{\|v\|} + \frac{\|\|v\| - \|u\|\|}{\|v\|} \le \frac{2\|u - v\|}{\|v\|}.$$

The first inequality follows by adding and subtracting $u/\|v\|$ and the second by the reverse triangle inequality. For the scaled inverse $f_2(A) = A^{-1}/(2\kappa)$, one has $\|f_2(A)|b\rangle\| = \|A^{-1}|b\rangle\|/(2\kappa) \ge 1/(2\kappa)$ because all singular values of A^{-1} are at least one under $\|A\|_2 = 1$. Thus normalized-state error can be of order $\kappa\delta$. Requiring final error at most ε motivates choosing $\delta = O(\varepsilon/\kappa)$ rather than simply $\delta = O(\varepsilon)$.

The displayed vector-norm estimate bounds a natural distance between normalized pure-state representatives; converting it to trace distance uses standard inequalities relating Euclidean vector distance and fidelity. Numerical constants may differ depending on the chosen state-distance convention, but the factor κ caused by normalization is the key asymptotic mechanism.

12.11.6 The basic QSVT query-count calculation

Assume the scaled reciprocal can be approximated on the promised spectral interval by a bounded, parity-compatible polynomial of degree

$$d = O\left(\kappa \log \frac{1}{\delta}\right) = O\left(\kappa \log \frac{\kappa}{\varepsilon}\right)$$

for the target unnormalized approximation accuracy $\delta = O(\varepsilon/\kappa)$. Under a unit-cost matrix block-encoding oracle, one application of the corresponding QSVT transformation uses $O(d)$ calls to U_A and $U_A^\dagger$. The smallest possible successful good-branch norm for scaled inversion is $\Omega(1/\kappa)$, hence success probability is $\Omega(1/\kappa^2)$. Standard amplitude amplification introduces a further factor $O(\kappa)$ in the worst case. Combining these terms yields a baseline query count

$$O\left(\kappa^2 \log \frac{\kappa}{\varepsilon}\right).$$

This is the cost of the *basic construction under stated assumptions*, not a universal lower bound on quantum linear-system algorithms. Other algorithmic techniques, more refined amplification procedures, and different input models can improve dependencies. The displayed count also omits state preparation, implementation of controlled block-encodings, precision of synthesized phase angles, and any classical readout required by the final application.

12.11.7 Non-Hermitian systems by Hermitian embedding

For a general invertible $A \in \mathbb{C}^{N\times N}$, define the block-Hermitian matrix

$$\widetilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}.$$

It satisfies $\widetilde{A} = \widetilde{A}^\dagger$ and

$$\widetilde{A}^2 = \begin{pmatrix} AA^\dagger & 0 \\ 0 & A^\dagger A \end{pmatrix}.$$

Consequently the eigenvalues of $\widetilde{A}$ are the positive and negative singular values of A, counted with their multiplicities. When A is invertible, $\widetilde{A}$ is invertible, and solving

$$\widetilde{A}\begin{pmatrix} 0 \\ x \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}$$

is equivalent to $Ax = b$. This embedding allows Hermitian spectral methods to address non-Hermitian systems at the cost of a doubled vector space and whatever overhead is required to block-encode $\widetilde{A}$. It is therefore incorrect to assert that inverse-polynomial techniques categorically cannot apply to a non-Hermitian input; the precise statement is that the direct Hermitian argument must be supplemented by an appropriate reduction.

12.11.8 When a speedup claim is and is not justified

Suppose the matrix size is $N = 2^n$ and a quantum routine has query complexity polynomial in n, κ, and $\log(1/\varepsilon)$. One must still specify how A and b are accessed. If assembling O_b or U_A from classical dense input requires $\Omega(N)$ gates, a formal logarithmic-dimensional query count does not imply logarithmic end-to-end time. Likewise, requesting a complete classical description of the solution generally incurs measurement overhead that scales with the amount of output information. A useful speedup comparison therefore fixes the same input access model and output task for the quantum and classical algorithms. Valid quantum tasks include sampling from the solution state and estimating a small number of well-defined observables; a claim about producing all N solution coordinates requires additional justification.

12.12 Further numerical linear-system experiments and error bounds

12.12.1 A non-Hermitian two-by-two system and its Hermitian reduction

Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The original classical equation $Ax = b$ has solution $x = (-1, 1)^T$, since $A^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$. The normalized quantum target state is $(-|0\rangle + |1\rangle)/\sqrt{2}$, up to global phase. Although $A \neq A^\dagger$, the Hermitian embedding

$$\widetilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

acts on a four-dimensional enlarged space. Then

$$\widetilde{A} \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Thus the original solution appears in the lower component of the Hermitian system. Scaling the embedded matrix by its spectral norm is still required before treating it as a contraction, and any block-encoding cost for the larger matrix must be included in an algorithmic analysis.

12.12.2 Singular values and condition number of the example

For the matrix above,

$$A^\dagger A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its eigenvalues solve $\lambda^2 - 3\lambda + 1 = 0$, so they are $(3 \pm \sqrt{5})/2$. The singular values are $\sigma_\pm = \sqrt{(3 \pm \sqrt{5})/2}$, and the spectral condition number is

$$\kappa(A) = \frac{\sigma_+}{\sigma_-} = \sqrt{\frac{3+\sqrt{5}}{3-\sqrt{5}}} = \frac{3+\sqrt{5}}{2}.$$

The last equality uses $(3 + \sqrt{5})(3 - \sqrt{5}) = 4$. The eigenvalues of the Hermitian embedding are $\pm\sigma_+$ and $\pm\sigma_-$, so its condition number matches that of A under the spectral-norm definition. This demonstrates why the Hermitian reduction is formulated through singular values rather than the generally complex eigenvalues of a nonnormal input matrix.

12.12.3 Success probability depends on the spectral components of the right-hand side

Let $A = \mathrm{diag}(1, 1/\kappa)$ and $|b\rangle = \cos\gamma|0\rangle + \sin\gamma|1\rangle$. Under the scaled inverse $f_1(A) = \mathrm{diag}(1/\kappa, 1)$, the good probability is $p(\gamma) = \cos^2\gamma/\kappa^2 + \sin^2\gamma$. Differentiating with respect to the input angle gives

$$p'(\gamma) = 2\sin\gamma\cos\gamma\left(1 - \frac{1}{\kappa^2}\right).$$

For $\kappa > 1$ and $\gamma \in [0, \pi/2]$, this derivative is nonnegative, so increasing the input overlap with the small-eigenvalue eigenspace increases postselection probability. At the same time the unnormalized classical solution $A^{-1}|b\rangle$ becomes more strongly amplified in that direction. This does not contradict the operator-norm bound: the circuit implements the *scaled* inverse, whose maximum singular value is one.

12.12.4 A concrete normalization-sensitivity calculation

Suppose the ideal good vector is $v = (\eta, 0)^T$ with $0 < \eta < 1$ and an approximation changes it to $u = (\eta, \delta)^T$, with $0 < \delta \ll \eta$. Their unnormalized distance is $\|u - v\| = \delta$. The normalized states are $v/\|v\| = (1, 0)^T$ and

$$\frac{u}{\|u\|} = \frac{1}{\sqrt{\eta^2 + \delta^2}}(\eta, \delta)^T.$$

The second normalized component equals $\delta/\sqrt{\eta^2 + \delta^2} \approx \delta/\eta$. Thus an unnormalized perturbation of size δ becomes a normalized-state deviation of order δ/η. Setting

$\eta \approx 1/\kappa$ gives the familiar $\kappa\delta$ scaling. This example is not merely a loose inequality: it shows that normalization sensitivity can be realized geometrically, motivating the stronger approximation precision required for a poorly conditioned inverse operation.

12.12.5 Classical readout and the information content of a state

A normalized state $|x\rangle = \sum_j x_j|j\rangle$ has an exponential number of complex amplitudes in n qubits, but measurement in the computational basis yields a single index J with probability $|x_J|^2$. From S independent samples, one can estimate a particular probability with standard error at most $1/(2\sqrt{S})$; obtaining its complex phase requires additional interference or measurement bases. Learning all amplitudes to specified accuracy is a tomographic task whose resources are not captured by the oracle query count for preparing one copy of $|x\rangle$. On the other hand, if the application needs only $\mathbb{E}_{J\sim|x|^2}[g(J)]$ for a bounded function g, sample averaging may be sufficient, and no full state tomography is necessary. This is the operational reason that the output task must be fixed before comparing classical and quantum linear solvers.